L503C-Y/L511C-Y AT Command User Guide

LTE Module Series

Version： V1.40

Date： 2023-12-13

Shanghai Mobiletek Communication Ltd.

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Revision History

Date	Version	Description of change	Author
2021-10-26	V1.00	Initial.	George&ym.shu
2022-03-25	V1.01	Remove some unuseful commands and modify some errors.	Leo
2022-05-06	V1.02	Add some new HTTP&MQTT AT commands	Bo.yang
2022-05-26	V1.03	Add AT+ECSNTP and AT+ECWIFISCAN command and modify some errors.	Leo
2022-06-02	V1.04	Add FTP AT commands	Bo.yang
2022-06-02	V1.05	Add LYNQ TCP/IP command	ym.shu
2022-06-10	V1.06	Update LYNQ MQTT command	Bo.yang
2022-06-30	V1.07	Add FTP error code	Bo.yang

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2022-07-01	V1.08	Add AT+CGNETLED, AT+RESET command and modify some errors.	Leo
2022-07-01	V1.09	Add AT+SERVERSTART and AT+SERVERSTOP command	ym.shu
2022-07-06	V1.10	Modify some LYNQ TCP/IP command format	ym.shu
2022-07-12	V1.11	Modify TCP/IP and SSL command errors, add AT+FOTA command.	Leo
2022-07-25	V1.12	Update HTTP and FTP commands	Bo.yang
2022-08-31	V1.13	Fix some errors	Bo.yang
2022-09-08	V1.14	Modify AT+SKTSTATUS number And add AT+ECDNS=? Modify AT+ ECBAND describe	Bo.yang
2022-09-14	V1.15	Add AT+GSN,AT+GMR and AT+IPADDR command	ym.shu
2022-09-22	V1.16	Modify AT+CGNETLED parameter	Bo.yang
2022-09-27	V1.17	Fix AT+HTTPCFG BUG	Bo.yang
2022-10	V1.18	Modify command para error	Leo

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2022-10-28	V1.19	Add AT+GMI,AT+GMM, AT+ISLKVRSCAN	Gn.tian
2022-11-09	V1.20	Modify some errors.	Leo
2023-02-01	V1.21	Modify some errors.	Leo
2023-02-09	V1.22	Add MQTTS/HTTPS Demo	ym.shu
2023-02-13	V1.23	Update HTTP error code and SKTCREATE max link number	ym.shu
2023-03-09	V1.24	Add AT+ECMTCFG="ssl","enable" command	ym.shu
2023-03-13	V1.25	Add the maximum response time for each command and Modify some errors	Bo.yang
2023-03-20	V1.26	Add CTWing command and AT+HVER command	ym.shu
2023-04-04	V1.27	Add CTM2M command and Demo	ym.shu
2023-04-07	V1.28	Add +SKTKEEPALIVE command	Bo.yang
2023-04	V1.29	Update CTM2MSETMOD command	Gn.tian

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2023-05-19	V1.30	Add OneNET command	ym.shu
2023-05-19	V1.31	Add L503C-Y7S not support note	Gn.tian
2023-05-24	V1.32	Modify AT+ECSNTP,AT+CGMI add AT+QNTP	Cj.zhou
2023-07-06	V1.33	ADD HTTPURL HTTPPARA IPV6 PROMPT,update L503C-Y7S not support note	Gn.tian
2023-09-05	V1.34	Update AT+SSLOPEN, AT+SSLSEND, AT+SSLCLOSE	Cj.zhou
2023-09-11	V1.35	ADD CTW errcode Update AT+CFTPDELE	Cj.zhou
2023-09-12	V1.36	Update AT+CTM2MSETPSK, AT+CTM2MSEND	Cj.zhou
2023-09-18	V1.37	AT+HTTPGET, AT+MIPLREADSP Update AT+HTTPURL,AT+FWRITE ,AT+ECMTSUB,AT+QNTP	Cj.zhou
2023-11-03	V1.38	Add LBS commands	Bo.yang

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2023-12-06	V1.39	Modify some errors and add AT+CFTPSIZE.	ym.shu
2023-12-13	V1.40	Add AT+HTTPFOTADL command	Bo.yang

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1 Introduction

1.1 Overview

This document introduces the supported AT command set of L503C-Y/L511C-Y product.

1.2 References

[1] 3GPP TS 27.007 V3.13.0 (2003-03)
[2] ETSI TS 27.005 V3.1.0 (2000-01)
[3] ITU-T V.25 ter (07/1997)

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2 V.25ter AT Commands

2.1 ATE - Set echo

This command determines whether or not the DCE echoes characters received from the DTE during command state and online command state.

2.1.1 Format

Command	Response
ATE[<value>]	OK
Max response time	300 ms

2.1.2 Field

Parameters	Description
<value>	0: DCE does not echo characters during command state and online command state. 1: DCE echoes characters during command state and online command state.

2.2 ATI - Request identification information

2.2.1 Format

Command	Response
ATI	<text> OK
Max response time	300 ms

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2.2.2 Field

Parameters	Description
<text>	Product information.

2.3 ATQ - Set result code suppression mode

2.3.1 Format

Command	Response
ATQ[<value>]	OK If value is 0. (none) If value is 1 (because result codes are suppressed). ERROR For unsupported values (if previous value was Q0). (none) For unsupported values (if previous value was Q1).
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

2.3.2 Field

Parameters	Description
<value>	0: DCE transmits result codes. 1: Result codes are suppressed and not transmitted.

2.4 ATO - Set Data mode

2.4.1 Format

Command	Response
ATO[<value>]	CONNECT
Max response time	300 ms

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2.4.2 Field

Parameters	Description
<value>	0 Switch from command mode to data mode.

2.5 AT+IPR - Set UE baud rate

2.5.1 Format

Command	Response
AT+IPR=<rate>	OK Or ERROR
AT+IPR=?	+IPR: (list of fixed-only <rate> values) OK
AT+IPR?	+IPR: <rate> OK
Max response time	300 ms

2.5.2 Field

Parameters	Description
<rate>	Baud rate at which the UE will accept commands. 600,1200,2400,4800,9600,19200,38400,57600,115200,230400,460800 ,921600

2.6 AT+CGMI - Request manufacturer identification

The command causes the phone to return one or more lines of information text <manufacturer> which is intended to permit the user of the ITAE/ETAE to identify the manufacturer of the phone to which it is connected to.

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2.6.1 Format

Command	Response
AT+CGMI	<manufacturer> OK
AT+CGMI=?	+CGMI:<manufacturer_ID> OK
Max response time	300 ms

2.7 AT+CGMM - Request model identification

The command causes the phone to return one or more lines of information text <model> which is intended to permit the user of the ITAE/ETAE to identify the specific model of phone to which it is connected to.

2.7.1 Format

Command	Response
AT+CGMM	<model> OK
AT+CGMM=?	+CGMM:<list of supported technologies>,<model> OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

2.8 AT+CGMR - Request revision identification

The command causes the phone to return a string containing information regarding SW version.

2.8.1 Format

Command	Response
AT+CGMR	<SW_version> OK
AT+CGMR=?	OK

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Max response time	300 ms

2.9 AT+HVER - Request hardware version

The command causes the phone to return a string containing information regarding hardware version.

2.9.1 Format

Command	Response
AT+HVER	<HW_version> OK
AT+HVER=?	OK
Max response time	300 ms

2.10 AT+CGSN - Request product serial number identification

2.10.1 Format

Command	Response
AT+CGSN Or AT+CGSN=<snt>	when <snt>=0 and command successful: <sn> OK when <snt>=1 (or omitted) and command successful: <imei> OK when <snt>=2 and command successful: <imeisv> OK when <snt>=3 and command successful: <sv> OK
AT+CGSN=?	+CGSN: (list of supported <snt>s) OK
Max response time	300 ms

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2.10.2 Field

Parameters	Description
<snt>	Integer type indicating serial number type that has been requested 0 returns <sn> 1 returns IMEI 2 returns IMEISV (IMEI & SV) 3 returns SV
<sn>	One or more lines of information text determined by the MT manufacturer (not support now)
<imei>	String type; in decimal format indicating the IMEI.
<imeisv>	String type; in decimal format indicating the IMEISV.
<sv>	String type; in decimal format indicating the current SV which is a part of IMEISV.

2.11 AT+GSN - Request to get IMEI

2.11.1 Format

Command	Response
AT+GSN	<IMEI> OK Or +CME ERROR: <err>
Max response time	300 ms

2.12 AT+GMR - Request revision identification

2.12.1 Format

Command	Response
AT+GMR	<SW_version> OK
AT+GMR=?	OK

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2.13 AT+GMI - Request manufacturer identification

2.13.1 Format

Command	Response
AT+GMI	<manufacturer> OK
AT+GMI=?	+CGMI:<manufacturer_ID> OK
Max response time	300 ms

2.14 AT+GMM - Request model identification

2.14.1 Format

Command	Response
AT+GMM	<model> OK
AT+GMM=?	+CGMM:<list of supported technologies>,<model> OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

2.15 AT+ISLKVRSCAN - Request revision identification

2.15.1 Format

Command	Response
AT+ISLKVRSCAN	+ISLKVRSCAN: <SW_version> OK
Max response time	300 ms

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2.16 AT+CIMI - Request international mobile subscriber identity

2.16.1 Format

Command	Response
AT+CIMI	<IMSI> OK Or +CME ERROR: <err>
AT+CIMI=?	OK
Max response time	300 ms

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3 Network service related command

3.1 AT+COPS - Operator selection

The set command forces an attempt to select and register the network operator using the USIM card installed in the currently selected card slot. <mode> is used to select whether the selection is done automatically by the MT or is forced by this command to operator <oper> (it shall be given in format <format>) to a certain access technology, indicated in <AcT>. If the selected operator is not available, no other operator shall be selected (except <mode>=4). If the selected access technology is not available, then the same operator shall be selected in other access technology. The selected operator name format shall also apply to further read commands (AT+COPS?). <mode>=2 forces an attempt to deregister from the network. The selected mode affects all further network registration (e.g. after <mode>=2, MT shall be unregistered until <mode>=0 or 1 is selected). This command should be abortable when registration/deregistration attempt is made.

The read command returns the current mode, the currently selected operator and the current access technology. If no operator is selected, <format>, <oper> and <AcT> are omitted.

The test command returns a set of five parameters, each representing an operator present in the network. A set consists of an integer indicating the availability of the operator <stat>, long and short alphanumeric format of the operator's name, numeric format representation of the operator and access technology. Any of the formats may be unavailable and should then be an empty field. The list of operators shall be in order: home network, networks referenced in USIM or active application in the UICC (GSM or USIM) in the following order: HPLMN selector, user controlled PLMN selector, operator controlled PLMN selector and PLMN selector (in the USIM or GSM application), and other networks. After the operator list MT returns lists of supported <mode>s and <format>s. These lists shall be delimited from the operator list by two commas.

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3.1.1 Format

Command	Response
AT+COPS=<mode>[,<format>[,<oper>[,<Act>]]]	OK Or +CME ERROR: <err>
AT+COPS?	+COPS: <mode>[,<format>,<oper>[,<AcT>]] OK Or +CME ERROR: <err>
AT+COPS=?	+COPS: [list of supported (<stat>, long alphanumeric <oper>, short alphanumeric <oper>,numeric <oper>[,<Act>])s],,(list of supported <mode>s),(list of supported <format>s) OK Or +CME ERROR: <err>
Max response time	60s, depend on network status

3.1.2 Field

Parameters	Description
<mode>	0: Automatic mode; <oper> field is ignored. 1: Manual operator selection; <oper> field shall be present. 2: Manual deregister from network. 3: Set only <format> (for +COPS?) – not shown in response. 4: Manual / automatic selected; if manual selection fails, automatic mode (<mode> = 0) is entered.
<format>	0: Long format alphanumeric <oper>; can be up to 16 characters long. 1: Short format alphanumeric <oper>. 2: Numeric <oper>; GSM Location Area Identification number.
<oper>	String type <format> indicates if the format is alphanumeric or numeric; long alphanumeric format can be up to 16 characters long and short format up to 8 characters; numeric format is the GSM location area identification number which consists of a three BCD digit ITU-T country code coded, plus a two or three BCD digit network code, which is administration specific.

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<stat>	0: Unknown. 1: Available. 2: Current. 3: Forbidden.
<Act>	7: LTE.

3.2 AT+CLCK - Facility lock

This command is used to lock, unlock or interrogate a ME or a network facility <fac>. Password is normally needed to do such actions. When querying the status of a network service (<mode>=2) the response line for 'not active' case (<status>=0) should be returned only if service is not active for any class. This command should be abortable when network facilities are set or interrogated.

3.2.1 Format

Command	**Response**
AT+CLCK=<fac>,<mode>[,<passwd>]	OK Or +CME ERROR: <err> when <mode>=2 and command successful: +CLCK: <status> OK
AT+CLCK=?	+CLCK: (list of supported <fac>s) OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

3.2.2 Field

Parameters	**Description**
<fac>	"SC" SIM (lock SIM/UICC card installed in the currently selected card slot) (SIM/UICC asks password in MT power-up and when this lock command issued)
<mode>	0: Unlock. 1: Lock. 2: Query status.

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<status>	0: Not active. 1: Active.
<passwd>	String type; shall be the same as password specified for the facility from the MT user interface or with command Change Password +CPWD

3.3 AT+CPWD - Change password

Action command sets a new password for the facility lock function defined by command facility lock +CLCK.

3.3.1 Format

Command	**Response**
AT+CPWD=<fac>,<oldpwd>,<newpwd>	OK Or +CME ERROR: <err>
AT+CPWD=?	+CPWD: list of supported(<fac>,<pwdlength>)s OK Or +CME ERROR: <err>
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

3.3.2 Field

Parameters	**Description**
<fac>	"SC" SIM (lock SIM/UICC card installed in the currently selected card slot) (SIM/UICC asks password in MT power-up and when this lock command issued)
<oldpwd> <newpwd>	<oldpwd> shall be the same as password specified for the facility from the MT user interface or with command Change Password +CPWD and <newpwd> is the new password, maximum length of password can be determined with <pwdlength>
<pwdlength>	Integer type, maximum length of the password for the facility.

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3.4 AT+CESQ - Received signal quality

Execution command returns received signal quality parameters. Since it only supports NB-IoT <ber> and <rxlev> are set to value 99, <rscp> and <ecno> is set to 255.

3.4.1 Format

Command	Response
AT+CESQ	+CESQ: <rxlev>,<ber>,<rscp>,<ecno>,<rsrq>,<rsrp> OK Or If error is related to wrong AT syntax: ERROR
AT+CESQ=?	+CESQ: (list of supports <rxlev>s), (list of supports <ber>s), (list of supports <rscp>s), (list of supports <ecno>s), (list of supports <rsrq>s), (list of supports <rsrp>s) OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

3.4.2 Field

Parameters	Description
<rxlex>	Integer type. Rx signal strength level. 99: Not known or not detectable.
<ber>	Integer type; channel bit error rate (in percent). 99: Not known or not detectable.
<rscp>	Integer type, received signal code power (see 3GPP 25.133 and 3GPP 25.123). 255: not known or not detectable.
<ecno>	Integer type, Ec/No (see 3GPP 25.133). 255: not known or not detectable.

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<rsrq>	Integer type, reference signal received quality (see 3GPP 36.133). 0: -19.5 dB or less. 1: -19.5dB <= rsrq < -19 dB. 2: -19dB <= rsrq < -18.5 dB. …… 32: -4 dB <= rsrq < -3.5 dB. 33: -3.5 dB <= rsrq < -3 dB. 34: -3 dB <= rsrq. 255: not known or not detectable.
<rsrp>	Integer type, reference signal received power (see 3GPP 36.133). 0: -140 dBm or less. 1: -140dBm <= rsrp < -139 dBm. 2: -139dBm <= rsrp < -138 dBm. …… 95: -46dBm <= rsrp < -45 dBm. 96: -45dBm <= rsrp < -44 dBm. 97: -44dBm <= rsrp. 255: not known or not detectable.

3.5 AT+CSQ - Signal quality

Execution command returns received signal strength indication <rssi> and channel bit error rate <ber> from the MT. Test command returns values supported as compound values.

3.5.1 Format

Command	**Response**
AT+CSQ	+CSQ: <rssi>,<ber> OK If error is related to wrong AT syntax: ERROR
AT+CSQ=?	+CSQ: (list of supports <rssi>s), (list of supported <ber>s) OK
Max response time	300 ms

3.5.2 Field

Parameters	**Description**

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<rssi>	Integer type 0: -113 dBm or less. 1: -111 dBm. 2...30: -109... -53 dBm. 31: -51 dBm or greater. 99: not known or not detectable.
<ber>	Integer type; channel bit error rate (in percent). 0…7: as RXQUAL values RXQUAL_0…RXQUAL_7 as defined in 45.008. 99: not known or not detectable.

3.6 AT+ECDNS –Get IP address for specific domain name

This command can be used to get the IP address for specific domain name. Because of the limitation, currently, only one IP address will be returned for the specific domain name. All the IP address numbers are decimalism.

3.6.1 Format

Command	**Response**
AT+ECDNS=<domain name>	+ECDNS: <IP addr> OK Or +SOCKET ERROR: <err>
AT+ECDNS=?	OK
Max response time	30s, depend on network status

3.6.2 Field

Parameters	**Description**
<domain name>	String format, server domain address or IP address.

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3.7 AT+ECDNSCFG – Set DNS Server

3.7.1 Format

Command	Response
AT+ECDNSCFG=?	OK
AT+ECDNSCFG=<ipadd1>,[<ipadd2>,[<ipadd3>,[<ipadd4>]]]	OK Or +SOCKET ERROR: <err>
AT+ECDNSCFG?	+ECDNSCFG: <ipaddr1>[,<ipaddr2>[,<ipaddr3>[,<ipaddr4>]]] OK
Max response time	300 ms

3.7.2 Field

Parameters	Description
<ipaddr>	The DNS Server IP address. If IPv4 type, output is dot-notation format, such as: “114.114.114.114” If IPv6 type, output is colon-notation format, such as: “2001:0DB8:0000:CD30:0000:0000:0000:0002” Note: IPv4 and IPv6 each hold up to two addresses

3.8 AT+ECPING - Test IP network connectivity to a remote host

This command sends an ICMP packet to the specified host address. AT+ECPING initiates the sending of a PING packet with payload size: <size> to the specified address. This will either cause a packet to be returned if the remote system is connected and responding to PING packets or no response will be received. If none of the response packet received within the timeout period <timeout>. It will continue to send PING packet until the <count> number of times.

3.8.1 Format

Command	Response

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AT+ECPING=[<ipaddr/Url>/<mode>[,<count>[,<size>[,<timeout>]]]]	OK If there is any error, response: SOCKET ERROR: <err>
AT+ECPING=?	+ECPING:(list of supported <ipaddr/Url/mode>s),(list of supported <count>s),(list of supported <size>s),(list of supported <timeout>s) OK
	Note: a) When one PING reply received in <timeout>, an unsolicited result code: +ECPING: SUCC, dest:<dest_ip_addr>, RTT: <rtt_time>ms will sent to TE. b) If no PING reply received in <timeout>, an unsolicited result code: +ECPING: FAIL, dest:<dest_ip_addr>, time out: <timeout>ms will sent to TE. c) If this is an ERROR meet during PING procedure, an unsolicited result code: +ECPING: ERROR, cause: <cause> will sent to TE. d) When PING procedure is done, an unsolicited result code: +ECPING: DONE<CR><LF>+ECPING:dest: <dest_ip_addr>, <count> packets transmitted, <reply_count> received, <lost_percent>% packet loss<CR> rtt min/avg/max = <rtt _min> / <rtt_avg> / <rtt_max> ms will sent to TE.
Max response time	5 s

3.8.2 Field

Parameters	**Description**
<ipaddr/Url>	String type, IP address or URL.
<mode>	Integer type 0 Stop ping
<count>	Integer type, ping times 1-255 Note: default value: 4. 255 means all the time.
<size>	Integer type, payload size 1-1500 Note: default value: 32
<timeout>	Integer type, UE ping reply timeout after ping request.(ms) Note: <timeout> values between 1000 to 600000 are supported, and default value: 20000

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4 MT control and status command

4.1 AT+CFUN - Set phone functionality

Set command selects the level of functionality in the MT. Level "full functionality" is where the highest level of power is drawn. "Minimum functionality" is where minimum power is drawn.

4.1.1 Format

Command	Response
AT+CFUN=<fun>[,<rst>]	OK Or +CME ERROR: <err>
AT+CFUN=?	+CFUN: (list of supported <fun>s), (list of supported <rst>s) OK
AT+CFUN?	+CFUN: <fun> OK
	Note: Settings 2 and 3 of <fun> are not supported. <fun> = 0 performs all system shutdown actions without removing power. <fun> = 1 performs a system startup. <rst> = 0 shall always be the default if the <rst> parameter is not given.
Max response time	60s, depend on network status

4.1.2 Field

Parameters	Description
<fun>	0: Minimum functionality. 1: Full functionality (Default). 4: Disable phone both transmit and receive RF circuits.
<rst>	0: Do not reset the MT before setting it to <fun> power level. This shall always be defaulted when <rst> is not given. 1: Reset the MT before setting it to <fun> power level. (not supported and will be ignored)

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4.2 AT+CPIN - Enter PIN

This command sends to the ME a password which is necessary before it can be operated (SIM PIN, SIM PUK, etc.). If the PIN is to be entered twice, the TA shall automatically repeat the PIN. If no PIN request is pending, no action is taken towards ME and an error message, +CME ERROR, is returned to TE.

If the PIN required is SIM PUK or SIM PUK2, the second pin is required. This second pin, <newpin>, is used to replace the old pin in the SIM.

4.2.1 Format

Command	Response
AT+CPIN=<pin>[,<new pin>]	OK Or +CME ERROR: <err>
AT+CPIN?	+CPIN: <code> OK
AT+CPIN=?	OK
Max response time	300 ms

4.2.2 Field

Parameters	Description
<pin>	String type values.
<newpin>	String type values.
<code>	READY: No further entry needed. SIM PIN: ME is waiting for SIM PIN. SIM PUK: ME is waiting for SIM PUK.

4.3 AT+CCLK - Clock

Read command returns the current setting of the clock.

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4.3.1 Format

Command	Response
AT+CCLK=<time>	OK Or +CME ERROR: <err>
AT+CCLK?	+CCLK: <time> OK
AT+CCLK=?	OK
Max response time	300 ms

4.3.2 Field

Parameters	Description
<time>	string type value; format is “yy/MM/dd,hh:mm:ss±zz”; where characters indicate year (two last digits), month, day, hour, minutes and seconds and time zone (indicates the difference, expressed in quarters of an hour, between the local time and GMT; range -47...+48). E.g. 6th of May 2014, 22:10:00 GMT+2 hours equals to "14/05/06,22:10:00+08"
	Note: the year should be after 2000 years, otherwise there will be asserted

4.4 AT+CGDCONT - Define PDP context

The set command specifies PDP context parameter values for a PDP context identified by the (local) context identification parameter, <cid>. It also allows the TE to specify whether security protected transmission of ESM information is requested, because the PCO can include information that requires ciphering. There can be other reasons for the UE to use security protected transmission of ESM information, e.g. if the UE needs to transfer an APN. The number of PDP contexts that may be in a defined state at the same time is given by the range returned by the test command.

For EPS the PDN connection and its associated EPS default bearer is identified herewith. For EPS the <PDP_addr> shall be omitted.

A special form of the set command, +CGDCONT=<cid> causes the values for context number <cid>

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to become undefined.

The read command returns the current settings for each defined context.

The test command returns values supported as a compound value.

4.4.1 Format

Command	Response
AT+CGDCONT=<cid>[,<PDP_type>[,<APN>[,<PDP_addr>[,<d_comp>[,<h_comp>[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>[,<NSLPI>[,<securePCO>[,<IPv4_MTU_discovery>][,<Local_Addr_Ind>]]]]]]]]]]	OK Or ERROR
AT+CGDCONT?	[+CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>[,<NSLPI>[,<securePCO>[,<IPv4_MTU_discovery>[,<Local_Addr_Ind>]]]]]]]][<CR><LF> +CGDCONT: <cid>,<PDP_type>,<APN>,<PDP_addr>,<d_comp>,<h_comp>[,<IPv4AddrAlloc>[,<request_type>[,<PCSCF_discovery>[,<IM_CN_Signalling_Flag_Ind>[,<NSLPI>[,<securePCO>[,<IPv4_MTU_discovery>[,<Local_Addr_Ind>]]]]]]][...]] OK
AT+CGDCONT=?	+CGDCONT: (range of supported <cid>s),<PDP_type>,,,(list of supported <d_comp>s),(list of supported <h_comp>s),(list of supported <IPv4AddrAlloc>s),(list of supported <request_type>s),(list of supported <PCSCF_discovery>s),(list of supported <IM_CN_Signalling_Flag_Ind>s),(list of supported <NSLPI>s),(list of supported <securePCO>s),(list of supported <IPv4_MTU_discovery>s),(list of supported <Local_Addr_Ind>s),(list of supported <Non-IP_MTU_discovery>s)[<CR><LF> +CGDCONT: (range of supported <cid>s),<PDP_type>,,,(list of supported <d_comp>s),(list of supported <h_comp>s),(list of supported <IPv4AddrAlloc>s),(list of supported <request_type>s),(list of supported

	<PCSCF_discovery>s),(list of supported <IM_CN_Signalling_Flag_Ind>s) ,(list of supported <NSLPI>s),(list of supported <securePCO>s,(list of supported <IPv4_MTU_discovery>s),(list of supported <Local_Addr_Ind>s), [...]] OK
Max response time	5 s

4.4.2 Field

Parameters	Description
<cid>	(PDP Context Identifier) a numeric parameter that specifies a particular PDP context definition. The parameter is local to the UE-TE interface and is used in other PDP context-related commands. The range of permitted values is returned by the test form of the command. <cid> values of 1 - 15 are supported.
<PDP_type>	IP: Internet Protocol (IETF STD 5). IPV6: Internet Protocol, version 6 (IETF RFC 2460). IPV4V6: Virtual <PDP_type> introduced to handle dual IP stack UE capability (see 3GPP Technical Specifications 24.301).
<APN>	(Access Point Name) a string parameter which is a logical name that is used to select the GGSN or the external packet data network. If the value is null or omitted, then the subscription value will be requested. Max length is 98.
<PDP_address>	A string parameter that identifies the MT in the address space applicable to the PDP. If the value is null or omitted, then a value may be provided by the TE during the PDP startup procedure or, failing that, a dynamic address will be requested. The read form of the command will continue to return the null string even if an address has been allocated during the PDP startup procedure. The allocated address may be read using the +CGPADDR command.
<d_comp>	A numeric parameter that controls PDP data compression (applicable for SNDCP only) (refer 3GPP TS 04.65). Integer type; don't need for NB-IOT
<h_comp>	A numeric parameter that controls PDP header compression (refer 3GPP TS 04.65). Integer type; don't need for NB-IOT
<IPv4_MTU_discovery>	Integer type; influences how the MT/TA requests to get the IPv4 MTU size, see 3GPP TS 24.008 sub-clause 10.5.6.3. 0: Preference of IPv4 MTU size discovery not influenced by +CGDCONT. 1: Preference of IPv4 MTU size discovery through NAS signaling.

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	Note: If the initial PDP context is supported, the context with <cid>=0 is automatically defined at startup. As all other contexts, the parameters for <cid>=0 can be modified with +CGDCONT. If the initial PDP context is supported, +CGDCONT=0 resets context number 0 to its particular default settings. Parameters: <IPv4AddrAlloc>,<request_type>,<PCSCF_discovery>, <IM_CN_Signalling_Flag_Ind>,<NSLPI>,<securePCO>,<Local_Addr_Ind> are not supported.

4.5 AT+CGATT - PS attach or detach

The execution command is used to attach the MT to, or detach the MT from, the packet domain service. If the MT is already in the requested state, the command is ignored and the OK response is returned. If the requested state cannot be achieved, +CME E RROR response is returned. Any active PDP contexts will be automatically deactivated when the attachment state changes to detached.

4.5.1 Format

Command	**Response**
AT+CGATT=<state>	OK Or ERROR
AT+CGATT?	+CGATT: <state> OK
AT+CGATT=?	+CGATT: (list of supported <state>s) OK
Max response time	45s, depend on network status

4.5.2 Field

Parameters	**Description**
<state>	Indicates the state of PS attachment 0: Detached 1: Attached

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4.6 AT+CGACT - PDP context activate or deactivate

The set command is used to activate or deactivate the specified PDP context. If any PDP context is already in the requested state, the state for that context remains unchanged. If the requested state for any specified context cannot be achieved, a +CME ERROR response is returned. If the MT is not PS attached when the activation form of the command is executed, the MT first performs a PS attach and then attempts to activate the specified contexts. If the attach fail, then the MT responds with +CME ERROR.

For EPS, if an attempt is made to disconnect the last PDN connection, then the MT responds with a +CME ERROR.

For EPS, the activation request for an EPS bearer resource will be answered by the network by either an EPS dedicated bearer activation or EPS bearer modification request. The request must be accepted by the MT before the PDP context can be set in to established state.

4.6.1 Format

Command	Response
AT+CGACT=<state>,<cid>	OK Or ERROR
AT+CGACT?	[+CGACT: <cid>,<state>] [<CR><LF>+CGACT:<cid>,<state> […]] OK
AT+CGACT=?	+CGACT: (list of supported <state>s) OK
Max response time	45s, depend on network status

4.6.2 Field

Parameters	Description

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<state>	Integer type; indicates the activation state of PDP context activation 0: Deactivated 1: Activated
<cid>	Integer type; specifies a particular PDP context definition. Only one <cid> can be activated or deactivated at the same time. <cid> values of 1-15 are supported.

4.7 AT+CGPADDR - Show PDP address

4.7.1 Format

Command	Response
AT+CGPADDR[=<cid>]	+CGPADDR: <cid>[,<PDP_addr_1>[,<PDP_addr_2]] [<CR><LF>+CGPADDR:<cid>[,<PDP_addr_1>[,<PDP_addr_2]]] […] OK Or ERROR
AT+CGPADDR=?	+CGPADDR: (list of defined and actived <cid>s) OK
Max response time	300 ms

4.7.2 Field

Parameters	Description
<cid>	A numeric parameter which specifies a particular PDP context definition (see +CGDCONT command). If no <cid> is specified, the addresses for all defined contexts are returned. <cid> values of 1 - 15 are supported.
<PDP_addr_1> <PDP_addr_2>	<PDP_addr_1> and <PDP_addr_ 2>: each is a string type that identifies the MT in the address space applicable to the PDP. Both <PDP_addr_1> and <PDP_addr_2> are omitted if none is available. Both <PDP_addr_1> and <PDP_addr_2> are included when both IPv4 and IPv6 addresses are assigned, with <PDP_addr_1> containing the IPv4 address and <PDP_addr_2> containing the IPv6 address. The string is given as dot-separated numeric (0-255) parameter of the form: a1.a2.a3.a4 for IPv4 and a1.a2.a3.a4.a5.a6.a7.a8.a9.a10.a11.a12.a13.a14.a15.a16 for IPv6.

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4.8 AT+CEREG - EPS network registration status

This command controls the presentation of an unsolicited result code +CEREG: <stat> when <n>=1 and there is a change in the UE's EPS network registration status, or result code +CEREG: <stat>[,<tac>,<ci>[,<AcT>]] when <n>=2 and there is a change of the network cell. The parameters <AcT>, <tac> and <ci> are provided only if available. The value <n>=3 further extends the unsolicited result code with [,<cause_type>,<reject_cause>], when available, when the value of <stat> changes.

If the UE requests PSM for reducing its power consumption, the set command controls the presentation of an unsolicited result code: +CEREG:<n>,<stat>[,[<tac>],[<ci>],[<AcT>][,[<cause_type>],[<reject_cause>][,[<Active Time>],[<Periodic TAU>]]]]. When <n>=4, the unsolicited result code will provide the UE with additional information for the active time value and the extended periodic TAU value if there is a change of the network cell in E-UTRAN. The value <n>=5 further enhances the unsolicited result code with <cause_type> and <reject_cause> when the value of <stat> changes. The parameters <AcT>,<tac>,<ci>,<cause_type>,<reject_cause>,<Active Time> and <Periodic TAU> are provided only if available.

The read command returns the status of result code presentation and an integer <stat> which shows whether the network has currently indicated the registration of the MT. Location information elements <tac>, <ci> and <AcT> if available, are returned only when <n>=2 and MT is registered in the network. The parameters [,<cause_type>,<reject_cause>], if available, are returned when <n>=3.

4.8.1 Format

Command	Response
AT+CEREG=?	+CEREG: (list of supported <n>s) OK
AT+CEREG=<n>	OK Or ERROR

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AT+CEREG?	when <n>=0, 1, 2 or 3 and command successful: +CEREG: <n>,<stat>[,[<tac>],[<ci>],[<AcT>[,<rac>][,<cause_type>,<reject_cause>]]] when <n>=4 or 5 and command successful: +CEREG: <n>,<stat>[,[<tac>],[<ci>],[<AcT>],[<rac>][,[<cause_type>],[<reject_cause>][,[<Active-Time>],[<Periodic-TAU>]]]] If error is related to wrong AT syntax or operation not allowed: +CME ERROR: <err>
Max response time	300 ms

4.8.2 Field

Parameters	Description
<n>	0: Disable network registration unsolicited result code. 1: Enable network registration unsolicited result code +CEREG: <stat>. 2: Enable network registration and location information unsolicited result code +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>],[<rac>]]. 3: Enable network registration, location information and EMM cause value information unsolicited result code +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>],[<rac>][,<cause_type>,<reject_cause>]]. 4: For a UE that wants to apply PSM, enable network registration and location information unsolicited result code +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>],[<rac>][,,[,[<Active-Time>],[<Periodic-RAU>],[<GPRS-READY-timer>]]]]. 5: For a UE that wants to apply PSM, enable network registration, location information and EMM cause value information unsolicited result code +CEREG: <stat>[,[<tac>],[<ci>],[<AcT>],[<rac>][,[<cause_type>],[<reject_cause>][,[<Active-Time>],[<Periodic-RAU>],[<GPRS-READY-timer>]]]].
<stat>	EPS registration status 0: Not registered, ME is not currently searching a new operator to register to. 1: Registered, home network. 2: Not registered, but ME is currently searching for a new operator to register to. 3: Registration denied. 4: Unknown. 5: Registered, roaming. 6: Registered for “SMS only”, home network (not applicable). 7: registered for “SMS only”, roaming (not applicable).

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<tac>	String type; two byte tracking area code in hexadecimal format (e.g. "00C3" equals 195 in decimal).
<ci>	String type; four byte GERAN/UTRAN/E-UTRAN cell ID in hexadecimal format.
<AcT>	Access technology of the registered network. 7: E-UTRAN 9: E-UTRAN (NB-S1 mode).
<cause_type>	Integer type; indicates the type of <reject_cause> 0: Indicates that <reject_cause> contains an EMM cause value, see 3GPP TS 24.301 Annex A. 1: Indicates that <reject_cause> contains a manufacturer-specific cause.
<reject_cause>	Integer type; contains the cause of the failed registration. The value is of type as defined by <cause_type>.
<Active-Time>	String type; one byte in an 8-bit format. Indicates the Active Time value (T3324) allocated to the UE in E-UTRAN. The code of Active Time to see the AT+CPSMS command.
<Periodic-TAU>	String type; one byte in an 8-bit format. Indicates the extended periodic TAU value (T3412) allocated to the UE in E-UTRAN. The code of Active Time to see the AT+CPSMS command.

4.9 AT+CSCON - Signaling Connection Status

The set command controls the presentation of an unsolicited result code +CSCON. If <n>=1, +CSCON: <mode> is sent from the MT when the connection mode of the MT is changed.

For NB-IoT, only <n>=1 is supported.

The mode of the MT refers to idle when no PS signaling connection and to connected mode when a PS signaling connection between UE and network is setup.

The read command returns the status of result code presentation and an integer <mode> which shows whether the MT is currently in idle mode or connected mode.

4.9.1 Format

Command	**Response**

AT+CSCON=?	+CSCON: (0,1) OK
AT+CSCON?	+CSCON: <n>,<mode> OK
AT+CSCON=<n>	OK If error is related to wrong AT syntax: +CME ERROR: <err>
	Unsolicited result code: +CSCON:<mode>
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

4.9.2 Field

Parameters	**Description**
<n>	Integer type: 0: Disable unsolicited result code. 1: Enable unsolicited result code +CSCON: <mode>.
<mode>	Integer type; indicates signaling connection status 0: Idle 1: Connected

4.10 AT+CGEREP - Packet domain event reporting

This command enables or disables sending of unsolicited result codes, +CGEV: XXX from UE to TE in the case of certain events occurring in the packet domain UE or the network. <mode> controls the processing of unsolicited result codes specified within this command. <bfr> controls the effect on buffered codes when <mode> 1 or 2 is entered.

4.10.1 Format

Command	**Response**
AT+CGEREP=?	+CGEREP: (0,1), (0) OK
AT+CGEREP?	+CGEREP: <mode>,<bfr> OK

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AT+CGEREP=<mode>	OK Or +CME ERROR: <err>
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

4.10.2 Field

Parameters	Description
<mode>	0: Buffer unsolicited result codes in the UE; if UE result code buffer is full, the oldest ones can be discarded. No codes are forwarded to the TE. 1: Discard unsolicited result codes when UE-TE link is reserved (e.g. in on-line data mode); otherwise forward them directly to the TE.
<bfr>	0: UE buffer of unsolicited result codes defined within this command is cleared when <mode> 1 or 2 is entered.

4.11 AT+CMEE - Disable or enable the use of result code

This command disables or enables the use of result code +CME ERROR: <err> as an indication of an error relating to the functionality of the MT. When enabled, MT related errors cause +CME ERROR: <err> final result code instead of the regular ERROR final result code. ERROR is returned normally when error is related to syntax, invalid parameters, or TA functionality.

Test command returns values supported as a compound value.

4.11.1 Format

Command	Response
AT+CMEE=<n>	TA disables or enables the use of result code +CME ERROR: <err> as an indication of an error relating to the functionality of the ME. OK
AT+CMEE?	+CMEE: <n> OK
AT+CMEE=?	+CMEE: (list of supported <n>s) OK

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Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

4.11.2 Field

Parameters	Description
<n>	0 Disable +CME ERROR: <err> result code and use ERROR instead. 1 Enable +CME ERROR: <err> result code and use numeric <err> values (refer next subclause). 2 Enable +CME ERROR: <err> result code and use verbose <err> values (refer next subclause).
<err>	<err> values (numeric format followed by verbose format): General errors 0 Phone failure. 1 No connection to phone. 2 Phone adaptor link reserved. 3 Operation not allowed. 4 Operation not supported. 5 PH SIM PIN required. 6 PH-FSIM PIN required. 7 PH-FSIM PUK required. 10 SIM not inserted. 11 SIM PIN required. 12 SIM PUK required. 13 SIM failure. 14 SIM busy. 15 SIM wrong. 16 Incorrect password. 17 SIM PIN2 required. 18 SIM PUK2 required. 20 Memory full. 21 Invalid index. 22 Not found. 23 Memory failure. 24 Text string too long. 25 Invalid characters in text string. 26 Dial string too long. 27 Invalid characters in dial string. 30 No network service. 31 Network timeout. 32 Network not allowed - emergency calls only. 40 Network personalization PIN required.

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	41 Network personalization PUK required. 42 Network subset personalization PIN required. 43 Network subset personalization PUK required. 44 Service provider personalization PIN required. 45 Service provider personalization PUK required. 46 Corporate personalization PIN required. 47 Corporate personalization PUK required. 48 Hidden key required (NOTE: This key is required when accessing hidden phonebook entries.). 50 Incorrect Parameters. 100 Unknown. GPRS-related errors: Errors related to a failure to perform an Attach: 103 Illegal MS (#3). 106 Illegal ME (#6). 107 GPRS service not allowed (#7). 111 PLMN not allowed (#11). 112 Location area not allowed (#12). 113 Roaming not allowed in this location area (#13). (Values in parentheses are TS 24.008 cause codes.) 9.2.2.2 Errors related to a failure to Activate a Context. 132 Service option not supported (#32). 133 Requested service option not subscribed (#33). 134 Service option temporarily out of order (#34). 149 PDP authentication failure. (Values in parentheses are TS 24.008 cause codes.) Other GPRS errors: 150 Invalid mobile class. 148 Unspecified GPRS error. Other values in the range 101-150 are reserved for use by GPRS. CRSM/CSIM CME errors: 749 Invaild command length provided to CSIM. 750 Invalid command string provided to CSIM. 751 SIM command not allowed on 3G SIM (Release 5 only). 752 Invalid pathid for SIM. 753 Command type parameter missing from CRSM command. 754 Command type parameter for CRSM invalid. 755 FileID parameter for CRSM invalid. 756 P1/2/3 for CRSM command missing. 757 P1/2/3 for CRSM command invalid. 758 Command Data for CRSM command missing. 759 Command Data for CRSM command invalid. +CSCS CME errors: 737 The CSCS mode specified is not supported.

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	738 The CSCS mode specified is not supported. +CPOL CME errors: 741 Operator format parameter is missing 742 Operator data is in incorrect format 743 Operator data is too long 744 PLMN data cannot be written as the PLMN store is full 745 The SIM PLMN list cannot be changed since CPOL cannot access it 746 Operator specified is not recognized 747 Specified access technology missing 748 Specified access technology not supported

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5 Customize AT Command

5.1 AT+ECICCID - Read the USIM ICCID field

This command is used to read the USIM ICCID field (EFIccId).

5.1.1 Format

Command	Response
AT+ECICCID	+ECICCID: <ICCID> OK If error is related to wrong AT syntax +CME ERROR: <err>
AT+ECICCID=?	OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

5.1.2 Field

Parameters	Description
<ICCID>	International Circuit Card (ICC) ID of the (U)SIM. Formatted as a numeric string without double quotes. The string can be up to a maximum of 20 digits long.

5.2 AT+ECBAND - Query current modem operating band

5.2.1 Format

Command	Response
AT+ECBAND=<band1>[,<band2>…]	OK If there is any error, response: +CME ERROR: <err>

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AT+ECBAND?	+ECBAND: <band1>,<band2>… OK
AT+ECBAND=?	+ECBAND: (list of supported <band>s) OK
Max response time	25s

5.2.2 Field

Parameters	Description
<band>	Integer value Band list in decimal number. <band> values of 1,3,5,8,34,38,39,40,41 are supported. The default value is decided by RF Calibration table

5.3 AT+ECBCINFO - Query current network status

This command is used to query current network status, and modem status information for serving cell information and neighbor cells information, mainly used for location service.

5.3.1 Format

Command	Response
AT+ECBCINFO[=<mode>[,<timeoutS>[,<save_for_later>[,<max_cell_number>[,<report_mode>]]]]]	Display serving cell and neighbor cell information: +ECBCINFOSC:<earfcn>,<pci>,<rsrp>,<rsrq>,<mcc>,<mnc>,<cellid>,<tac>[<CR><LF>+ECBCINFONC:<earfcn>,<pci>,<rsrp>,<rsrq>[,<mcc>,<mnc>,<cellid>,<tac>][…]] OK
AT+ECBCINFO=?	+ECBCINFO:(list of supported <mode>s),(range of supported <timeoutS>),(list of supported <save_for_later>s), (range of supported <max_cell_number>s),(list of supported <report_mode>s) OK
Max response time	305s, depend on network status

5.3.2 Field

Parameters	Description

<mode>	Integer type 0 Only search/measure neighbor cell signal quality: RSRP and RSRQ, don't acquire neighbor cell SIB1, don't return cell location information. In this mode, neighbor cell information in AT response is: +ECBCINFONC:<earfcn>,<pci>,<rsrp>,<rsrq> Note: default value 0 1 Search cell, including measure cell and acquire cell SIB1 to get cellID, PLMN, and tac info. In this mode, neighbor cell information in AT response is: +ECBCINFONC: <earfcn>,<pci>,<rsrp>,<rsrq>,<mcc>,<mnc>,<cellid>,<tac> Note: If cell locked is enabled, UE will only response serving cell information. If frequency locked is enabled, UE will response serving cell and intra-frequency information. 2 Read saved cell information, <save_for_later> set to 1 should be called before, otherwise, no cell information will be returned.
<timeoutS>	Integer type MAX measurement time in unit of seconds, cell information should be acquired/returned in this time period. Note: 1. Range: (4-300) 2. Default value: 8
<save_for_later>	Integer type Whether need to save cell information, which could be returned in case of AT+ECBCINFO=2 later. Note: 1. As the cell information should be saved into flash when enter deep sleep mode: sleep2 and HIB, don't suggest to save it frequently. 2. List value: (0,1) 3. Default value: 0, not save.
<max_cell_number>	Integer type Max number of cells to be measured or searched, including serving cell. Note: 1. The more cells needed, the longer measurement time it takes. 2. Range: (1-7) 3. Default value: 7

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<report_mode>	Integer type 0 Synchronous mode. Report cell info in AT response. Note: Default value: 0 1 Asynchronous mode. AT response immediately, measured cell information returned in URC: [+ECBCINFOSC:<earfcn>,<pci>,<rsrp>,<rsrq>,<mcc>,<mnc>,<cellid>,<tac>[<CR><LF>+ECBCINFONC:<earfcn>,<pci>,<rsrp>,<rsrq>[,<mcc>,<mnc>,<cellid>,<tac>]] […]]
<earfcn>	Integer value; indicating the EARFCN for serving cell. Range 0-262143.
<pci>	Integer value; indicating the serving cell physical cell ID. Range 0-503.
<cellid>	String type; four bytes (28 bit) cell ID in hexadecimal format for serving cell.
<rsrp>	Signed integer; indicating serving cell RSRP value in units of dBm (can be negative value). Range -156 ~ -44. Available only in RRC-IDLE state.
<rsrq>	Signed integer; indicating serving cell RSRQ value in units of dB (can be negative value). Range -34 ~ 25. Available only in RRC-IDLE state.
<mcc>	String type; Indicate the mobile country code.
<mnc>	String type; Indicate the mobile network code.
<tac>	String type; Two byte tracking area code in hexadecimal format.

5.4 AT+ECFREQ - Lock UE to cell ID

This command is used to set prefer EARFCN list, lock or unlock cell, after set AT+CFUN=0.

5.4.1 Format

Command	**Response**

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If cell unlock or remove prefer EARFCN (mode = 0): AT+ECFREQ=<mode> If set prefer EARFCN list (mode = 1): AT+ECFREQ=<mode>[,<earfcn1>[,<earfcn2>…]] If cell lock (mode = 2): AT+ECFREQ=<mode>,<earfcn>[,<phyCellId>]	OK If error is related to wrong AT syntax or incorrect parameters. +CME ERROR: <err>
AT+ECFREQ?	If neither set prefer EARFCN list nor lock EARFCN/cell : OK If set prefer EARFCN list: +ECFREQ:<1>,<arfcn1>,<arfcn2>,… If lock EARFCN or lock cell: +ECFREQ:<2>,<arfcn>,<phyCellId> OK If both set prefer EARFCN list and lock EARFCN/cell: +ECFREQ:<1>,<arfcn1>,<arfcn2>,… +ECFREQ:<2>,<arfcn>,<phyCellId> OK
AT+ECFREQ=?	+ECFREQ: (list of supported <mode>s) OK
Max response time	300 ms

5.4.2 Field

Parameters	**Description**
<mode>	Integer value 0 Cancel cell lock 1 Set prefer EARFCN list 2 EARFCN lock, or cell lock 3 Clear prefer EARFCN
<earfcn>	Integer value indicating requested EARFCN on which to lock. Range 0-262143.
<phyCellId>	Integer value: Physical cell ID. Range: 0-503.

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5.5 AT+CPSMS - Control the setting of UE's power saving mode

This command controls the setting of the UE's power saving mode (PSM) parameters.

5.5.1 Format

Command	Response
AT+CPSMS=<mode>[,<Requested_Periodic-RAU>[,<Requested_GPRS-READY-timer>[,<Requested_Periodic-TAU>[,<Requested_Active-Time>]]]]	OK If error is related to wrong AT syntax: +CME ERROR: <err>
AT+CPSMS?	+CPSMS: <mode>,[<Requested_Periodic-RAU>],[<Requested_GPRS-READY-timer>],[<Requested_Periodic-TAU>],[<Requested_Active-Time>] If error is related to wrong AT syntax or operation not allowed: +CME ERROR: <err>
AT+CPSMS=?	+CPSMS: (list of supported <mode>s),(list of supported <Requested_Periodic-RAU>s),(list of supported <Requested_GPRS-READY-timer>s),(list of supported <Requested_Periodic-TAU>s),(list of supported <Requested_Active-Time>s) OK
Max response time	300 ms

5.5.2 Field

Parameters	Description
<mode>	Integer type; Indication to disable or enable the use of PSM in the UE: 0: Disable the use of PSM. 1: Enable the use of PSM. 2: Disable the use of PSM and discard all parameters for PSM or, if available reset to the manufacturer specific default values.
<Requested_Periodic-RAU>	String type; not supported by NB-IoT.
<Requested_GPRS-READY-timer>	String type; not supported by NB-IoT.

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<Requested_Periodic-TAU>	String type; one byte in an 8-bit format. Requested extended periodic TAU value (T3412) to be allocated to the UE in E-UTRAN. The requested extended periodic TAU value is coded as one byte (octet 3) of the GPRS Timer 3 information element coded as bit format (e.g. "01000111" equals 70 hours). For the coding and the value range, see the GPRS Timer 3 IE in 3GPP TS 24.008 Table 10.5.163a / 3GPP TS 24.008. See also 3GPP TS 23.682 and 3GPP TS 23.401. The default value, if available, is manufacturer specific.
<Requested_Active-Time>	String type; one byte in an 8-bit format. Requested Active Time value (T3324) to be allocated to the UE. The requested Active Time value is coded as one byte (octet 3) of the GPRS Timer 2 information element coded as bit format (e.g. "00100100" equals 4 minutes). For the coding and the value range, see the GPRS Timer 2 IE in 3GPP TS 24.008 Table 10.5.163 / 3GPP TS 24.008. See also 3GPP TS 23.682, 3GPP TS 23.060 and 3GPP TS 23.401. The default value, if available, is manufacturer specific.
Reference	GPRS Timer 3 IE in 3GPP TS 24.008 Table 10.5.163a/3GPP TS 24.008: Bits 5 to 1 represent the binary coded timer value. Bits 6 to 8 defines the timer value unit for the GPRS timer as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 10 minutes. 0 0 1 value is incremented in multiples of 1 hour. 0 1 0 value is incremented in multiples of 10 hours. 0 1 1 value is incremented in multiples of 2 seconds. 1 0 0 value is incremented in multiples of 30 seconds. 1 0 1 value is incremented in multiples of 1 minute. 1 1 1 value indicates that the timer is deactivated. Other values shall be interpreted as multiples of 1 hour in this version of the protocol. GPRS Timer 2IE in 3GPP TS 24.008 Table 10.5.163 / 3GPP TS 24.008: Timer value (octet 2) Bits 5 to 1 represents the binary coded timer value. Bits 6 to 8 defines the timer value unit for the GPRS timer as follows: Bits 8 7 6 0 0 0 value is incremented in multiples of 2 seconds. 0 0 1 value is incremented in multiples of 1 minute. 0 1 0 value is incremented in multiples of decihours.

	1 1 1 value indicates that the timer is deactivated. Other values shall be interpreted as multiples of 1 minute in this version of the protocol.

5.6 AT+CEDRXS - Control the setting of the UE eDRX parameters

The set command controls the setting of the UE's eDRX parameters. The command controls whether the UE wants to apply eDRX or not, as well as the requested eDRX value for each specified type of access technology.

The also controls the presentation of an unsolicited result code +CEDRXP: <AcTtype>[,<Requested_eDRX_value>[,<NW-provided_eDRX_value>[,<Paging_time_window>]]] when <n>=2 and there is a change in the eDRX parameters provided by the network.

A special form of the command can be given as +CEDRXS=3. In this form, eDRX will be disabled and data for all parameters in the command +CEDRXS will be removed or, if available, set to the manufacturer specific default values.

5.6.1 Format

Command	**Response**
AT+CEDRXS=<mode>[,<AcT-type>[,<Requested_eDRX_value>]]	OK If error is related to wrong AT syntax: +CME ERROR: <err>
AT+CEDRXS?	+CEDRXS: <AcT-type>,<Requested_eDRX_value> OK Or +CME ERROR: <err>
AT+CEDRXS=?	+CEDRXS: (list of supported <mode>s),(list of supported <AcT-type>s),(list of supported <Requested_eDRX_value>s) OK
	Unsolicited result code: +CEDRXP:<AcT-type>[,<Requested_eDRX_value>[,<NW-provided_eDRX_value>[,<Paging_time_window>]]]
Max response time	300 ms

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Note	Y7S/Y7SW/Y6 not support

5.6.2 Field

Parameters	Description
<mode>	Integer type; indicates to disable or enable the use of eDRX in the UE. This parameter is applicable to all specified types of access technology, i.e. the most recent setting of <mode> will take effect for all specified values of <AcT>. 0: Disable the use of eDRX. 1: Enable the use of eDRX. 2: Enable the use of eDRX and enable the unsolicited result code +CEDRXP: <AcT-type>[,<Requested_eDRX_value>[,<NW-provided_eDRX_value>[,<Paging_time_window>]]]. 3 Disable the use of eDRX and discard all parameters for eDRX or, if available, reset to the manufacturer specific default values.
<AcT-type>	Integer type; indicates the type of access technology. This AT command is used to specify the relationship between the type of access technology and the requested eDRX value. 0: Access technology is not using eDRX. This parameter value is only used in the unsolicited result code. 4: E-UTRAN (WB-S1 mode).
<Requested_eDRX_value>	String type; half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see sub-clause 10.5.5.32of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008. The default value, if available, is manufacturer specific.
<NW-provided_eDRX_value>	String type; half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see subclause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008.
<Paging_time_window>	String type; half a byte in a 4-bit format. The paging time window refers to bit 8 to 5 of octet 3 of the Extended DRX parameters information element (see sub-clause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see the Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008.

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5.7 AT+CEDRXRDP - Return eDRX parameters

The execution command returns <AcT-type>, <Requested_eDRX_value>, <NW-provided_eDRX_value> and <Paging_time_window>. If eDRX is used for the cell that the MS is currently registered to. If the cell that the MS is currently registered to is not using eDRX, AcT-type=0 is returned.

5.7.1 Format

Command	Response
AT+CEDRXRDP	+CEDRXRDP: <AcT-type>[,<Requested_eDRX_value>[,<NW-provided_eDRX_value>[,<Paging_time_window>]]] OK
AT+CEDRXRDP=?	OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

5.7.2 Field

Parameters	Description
<AcT-type>	Integer type; indicates the type of access technology. This AT command is used to specify the relationship between the type of access technology and the requested eDRX value. 0: Access technology is not using eDRX. 4: E-UTRAN (WB-S1 mode).
<Requested_eDRX_value>	String type; half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see sub-clause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008.
<NW-provided_eDRX_value>	String type; half a byte in a 4-bit format. The eDRX value refers to bit 4 to 1 of octet 3 of the Extended DRX parameters information element (see sub-clause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008.

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<Paging_time_window>	String type; half a byte in a 4-bit format. The paging time window refers to bit 8 to 5 of octet 3 of the Extended DRX parameters information element (see sub-clause 10.5.5.32 of 3GPP TS 24.008). For the coding and the value range, see the Extended DRX parameters information element in 3GPP TS 24.008 Table 10.5.5.32 / 3GPP TS 24.008.

5.8 AT+ECPMUCFG - Set PMU mode

5.8.1 Format

Command	Response
AT+ECPMUCFG=<enable>[,<mode>]	OK If there is any error, response: +CME ERROR: <err>
AT+ECPMUCFG=?	+ECPMUCFG:(range of supported <enable>s),(list of supported <mode>s) OK
AT+ECPMUCFG?	+ECPMUCFG:<enable>[,<mode>] OK
Max response time	300 ms

5.8.2 Field

Parameters	Description
<enable>	Integer type; specifies to enable PMU or not 0 Disable the PMU 1 Enable the PMU The default value is 1.
<mode>	Integer type; specifies to depth of sleep mode 0 Active 1 Idle 2 Sleep1 3 Sleep2 4 Hibernate The default value is 1.

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5.9 AT+ECLEDMODE – Enable/Disable the netlight function

The command enable/disable the netlight function.

5.9.1 Format

Command	Response
AT+ECLEDMODE?	+ECLEDMODE:<mode> OK
AT+ECLEDMODE=?	+ECLEDMODE:(0,1) OK
AT+ECLEDMODE=<mode>	OK If there is any error, response: +CME ERROR: <err>
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

5.9.2 Field

Parameters	Description
<mode>	0: Netlight function disable 1: Netlight function enable

5.10 AT+MOPENLIGHT – Close/Open network light

The command close/open network light.

5.10.1 Format

Command	Response
AT+MOPENLIGHT?	+MOPENLIGHT:<mode> OK
AT+MOPENLIGHT=?	+MOPENLIGHT:(0,1) OK
AT+MOPENLIGHT=<mode>	OK If there is any error, response: +CME ERROR: <err>

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Max response time	300 ms

5.10.2 Field

Parameters	Description
<mode>	Integer type; 0 disable the network light. 1 enable the network light

5.11 AT+CGNETLED – Close/Open network light

The command close/open network light.

5.11.1 Format

Command	Response
AT+CGNETLED?	+CGNETLED:<mode> OK
AT+CGNETLED=?	+CGNETLED:(0,1) OK
AT+CGNETLED=<mode>	OK If there is any error, response: +CME ERROR: <err>
Max response time	300 ms

5.11.2 Field

Parameters	Description
<mode>	Integer type; 0 disable the network light. 1 enable the network light

5.12 AT+POWEROFF - Shutdown UE

The command trigger shutdown UE.

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5.12.1 Format

Command	Response
AT+POWEROFF	OK
Max response time	300 ms

5.13 AT+RESET - Reboot UE

The command trigger reboot UE.

5.13.1 Format

Command	Response
AT+RESET	OK If there is any error, response: +CME ERROR: <err>
Max response time	300 ms

5.14 AT+EWDT - Reboot UE

The command trigger reboot UE.

5.14.1 Format

Command	Response
AT+EWDT=<mode>	OK If there is any error, response: +CME ERROR: <err>
Max response time	300 ms

5.14.2 Field

Parameters	Description
<mode>	Integer type; 1 Trigger reboot UE.

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5.15 AT+MAONIOCFG - Config VDD EXT Aonio GPIO

The command is used to config VDD EXT Aonio GPIO.

5.15.1 Format

Command	Response
AT+MAONIOCFG=<mode>	OK If there is any error, response: +CME ERROR: <err>
AT+MAONIOCFG?	+MAONIOCFG:<mode> OK
AT+MAONIOCFG=?	+MAONIOCFG: (0-1) OK
Max response time	300 ms
Note	Y7S/Y7SW/Y6 not support

5.15.2 Field

Parameters	Description
<mode>	Integer type; 0 Close VDD EXT Aonio GPIO output 1 Open VDD EXT Aonio GPIO output (default).

5.16 AT+CBC – Query Vbat voltage

The command is used to query Vbat voltage.

5.16.1 Format

Command	Response
AT+CBC	+CBC: <voltage> OK
AT+CBC=?	OK
Max response time	300 ms

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5.16.2 Field

Parameters	Description
<voltage>	Integer type; Vbat voltage, unit is mV.

5.17 AT+MADC – Query ADC channel voltage

The command is used to query ADC channel voltage.

5.17.1 Format

Command	Response
AT+MADC=<channel>	+MADC: <channel>,<voltage> OK If there is any error, response: ERROR
AT+MADC	+MADC: 1,<voltage> +MADC: 2,<voltage> OK
AT+MADC=?	+MADC: (1-2) OK
Max response time	300 ms

5.17.2 Field

Parameters	Description
<voltage>	Integer type; ADC channel voltage, unit is mV.
<channel>	Integer type; ADC channel, 1-2

5.18 AT+ECWIFISCAN – Detect Wi-Fi SSID information

The execution command is used for detect Wi-Fi SSID information.

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5.18.1 Format

Command	Response
AT+ECWIFISCAN=[<time>],[<round>],[<maxbssidnum>],[<scantimeout>],[<priority>],[<channelRecLen>],[<channelCount>],[<channelId1>],[<channelId2>],[<channelId3>],[<channelId4>],[<channelId5>],[<channelId6>],[<channelId7>],[<channelId8>],[<channelId9>],[<channelId10>],[<channelId11>,[<channelId12>],[<channelId13>],[<channelId14>]]	+ECWIFISCAN:<ecn>,<ssid>,<rssi>,<mac>,< channel> OK If there is any error, response: ERROR Note: wifiscan result also provide ssid Name and the length of ssid Name.
AT+ECWIFISCAN=?	+ECWIFISCAN:(4000-255000),(1-3),(4-40),(1-255),(0-1),(100-280),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14),(1-14) OK
AT+ECWIFISCAN?	+ECWIFISCAN:<time>,<round>,<maxbssidnum>,<scantimeout>,<priority>,<channelRecLen>,<channelCount>,<channelId1>,<channelId2>,<channelId3>,<channelId4>,<channelId5>,<channelId6>,<channelId7>,<channelId8>,<channelId9>,<channelId10>,<channelId11>,<channelId12>,<channelId13>,<channelId14> OK
Max response time	255s, depend on network status
Note	Y7S/Y7SW/Y6 not support

5.18.2 Field

Parameters	Description

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<time>	Integer type; unit: milliseconds The value is max response time of Wi-Fi scan; Default value is 12000
<maxbssidnum>	Integer type; The value is the max required Wi-Fi SSID number. Default value is 5
<round>	Integer type; The value is max number of Wi-Fi scan rounds. Default value is 1
<scantimeout>	Integer type; unit : second The value is max search time per Wi-Fi scan round. Default value is 5
<priority>	Integer type; The value is the priority of Wi-Fi scan. [0 means data preferred; 1 means Wi-Fi preferred] Default value is 0
<channelreclen>	Integer type; The value is the max scan time on each frequency channel. [unit is ms] Default value is 280 ms
<channelcount>	Integer type; The count of the frequency channel to be scanned. Default value is 1 If channelCount is 1 and channelId1 is 0, UE will scanning all frequency.
<channelid1> ~ <channelid14>	Integer type; The value is the specific frequency channel of Wi-Fi scan. If channel ID is configured, the range of channel ID value is [1-14]. [If all channel ID value 0, means no channel specified, all frequency channels will be scanned] Default value is 0

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	Note: 1> ECWIFISCAN AT could be executed in CFUN0 / CFUN4 / NO-USIM / Normal state. 2> Wi-Fi scan only could be executed while PHY in Idle state. If the priority is set to 0 and UE in connection state, UE will not start the Wi-Fi scan procedure until UE return back to Idle state. If the priority is set to 1 and UE in connection state, UE will release connection immediately and start Wi-Fi scan procedure. 3> <time> value should not be less than <round> multiplied by <scantimeout>. 4> <maxbssidnum> means the max required Wi-Fi SSID number. Case 1 - If UE has detected the required Wi-Fi SSID, UE will stop Wi-Fi procedure and report the detected Wi-Fi result. Case 2 - If UE has not detected the required Wi-Fi SSID before execute time out, UE will abort Wi-Fi procedure and report the detected Wi-Fi result. Case 3 - The round is set to 3 and maxbssidnum is set to 10, if in the first round UE could detect five Wi-Fi SSID, in the second round UE could detect three Wi-Fi SSID, and in the third round UE could detect six Wi-Fi SSID. UE will delete the duplicate Wi-Fi SSID and sort the Wi-Fi SSID by descending order, then report the ten Wi-Fi SSID result.

5.19 AT+ECURC - Close/open URC (unsolicited result code) report

The command close/open URC (unsolicited result code) report.

5.19.1 Format

Command	Response
AT+ECURC?	+ECURC: "CREG":<value>,"CEREG":<value>,"CEDRXP":<value>,"CCIOTOPTI":<value>,"CSCON":<value>,"CTZEU":<value>,"ECCESQ":<value>,"CGEV":<value>,"ECPSMR":<value>,"ECPTWEDRXP":<value>,"ECPIN":<value>,"ECPADDR":<value>,"ECPCFUN":<value>,"ECSMSFULL":<value> OK

AT+ECURC=?	+ECURC: "ALL":(0-1),"CREG":(0-1),"CEREG":(0-1),"CEDRXP":(0-1),"CCIOTOPTI":(0-1),"CSCON":(0-1),"CTZEU":(0-1),"ECCESQ":(0-1),"CGEV":(0-1),"ECPSMR":(0-1),"ECPTWEDRXP":(0-1),"ECPIN":(0-1),"ECPADDR":(0-1),"ECPCFUN":(0-1),"ECSMSFULL":(0-1) OK
AT+ECURC=[<urcStr>,]<value>	OK If there is any error, response: +CME ERROR: <err>
Max response time	300 ms

5.19.2 Field

Parameters	Description
<urcStr>	String type; “ALL” All unsolicited result codes included as below “CREG” unsolicited result code +CREG “CEREG” unsolicited result code +CEREG “CEDRXP” unsolicited result code +CEDRXP “CCIOTOPTI” unsolicited result code +CCIOTOPTI “CSCON” unsolicited result code +CSCON “ECCESQ” unsolicited result code +ECCESQ “CGEV” unsolicited result code +CGEV “ECPSMR” unsolicited result code +ECPSMR “PTWEDRX” unsolicited result code +PTWEDRX “ECPTWEDRXP” unsolicited result code +ECPTWEDRXP “ECPIN” unsolicited result code +ECPIN “ECPADDR” unsolicited result code +ECPADDR “ECPCFUN” unsolicited result code +ECPCFUN
<value>	Integer type; 0 disable unsolicited result code report 1 enable unsolicited result code report

5.20 AT+ECNETDEVCTL

The command set data path control for the network adapter. Control data path open by setting <op> to bind LWIP to PDN context with <cid>. Control data path close by setting <op> to unbind LWIP to PDN context with <cid>.

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5.20.1 Format

Command	Response
AT+ECNETDEVCTL=<op>,<cid>,[,<urc_en>]	OK If there is any error, response: +CME ERROR: <err>
AT+ECNETDEVCTL?	+ECNETDEVCTL: <op>,<cid>,<urc_en>,<state> OK
AT+ECNETDEVCTL=?	+ECNETDEVCTL: (list of supported (<OP>,<cid>,<urc_en>)s) OK
Max response time	300 ms

5.20.2 Field

Parameters	Description
<op>	INTEGER TYPE; SPECIFIES DEVICE CONTROL OPTION. 0 Unbind cid for LWIP 1 Bind cid for LWIP once and no rebind if re-activate PDN context with same cid after deactivation. 2 Bind cid for LWIP and rebind if re-activate PDN context with the same cid after deactivation. 3 Auto dial and bind cid when power on, saved in NVM after power down.
<cid>	Integer type; specifies a particular non secondary PDP context definition. <cid> values of 1-15 are supported.
<urc_en>	Integer type; specifies whether report URC +ECNETDEVCTL:<state> 0 Disable URC +ECNETDEVCTL:<state> 1 Enable URC +ECNETDEVCTL:<state>
<state>	Integer type; specifies cid bound state. 0 Bind cid for LWIP failure 1 Bind cid for LWIP success

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	Note: If the <cid> for PDN context defined but deactivated, bind this <cid> will auto activate PDN context. If the <cid> for PDN context undefined, bind this <cid> will cause activation failed.

5.21 AT+CEMODE - UE modes of operation for EPS

The set command is used to set the MT to operate according to the specified mode of operation for EPS, see 3GPP TS 24.301 [83]. If the requested mode of operation is not supported, an ERROR or +CME ERROR response is returned. Extended error responses are enabled by the +CMEE command.
The read command returns the mode of operation set by the TE, independent of the current serving cell capability and independent of the current serving cell Access Technology.
The test command is used for requesting information on the supported MT modes of operation as a compound value

5.21.1 Format

Command	**Response**
AT+CEMODE=<mode>	OK If there is any error, response: +CME ERROR: <err>
AT+CEMODE?	+CEMODE: <mode> OK
AT+CEMODE=?	+CEMODE: (list of supported <mode>s) OK
Max response time	5s
Parameter Saving Mode	NO_SAVE

5.21.2 Field

Parameters	**Description**
<mode>	INTEGER TYPE; INDICATES THE MODE OF OPERATION. THE DEFAULT VALUE IS MANUFACTURER SPECIFIC 0 PS mode 2 of operation 1 CS/PS mode 1 of operation 2 CS/PS mode 2 of operation 3 PS mode 1 of operation

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5.22 AT+CGCONTRDP - PDP context read dynamic parameters

The execution command returns the relevant information for an active non-secondary PDP context with the context identifier <cid>. If the MT has dual stack capabilities, at least one pair of lines with information is returned per <cid>. First one line with the IPv4 parameters followed by one line with the IPv6 parameters. If this MT with dual stack capabilities indicates more than more than two IP addresses of DNS servers, multiple of such pairs of lines are returned.
If the parameter <cid> is omitted, the relevant information for all active non secondary PDP contexts is returned.
The test command returns a list of <cid>s associated with active non secondary contexts.

5.22.1 Format

Command	Response
AT+CGCONTRDP[=<cid>]	+CGCONTRDP: <cid>,<bearer_id>,<apn>[,<local_addr and subnet_mask>[,<gw_addr>[,<DNS_prim_addr>[,<DNS_sec_addr>[,<PCSCF_prim_addr>[,<PCSCF_sec_addr>[,<IM_CN_Signalling_Flag>[,<LIPA_indication>[,<IPv4_MTU>[,<WLAN_Offload>[,<Local_Addr_Ind>[,<Serving_PLMN_rate_control_value>]]]]]]]]]]]] OK or +CGCONTRDP: <cid>,<bearer_id>,<apn>[,<local_addr and subnet_mask>[,<gw_addr>[,<DNS_prim_addr>[,<DNS_sec_addr>[,<PCSCF_prim_addr>[,<PCSCF_sec_addr>[,<IM_CN_Signalling_Flag>[,<LIPA_indication>[,<IPv4_MTU>[,<WLAN_Offload>[,<Local_Addr_Ind>[,<Serving_PLMN_rate_control_value>]]]]]]]]]]]] OK If there is any error, response: +CME ERROR: <err>
AT+CGCONTRDP=?	+CGCONTRDP: (list of <cid>s associated with active contexts) OK

Max response time	5s
Parameter Saving Mode	NO_SAVE
Note	Y7S/Y7SW/Y6 not support

5.22.2 Field

Parameters	Description
<cid>	INTEGER TYPE; SPECIFIES A PARTICULAR NON SECONDARY PDP CONTEXT DEFINITION. <CID> VALUES OF 1-15 ARE SUPPORTED.
<bearer_id>	Integer type; identifies the bearer.
<apn>	String type; a logical name that was used to select the GGSN or the external packet data network.
<local_addr_and_subnet_mask>	String type; shows the IP address and subnet mask of the MT. The string is given as dot-separated numeric (0-255)
<dns_prim_addr>	String type; the IP address of the primary DNS server.
<dns_sec_addr>	String type; the IP address of the secondary DNS server.
<p_cscf_prim_addr>	String type; shows the IP address of the primary P-CSCF server. (Not displayed)
<p_cscf_sec_addr>	String type; shows the IP address of the secondary P-CSCF server. (Not displayed)
<im_cn_singalling_flag>	Integer type; shows whether the PDP context is for IM CN subsystem-related signalling only or not. (Not displayed)
<lipa_indication>	Integer type; indicates that the PDP context provides connectivity using a LIPA PDN connection. (Not displayed)
<ipv4_mtu>	Integer type; shows the IPv4 MTU size in octets
<wlan_offload>	Integer type; indicates whether traffic can be offloaded using the specified PDN connection via a WLAN or not. (Not displayed)
<local_addr_ind>	integer type; indicates whether or not the MS and the network support local IP address in TFTs. (Not displayed)
<serving_plmn_rate_control_value>	Integer type; indicates the maximum number of uplink messages the UE is allowed to send in a 6 minute interval.

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6 TCP/IP(Y7S/Y7SW/Y6 not support)

6.1 AT+SKTCFG - Config socket data

The command is used to config socket data.

6.1.1 Format

Command	Response
AT+SKTCFG=?	+SKTCFG:(list of supported <send_type>s) OK
AT+SKTCFG=<send_type>	OK If there is any error, response: +SOCKET ERROR: <err>
AT+SKTCFG?	+SKTCFG:<send_type> OK
Max response time	300 ms

6.1.2 Field

Parameters	Description
<send_type>	Integer type, send data type 0 Hex type 1 String type

6.2 AT+SKTCREATE - Create a socket

The command creates a socket on the UE and associates with specified protocol. UE supports up to 12 sockets(TCP or UDP) at the same time. And will return error if it is exceeded.

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6.2.1 Format

Command	Response
AT+SKTCREATE=?	+SKTCREATE:(list of supported <domain>s), (list of supported <type>s), (list of supported <protocol>s) OK
AT+SKTCREATE=<domain>,<type>,<protocol>	+SKTCREATE: <fd> OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	300 ms

6.2.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description
<domain>	Integer type 1 IPV4 2 IPV6
<type>	Integer type 1 TCP 2 UDP
<protocol>	Integer type; standard internet protocol definition 6 IPPROTO_TCP 17 IPPROTO_UDP

6.3 AT+SKTKEEPALIVE – Set heartbeat

The command is used to set the heartbeat packet function of the socket

6.3.1 Format

Command	Response
AT+SKTKEEPALIVE=?	+SKTCREATE:(list of supported <fd>s), (list of <mode>), (list of timeout <time>) OK

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AT+SKTKEEPALIVE=<fd>,<mode>,<time>	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	300 ms

6.3.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<mode>	Integer type 0 Disable TCP heartbeat 1 Enable TCP heartbeat
<time>	Integer type Heartbeat timeout time, 1-65535

6.4 AT+SKTCONNECT - Connect socket

For TCP, the command connect socket with remote address and port.

For UDP, the command save remote address and port for send.

6.4.1 Format

Command	Response
AT+SKTCONNECT=?	+SKTCONNECT:(list of supported <fd>s), (<addr>), (list of supported <port>s) OK
AT+SKTCONNECT=<fd>,<addr>,<port>	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	30 s

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6.4.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<addr>	string type Remote address to connect or send to
<port>	Integer type Remote port to connect or send to, 1-65535

6.5 AT+SKTBIND - Bind socket

The command bind socket with local address and port. If the address is default, it means any address.

6.5.1 Format

Command	Response
AT+SKTBIND=?	+SKTBIND:(list of supported <fd>s),(<addr>),(list of supported <port>s) OK
AT+SKTBIND=<fd>,<addr>,<port>	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	5 s

6.5.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<addr>	string type address to bind. If address is defaults means any address.
<port>	Integer type port to bind

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6.6 AT+SKTSEND – Send data

Send a containing length bytes of data to remote port on remote addr.

6.6.1 Format

Command	Response
AT+SKTSEND=?	+SKTSEND:(list of supported <fd>s), (list of supported <data len>s), (<data>), (list of supported <rai info>s), (list of supported <expect info>s) OK
AT+SKTSEND=<fd>,<data len>,<data>[,<rai info>[,<except info>]]	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	5 s

6.6.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<data len>	Integer type length of data in hex string format, the max length is 1400
<data>	Integer type Data in hex string
<rai info>	Integer type (option) 0-2 release assistance indication 0 no rai info 1 no further uplink or downlink data transmission subsequent to the uplink data transmission subsequent to the uplink data transmission is expected 2 only a single downlink data transmission and no further uplink data transmission subsequent to the uplink data transmission is expected Note: default value: 0
<except info>	Integer type (option) 0-1 expect data indication 0 disable expect data indication 1 enable expect data indication Note: default value: 0

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6.7 AT+SKTSENDT – Send data with transparent mode

Send a containing length bytes of data to remote port on remote address with transparent mode.

If the command contain the parameter of “data len”, It need input enough data with hex string format after “>” output. If the command did not contain the parameter of “data len”, the input of data must end with “CTRL+Z”.

6.7.1 Format

Command	Response
AT+SKTSENDT=?	+SKTSEND:(list of supported <fd>s), (list of supported <data len>s), (list of supported <rai info>s), (list of supported <expect info>s) OK
AT+SKTSENDT=<fd>[,<data len>[,<rai info>[,<except info>]]]	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	60 s

6.7.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<data len>	Integer type length of data in hex string format, the max length is 1400
<rai info>	Integer type (option) 0-2 release assistance indication 0 no rai info 1 no further uplink or downlink data transmission subsequent to the uplink data transmission subsequent to the uplink data transmission is expected 2 only a single downlink data transmission and no further uplink data transmission subsequent to the uplink data transmission is expected Note: default value: 0

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<except info>	Integer type (option) 0-1 expect data indication 0 disable expect data indication 1 enable expect data indication Note: default value: 0

6.8 AT+SKTSTATUS - Get the status of a socket

Get the status of a socket by file description.

6.8.1 Format

Command	Response
AT+SKTSTATUS=?	+SKTSTATUS: (list of supported <fd>s) OK
AT+SKTSTATUS=<fd>	+SKTSTATUS: <status> OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	300 ms

6.8.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<status>	Integer type 0 Closed 1 socket create 2 Connecting 3 Connected 4 DNS resolving 5 Closing

6.9 AT+SKTRXGET – Set receive mode and get buffer data

Set receive mode and get buffer data.

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6.9.1 Format

Command	Response
AT+SKTRXGET=?	+SKTRXGET: (list of supported <mode>s), (list of supported <fd>s), (list of supported <req_len>s) OK
AT+SKTRXGET=<mode>[,<fd>,<req_len>]	OK Or +SKTRXGET:<fd>,<got_len>,<buff_len>,<data> OK Or +SOCKET ERROR: <err>
AT+SKTRXGET?	+SKTRXGET: <mode> OK
Max response time	5 s

6.9.2 Field

Parameters	Description
<mode>	Integer type 0: Set received data direct output. 1: Set received data save to buffer. 2: Get data from buffer.
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE
<req_len>	Integer type The length of request data. Range: 1~512.
<got_len>	Integer type The got data length.
<buff_len>	Integer type The Remaining data length of buffer data.
<data>	String type When send_type set to 0, data in hex string When send_type set to 1, data in string string

6.10 AT+SKTDELETE - Delete a socket

Delete a socket by file description.

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6.10.1 Format

Command	Response
AT+SKTDELETE=?	+SKTDELETE: (list of supported <fd>s) OK
AT+SKTDELETE=<fd>	OK If there is any error, response: +SOCKET ERROR: <err>
Max response time	5 s

6.10.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description returned by +SKTCREATE

6.11 +SKTRECV - Received data on a socket

This is an unsolicited message to show data has been received on a socket.

6.11.1 Format

Command	Response
	+SKTRECV: <fd>,<len>[,<data>]

6.11.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description that data from
<len>	String type Received data length(bytes)
<data>	String type When send_type set to 0, received data in hex string When send_type set to 1, received data in string string

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6.12 +SKTERR - Show the error number

This is an unsolicited message to show the error number when error occur.

6.12.1 Format

Command	Response
	+SKTERR:<fd>,<errno>

6.12.2 Field

Parameters	Description
<fd>	Integer type 0-11 Socket file description that data from
<errno>	Integer type Posix Errno defines 11 Operation would block 12 Out of memory error 22 Invalid argument 62 Timer expired 103 Software caused connection abort 104 Connection reset by peer 105 No buffer space available 107 Transport endpoint is not connected 113 No route to host 115 Operation now in progress

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7 ECMQTT(Y7S/Y7SW/Y6 not support)

7.1 AT+ECMTCFG – Config MQTT client

7.1.1 Format

Command	Response
AT+ECMTCFG=”keepalive”,<tcpconnectID>[,<keep-alive time>]	OK If <keep-alive time> is omitted, query the keep-alive time: +ECMTCFG: "keepalive",<keep-alive time> OK If there is any error, response: ERROR
AT+ECMTCFG=”session”,<tcpconnectID>[,<clean_session>]	OK If <clean_session> is omitted, query the session type: +ECMTCFG:"session",<clean_session> OK If there is any error, response: ERROR
AT+ECMTCFG=”timeout”,<tcpconnectID>[,<pkt_timeout>,[<retry_timeout>,[<timeout_notice>]]]	OK If <pkt_timeout>, <retry_timeout> and <timeout_notice> are omitted, query the timeout value of message delivery: +ECMTCFG:"timeout",<pkt_timeout>,<retry_timeout>,<timeout_notice> OK If there is any error, response: ERROR
AT+ECMTCFG=”will”,<tcpconnectID>[,<will_fg>[,<will_qos>,<will_retain>,“<will_topic>”,“<will_msg>”]]	OK If <will_fg>, <will_qos>, <will_retain>, <will_topic> and <will_msg> are omitted, query the Will information: +ECMTCFG:"will",<will_fg>[,<will_retain>,<will_topic>,<will_msg>] OK If there is any error, response: ERROR

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AT+ECMTCFG=”version”,<tcpconnectID>[,<version>]	OK If <version> is omitted, query the MQTT protocol version: +ECMTCFG:"version",<version> OK If there is any error, response: ERROR
AT+ECMTCFG=”aliauth”,”,<tcpconnectID>[,“<product_key>”,“<device_name>”,“<device_secret>”]	OK If <product_key>, <device_name>, <device_secret> are omitted, query the device information: +ECMTCFG:"aliauth",<product_key>,<device_name>,<device_secret> OK If there is any error, response: ERROR
AT+ECMTCFG=”cloud”,<tcpconnectID>[,<cloud type>,<data type>]	OK If <cloud type> and <data type> are both omitted, query the cloud type and data type: +ECMTCFG: “cloud”,<cloud type>,<data type> OK If there is any error, response: ERROR
AT+ECMTCFG=”ssl”,”enable”	OK If there is any error, response: ERROR Note:this command is used to enable SSL.After AT+ECMTCLOSE, SSL is unenabled
AT+ECMTCFG=?	+ECMTCFG:"keepalive",(0),(0-3600) +ECMTCFG:"session",(0),(0,1) +ECMTCFG:"timeout",(0),(1-60),(1-10),(0,1) +ECMTCFG:"will",(0),(0,1),(0-2),(0,1),"will_topic","will_msg" +ECMTCFG:"version",(0),(3,4) +ECMTCFG:"aliauth",(0),"productkey","devicename","devicesecret" +ECMTCFG:"cloud",(0),(0-255),(0-255) OK
Max response time	300 ms

7.1.2 Field

Parameters	**Description**
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0

<keep-alive time>	Integer type. The range is 0-3600. The default value is 120. Unit: second. It defines the maximum time interval between messages received from a client. If the server does not receive a message from the client within 1.5 times of the keep-alive time period, it disconnects the client as if the client has sent a DISCONNECT message.
<clean_session>	Integer type. Configure the session type 0 The server must store the subscriptions of the client after it disconnects. 1 The server must discard any previously maintained information about the
<pkt_timeout>	Integer type. Timeout of the packet delivery. The range is 1-60. The default value is 10. Unit: second.
<retry_timeout>	Integer type. Timeout of the retry packet delivery. The range is 1-10. The default value is 3. Unit: second.
<timeout_notice>	Integer type. Timeout notice. The default value is 0. 0 no notice 1 notice
<will_fg>	Integer type. Configure the Will flag 0 Ignore the Will flag configuration 1 Require the Will flag configuration
<will_qos>	Integer type. Quality of service for message delivery 0 At most once 1 At least once 2 Exactly once
<will_retain>	Integer type. The Will retain flag is only used on PUBLISH messages. 0 When a client sends a PUBLISH message to a server, the server will not hold on to the message after it has been delivered to the current subscribers 1 When a client sends a PUBLISH message to a server, the server should hold on to the message after it has been delivered to the current subscribers
<will_topic>	Integer type. Will topic string. The maximum size is 255 bytes.
<will_msg>	Integer type. The Will message defines the content of the message that is published to the will topic if the client is unexpectedly disconnected. It can be a zero-length message. The maximum size is 255 bytes.

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<version>	Integer type. Version of MQTT protocol, the default is MQTT v3.1.1 3 MQTT v3.1 4 MQTT v3.1.1
<product_key>	Integer type. Product key issued by AliCloud. The maximum size is 32 bytes.
<device_name>	Integer type. Device name issued by AliCloud. The maximum size is 32 bytes.
<device_secret>	Integer type. Device secret key issued by AliCloud. The maximum size is 64 bytes.
<cloud type>	Integer type 0 mosquitto 1 OneNet 2 alibaba 3-255 is reserved for customers
<data type>	Integer type, the range is 0-255 For OneNet cloud 1 OneNet data type1 2 OneNet data type2 3 OneNet data type3 4 OneNet data type4 5 OneNet data type5 6 OneNet data type6 For Ali Cloud 1 Json data 2 string data For other, data format insensitive

7.2 AT+ECMTOPEN – Open a network for MQTT client

This command is used to new MQTT over TCP connection with server.

7.2.1 Format

Command	**Response**
AT+ECMTOPEN=<tcpconnectID>,“<host_name>”,<port>	OK +ECMTOPEN:<tcpconnectID>,<result> If there is any error, response: ERROR

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AT+ECMTOPEN=?	+ECMTOPEN: (list of supported <tcpconnectID>s),“<host_name>”,(list of supported <port>s) OK
AT+ECMTOPEN?	+ECMTOPEN: <tcpconnectID>,“<host_name>”,<port> OK
Max response time	5 s

7.2.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<host_name>	String type The address of the server. It could be an IP address or a domain name. The maximum size is 127 bytes
<port>	Integer type The port of the server. The range is 1-65535
<result>	Integer type, Result of the command execution -1 Failed to open network 0 Opened network successfully

7.3 AT+ECMTCONN– Send MQTT connection packet

This command is used to send MQTT connection packet.

7.3.1 Format

Command	Response
AT+ECMTCONN=<tcpconnectID>,“<clientID>”[,“<username>”[,“<password>”]]	OK +ECMTCONN:<tcpconnectID>,<result>[,<ret_code>] If there is any error, response: ERROR
AT+ECMTCONN=?	+ECMTCONN: (list of supported <tcpconnectID>s),“<clientID>”[,“<username>”[,“<password>”]] OK
AT+ECMTCONN?	[+ECMTCONN:<tcpconnectID>,<state>] OK
Max response time	5 s

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7.3.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<clientID>	String, The client identifier, The maximum size is 48 bytes.
<username>	String, User name of the client. It can be used for authentication, The maximum size is 48 bytes (optional).
<password>	String, Password corresponding to the user name of the client. It can be used for authentication, The maximum size is 128 bytes (optional).
<result>	0 Send connect packet success 2 Send connect packet fail
<ret_code>	0 Connection Accepted 1 Connection Refused, unacceptable protocol version 2 Connection Refused, identifier rejected 3 Connection Refused, Server unavailable 4 Connection Refused, bad user name or password 5 Connection Refused, not authorized 6 CONNACK packet is incorrect
<state>	Connection state, 1 Other state 2 Connecting 3 Connected 4 Disconnected or closed

7.4 AT+ECMTCLOSE – Close MQTT

This command is used to send MQTT close packet, disconnect TCP connection with MQTT server and delete MQTT.

7.4.1 Format

Command	Response

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AT+ECMTCLOSE=<tcpconnectID>	OK +ECMTCLOSE: <tcpconnectID>,<result> If there is any error, response: ERROR
AT+ECMTCLOSE=?	+ECMTCLOSE: (list of supported <tcpconnectID>s) OK
Max response time	5 s

7.4.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<result>	Integer type, Result of the command execution -1 Failed to close mqtt 0 Close mqtt successfully

7.5 AT+ECMTDISC – Disconnect MQTT

This command is used to send MQTT disconnection packet, disconnect TCP connection with MQTT server and delete MQTT.

7.5.1 Format

Command	Response
AT+ECMTDISC=<tcpconnectID>	OK +ECMTDISC: <tcpconnectID>,<result> If there is any error, response: ERROR
AT+ECMTDISC=?	+ECMTDISC: (list of supported <tcpconnectID>s) OK
Max response time	5 s

7.5.2 Field

Parameters	Description

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<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<result>	Integer type, Result of the command execution -1 Failed to disconnect mqtt 0 Disconnect mqtt successfully

7.6 AT+ECMTSUB – Send MQTT subscribe packet

This command is used to send MQTT subscribe packet.

7.6.1 Format

Command	Response
AT+ECMTSUB=<tcpconnectID>,<msgID>,“<topic>”,<qos>	OK +ECMTSUB:<tcpconnectID>,<msgID>,<result>,<value> If there is any error, response: ERROR
AT+ECMTSUB=?	+ECMTSUB: (list of supported <tcpconnectID>s),(list of supported <msgID>s),“<topic>”,(list of supported <qos>s) OK
Max response time	5 s

7.6.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 1-65535
<result>	Integer type, Result of the command execution 0 Send subscribe packet successfully 1 Received SUBACK packet is incorrect 2 Send subscribe packet fail
<value>	Integer type It is granted qos levels
<topic>	String, topic of subscribe message. Max topic length is 255 bytes.
<qos>	Integer, message QoS, can be 0, 1 or 2.

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7.7 AT+ECMTUNS – Send MQTT unsubscribe packet

This command is used to send MQTT unsubscribe packet.

7.7.1 Format

Command	Response
AT+ECMTUNS=<tcpconnectID>,<msgID>,“<topic>”	OK +ECMTUNS:<tcpconnectID>,<msgID>,<result> If there is any error, response: ERROR
AT+ECMTUNS=?	+ECMTUNS: (list of supported <tcpconnectID>s),(list of supported <msgID>s),“<topic>” OK
Max response time	5 s

7.7.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 1-65535
<topic>	String, topic of subscribe message. Max topic length is 255 bytes.
<result>	Integer type, Result of the command execution 0 Send unsubscribe packet successfully 1 Received UNSUBACK packet is incorrect 2 Send unsubscribe packet fail

7.8 AT+ECMTPUB – Send MQTT publication packet

This command is used to send MQTT publication packet.

7.8.1 Format

Command	Response

AT+ECMTPUB=<tcpconnectID>,<msgID>,<qos>,<retain>,“<topic>”[,“<payload>”]	OK +ECMTPUB:<tcpconnectID>,<msgID>,<result> If there is any error, response: ERROR Note: If the command did not contain the parameter of “payload”, the input of data must end with “CTRL+Z”.
AT+ECMTPUB=?	+ECMTPUB: (list of supported <tcpconnectID>s),(list of supported <msgID>s),(list of supported <qos>s),(list of supported <retain>s),“<topic>”,“<payload>” OK
Max response time	20 s

7.8.2 Field

Parameters	**Description**
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet. The range is 1-65535. It will be 0 only when <qos>=0.
<topic>	String, topic of subscribe message. Max topic length is 255 bytes.
<qos>	Integer, message QoS, can be 0, 1 or 2.
<retain>	Integer type 0 Server should not retain the message 1 Server should retain the message
<payload>	String type Message that needs to be published. Maximum length is 1024 bytes
<result>	Integer type, Result of the command execution 0 Send publish packet successfully 1 Received PUBACK packet is incorrect 2 Send publish packet fail

7.9 +ECMTSTAT – Receive MQTT status packet

This is an unsolicited message to indicate receive MQTT status packet.

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7.9.1 Format

Command	Response
	+ECMTSTAT: <tcpconnectID>,<err_code>

7.9.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<err_code>	Integer type. Error code 1 MQTT_CONN_NOT_OPEN 2 MQTT_CONN_IS_OPENING 3 MQTT_CONN_OPENED 4 MQTT_CONN_OPEN_FAIL 5 MQTT_CONN_IS_CONNECTING 6 MQTT_CONN_CONNECTED

7.10 +ECMTRECV – Receive data from MQTT server

This is an unsolicited message to indicate MQTT client receive data from MQTT server.

7.10.1 Format

Command	Response
	+ECMTRECV: <tcpconnectID>,< msgID>,<topic>,<data>

7.10.2 Field

Parameters	Description
<tcpconnectID>	Integer type MQTT socket identifier. The value is 0
<msgID>	Integer type Message identifier of packet.
<topic>	String, topic of subscribe message. Max topic length is 255 bytes.

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<data>	String type Receive data from server

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8 HTTP Client

8.1 AT+HTTPCFG – Configure HTTP(S) related parameters

This command is used to configure HTTP(S) related parameters, including custom request headers, and output HTTP response headers. If the setting command has only the first parameter, it means to query the current configuration.

8.1.1 Format

Command	Response
AT+HTTPCFG=<CFG_ITEM>[,<CFG_VALUE>]	If optional parameters are omitted, query the current configuration: +HTTPCFG: <CFG_ITEM>,<CFG_VALUE>] OK If optional parameters are specified: OK If there is any error, response: +HTTP ERROR:
AT+HTTPCFG=?	+HTTPCFG: "contextid":(1-15) +HTTPCFG: "requestheader":(0-1) +HTTPCFG: "responseheader":(0-1) +HTTPCFG: "sslctxid":(0-5) +HTTPCFG: "contenttype":(0-3) OK
AT+HTTPCFG?	+HTTPCFG: "contextid",1 +HTTPCFG: "requestheader",0 +HTTPCFG: "responseheader",0 +HTTPCFG: "sslctxid":0 +HTTPCFG: "contenttype",0
Max response time	300 ms

8.1.2 Field

Parameters	Description
<CFG_ITEM>	"contextid"/"requestheader"/"response_header"/"sslctxid""contenttype"

<CFG_VALUE>	contextid: Http client context id, Range: 1~15. requestheader: 0: disable 1: enable responseheader: 0: disable 1: enable sslctxid: SSL context ID, Range: 0~5. Contenttype: 0 application/x-www-form-urlencoded 1 text/plain 2 application/octet-stream 3 multipart/form-data AT+HTTPREAD and AT+HTTPREADFILE will fail to execute 0: disable 1: enable

8.2 AT+HTTPURL – Set HTTP URL

The URL must start with http:// or https://, which means access to an HTTP or HTTPS server.

8.2.1 Format

Command	Response
AT+HTTPURL=?	+HTTPURL: (1-2048),(1-65535) OK

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AT+HTTPURL=<URL_Length>[,<timeout>]	If parameter valid: CONNECT TA switches to transparent transmission mode and can enter URL. When the total size of the input data reaches , TA will switch back to command mode and report the following results: OK If the input time reaches , but the received URL length is less than , TA will switch back to command mode and report the following results: +HTTP ERROR:
AT+HTTPURL?	+HTTPURL: <URL> OK
Max response time	300 ms

8.2.2 Field

Parameters	Description
<URL_Length>	Integer type URL length. Range: 1~2048. Unit: byte.
<timeout>	Integer type The maximum input time of the URL. Range: 1~65535. Default value: 60. Unit: seconds.
<URL>	String type, The URL of an HTTP or HTTPS server.If set IPV6 address,we need to add [ ],such as : HTTP://[IPV6 PATH]:PORT

8.3 AT+HTTPGET - Send HTTP GET request to HTTP(S) server

According to in the AT+HTTPCFG="requestheader"[,<request_header>], the AT+HTTPGET setting command has two forms:

If is 0, after sending AT+HTTPGET, if CONNECT is output within 20 seconds, it means the HTTP(S) server connection is successful, if not, it will output +HTTP ERROR:<err>. If <request_header> is 1, it will immediately output CONNECT after sending AT+HTTPGET, and switch to transparent transmission mode, and receive custom request headers. After input custom request headers, if CONNECT is output within 20 seconds, it means successfully connect to HTTP server, if not, it will output +HTTP ERROR:<err> .

After send the AT+HTTPGET and get OK, it will output URC +HTTPGET:<err>,<httprspcode>[,content_length] after the data download is complete. If the HTTP response headers includes Content-Length, <content_length>will display.

When the GET request is executed and the download is interrupted by network problems, it will display +HTTPGET: <err>,<httprspcode>,<content_length>, the value of <err> is 11, and <content_length> indicates the length of data already download. Use AT+HTTPREAD or AT+HTTPREADFILE to read the downloaded part. After the network is restored, use the AT+HTTPGETEX command to download the remaining part.

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8.3.1 Format

Command	Response
AT+HTTPGET=?	+HTTPGET: (1-2048),(1-65535) OK
AT+HTTPGET[=<data_length>[,<input_time>]]	For <request_header> is 0: OK If connect success: CONNECT after get response: +HTTPGET: <err>,<httprspcode>[,<content_length>] For <requestheader> is 1: If parameter valid: CONNECT TA switch to transparent transmission mode, and receive customized request headers: OK Success connect with HTTP server: CONNECT After receive the response: +HTTPGET: <err>,<httprspcode>[,<content_length>] If input time is out of <input_time> and receive data less than <data_length>, TA will turn to command mode and display: +HTTP ERROR:<err>
Max response time	5 s

8.3.2 Field

Parameters	Description
<data_length>	Integer type HTTP request headers' length. Range: 1~2048. Unit: byte
<input_time>	Integer type The maximum input time of the HTTP request headers. Range: 1~65535. Default value: 60. Unit: second
<httprspcode>	Integer type HTTP response code.

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<request_header>	Integer type, Disable or enable custom HTTP headers. 0 disable 1 enable
<content_length>	Integer type HTTP response body length. Unit: byte.

8.4 AT+HTTPGETEX - Send range GET request to HTTP(S) server

Similar to reading files, MCU can get data of specified location and specified length from HTTP(S) server through AT+HTTPGETEX. This command can only be executed when AT+HTTPCFG="requestheader",0. HTTP(S) server will respond to the range GET request with 206.

8.4.1 Format

Command	Response
AT+HTTPGETEX=?	+HTTPGETEX: <start_postion>,<read_len> OK
AT+HTTPGETEX=<start_position>,<read_len>	OK If receive the response: +HTTPGET: <err>,<httprspcode>[,<content_length>] If there is any error: +HTTP ERROR:<err> Note: The sum of the starting position and the read length cannot exceed 2147483647.
Max response time	5 s

8.4.2 Field

Parameters	Description
<start_position>	Integer type. Start position of data want get, start from 0, max length is 2147483647.
<read_len>	Integer type Length of data want get, max length is 2147483647.
<httprspcode>	Integer type HTTP response code.

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<err>	Integer type Error code

8.5 AT+HTTPPOST - Send POST request to HTTP(S) Server

This command is used to send HTTP POST request. After AT+HTTPPOST is input, TA will output CONNECT and enter the transparent transmission mode. The user enters the information that the POST request needs to carry through the UART port, and once the input length reaches the <data_length>, TA exit the transparent transmission mode, return OK, and begin connect. If CONNECT is output within 20 seconds, it means the HTTP(S) server connection is successful; if not, +HTTP ERROR: <err> will be output.

The following URC will output after the response are downloaded:

+HTTPPOST:<err>,<httprspcode>[,<content_length>]. .

According to the <request_header> in the AT+HTTPCFG="requestheader"[,<request_header>], the request data has two forms. If <request_header> is 0, the request data only need to include POST request body, If <request_header> is 1, the request data need to include POST request headers and POST request body.

8.5.1 Format

Command	**Response**
AT+HTTPPOST=?	+HTTPPOST: (1-4096),(1-65535) OK

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AT+HTTPPOST=<data_length>[,<input_time>]	If <request_header> is 0 (disable custom HTTP request headers) CONNECT TA switch to transparent transmission mode, and input request data: HTTP POST request body. When the data length reaches the <data_length>,TA back to command mode and response: OK receive the HTTP(S) server's response: +HTTPPOST:<err>,<httprspcode>[,<content_length>] If input time is out of <input_time> and receive data less than <data_length>,TA will turn to command mode and response: +HTTP ERROR: <err> If there is any error, response: +HTTP ERROR:<err> If <request_header> is 1 (enable custom HTTP request headers) Response CONNECT TA switch to transparent transmission mode, and input request data: HTTP POST request headers and HTTP POST request body. When the data length reaches the <data_length>,TA back to command mode: OK receive the HTTP(S) server's response: +HTTPPOST:<err>,<httprspcode>[,<content_length>] If input time is out of <input_time> and receive data less than <data_length>,TA will turn to command mode and response: +HTTP ERROR: <err> If there is any error, response: +HTTP ERROR:<err>
Max response time	25 s

8.5.2 Field

Parameters	Description
<data_length>	Integer type. If <request_header> is 0, it's POST request body's length, If <request_header> is 1, It's HTTP POST request headers and body's length. Range: 1~4096. Unit: byte.

<input_time>	Integer type The maximum input time of the HTTP POST request. Range: 1~65535. Default value: 60. Unit: second.
<httprspcode>	Integer type. HTTP response code. Refer to chapter 6.
<content_length>	Integer type HTTP response content length. Unit: byte.
<err>	Integer type Error code

8.6 AT+HTTPPOSTFILE - Send POST request in file

This command can send the POST request that save in the file. According the <request_header> in the AT+HTTPCFG="requestheader"[,<request_header>], the command has two forms:
If <request_header> is 0, the content of the file is POST content; if <request_header> is 1,the content of the file is POST request headers and POST content. After the server's response is downloaded successfully or the download is interrupted due to network reasons, TA will send URC +HTTPPOSTFILE: <err>,<httprspcode>[,<content_length>].

8.6.1 Format

Command	Response
AT+HTTPPOSTFILE=?	+HTTPPOSTFILE: <file_name> OK
AT+HTTPPOSTFILE=<file_name>	connect with HTTP server success: CONNECT OK receive the HTTP(S) server's response: +HTTPPOSTFILE:<err>,<httprspcode>[,<content_length>] If there is any error, response: +HTTP ERROR:<err>
Max response time	5 s

8.6.2 Field

Parameters	Description
<file_name>	String type File name. The maximum length: 64 byte.

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<err>	Integer type Error code

8.7 AT+HTTPSTOP - Cancel HTTP request

MCU can cancel the HTTP GET/POST request through this command, and disconnect the connection with HTTP(S) server.

8.7.1 Format

Command	**Response**
AT+HTTPSTOP=?	OK
AT+HTTPSTOP	OK If there is any error, response: +HTTP ERROR:<err>
Max response time	10 s

8.7.2 Field

Parameters	**Description**
<err>	Integer type Error code

8.8 AT+HTTPREADFILE - Read HTTP(S) response to file

After receive the response of HTTP(S) server, this command can write the response data to file. It should after get the URC:

+HTTPGET: <err>,<httprspcode>[,<content_length>],

+HTTPPOST:<err>,<httprspcode>[,<content_length>]

+HTTPPOSTFILE:<err>,<httprspcode>[,<content_length>].

8.8.1 Format

Command	**Response**

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AT+HTTPREADFILE=?	+HTTPREADFILE: <file_name> OK
AT+HTTPREADFILE=<file_name>	OK If there is any error, response: +HTTP ERROR: <err>
Max response time	25 s

8.8.2 Field

Parameters	Description
<file_name>	String type File name. The maximum length: 64 byte.
<err>	Integer type Error code

8.9 AT+HTTPREAD - Read the response of HTTP(S) server

After receive the response of HTTP(S) server, this command can read the response data from UART/USB. It should after get the URC:
+HTTPGET:<err>,<httprspcode>[,<content_length>],
+HTTPPOST: <err>,<httprspcode>[,<content_length>]
+HTTPPOSTFILE: <err>,<httprspcode>[,<content_length>].
According the <response_header> in AT+HTTPCFG="responseheader"[,<response_header>], If <response_header> is 0, this command's response does not include the response headers, if <response_header> is 1, this command's response include the response headers.

8.9.1 Format

Command	Response
AT+HTTPREAD=?	OK

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AT+HTTPREAD	OK CONNECT <HTTP response data> If there is any error, response: +HTTP ERROR:<err>
Max response time	25 s

8.9.2 Field

Parameters	Description
<HTTP response data>	The data received from server.
<err>	Integer type Error code

8.10 AT+ HTTPFOTADL - Update Firmware

8.10.1 Format

Command	Response
AT+HTTPFOTADL =<URL>,<download_URC_max>	OK or ERROR
AT+ HTTPFOTADL?	+ HTTPFOTADL: <percent> OK
AT+ HTTPFOTADL =?	+HTTPFOTADL: <URL>,(0-100) OK
Max response time	25 s

8.10.2 Field

Parameters	Description

<URL>	String type. Maximum length: 255 bytes. The URL should start with "http: / /" or "HTTPS: / /".
<download_URC_max>	Then 50 download progress URCs will be reported, among which the 25th URC means half of the download is completed and the 50th URC means half of the download is completed The URC indicates that the download is complete. 0 prohibit reporting download progress URC 5 – 100 maximum number of reports for download progress URC
<percent>	Integer. Download or upgrade progress.
<err>	Integer. 0 indicates the upgrade was successful, and other values indicate errors

8.11 HTTP error codes

Parameters	Description

<err_code>	Integer type 0 Operation successful 1 Parameter error 2 Input timeout 3 URL is not set 4 Operation not support 5 No GET/POST requests 6 Request processing 7 Filename too long 8 File not found 9 Memory not enough 10 Ssl config error 11 URL parse error 12 DNS resolution failed 13 Protocol error 14 Socket error 15 PDP context bind failed 16 Socket connect timeout 17 Socket connect error 18 Socket connect close 19 Ssl error 20 Request timeout 21 Internal error 22 URL format error 23 Ssl is not supported 100 output complete 300 not init http

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9 Network Command – SNTP

9.1 AT+ECSNTP – Query network time

This command is used to synchronize the local time with the Universal Time Coordinated (UTC) via the SNTP server.

9.1.1 Format

Command	Response
AT+ECSNTP=<server>[,<port>,<autosync>]	OK +ECSNTP:<time> Or ERROR
AT+ECSNTP=?	+ECSNTP: ("IP ADDR\URL"),(0-65535),(0,1) OK
Max response time	300 ms

9.1.2 Field

Parameters	Description
<server>	A string of SNTP server IP address or domain name.
<port>	Integer type The NTP server port number, default is 123.
<autosync>	Integer type Whether to automatically set synchronized time to local UTC. Default is 0. 0 Not set 1 Set
<time>	String type Time synchronized from NTP server. The format is: “yyyy/mm/dd, hh:mm:ss”, where characters indicate year , month, day, hour, minutes and seconds, E.g. 6th of May 2014, 22:10:00 equals to "2014/05/06,22:10:00".

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9.2 AT+QNTP – Query network time

This command is used to synchronize the local time with the Universal Time Coordinated (UTC) via the QNTP server.

9.2.1 Format

Command	Response
AT+QNTP=<contextid>,<server>[,<port>,<autosync>]	OK +QNTP:0,<time> Or ERROR
AT+QNTP=?	+QNTP: (1-15),("IP ADDR\URL"),(1-65535),(0,1) OK
AT+QNTP?	+QNTP:(IP ADDR\URL),(1-65535) OK
Max response time	300 ms

9.2.2 Field

Parameters	Description
<contextid>	Int Type <1-15>
<server>	A string of QNTP server IP address or domain name.
<port>	Integer type The NTP server port number, default is 123.
<autosync>	Integer type Whether to automatically set synchronized time to local UTC. Default is 0. 0 Not set 1 Set
<time>	String type Time synchronized from NTP server. The format is: “yyyy/mm/dd, hh:mm:ss”, where characters indicate year , month, day, hour, minutes and seconds, E.g. 6th of May 2014, 22:10:00 equals to "2014/05/06,22:10:00".

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10 SSL Command(Y7S/Y7SW/Y6 not support except for MF)

10.1 AT+SSLCFG - Configure SSL related parameters

This command is used to configure the cipher suite of the SSL context, CA certificate path, authentication mode, client certificate path, client key and other SSL configuration. These parameters will be used in the handshake process. If the third parameter is default, it means to query the current configuration.

<SSL_ctxID> is the index of the SSL context. The module supports up to six SSL contexts. Multiple SSL connections can be established based on one SSL context.

10.1.1 Format

Command	Response

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AT+SSLCFG=?	+SSLCFG: ”sslversion”:(range of supported <SSL_ctxID>),(range of supported <sslversion>) +SSLCFG: “ciphersuite”:(range of supported <SSL_ctxID>),(range of supported <cipher_suites>) +SSLCFG: “cacert”:(range of supported <SSL_ctxID>),<cacertpath> +SSLCFG: ”clientcert”:(range of supported <SSL_ctxID>),<client_cert_path> +SSLCFG: ”clientkey”:(range of supported <SSL_ctxID>),<client_key_path> +SSLCFG: ”seclevel”:(range of supported <SSL_ctxID>),(range of supported <serclevel>) +SSLCFG: ”ssession_cache”:(range of supported <SSL_ctxID>),(range of supported <session_cache_enable>) +SSLCFG: ”sni”:(range of supported <SSL_ctxID>),(range of supported <sni>) +SSLCFG: "ignorelocaltime":(range of supported <SSL_ctxID>), (range of supported <ignore_ltime >) OK
AT+SSLCFG=”sslversion”,<SSL_ctxID>[,<SSL_version>]	If the optional parameter is omitted, query the SSL version of the specified SSL context: +SSLCFG: “sslversion”,<SSL_ctxID>,<SSL_version> OK If optional parameters are specified, configure the SSL version of the specified SSL context: OK or +SSL ERROR: <err>
AT+SSLCFG=”ciphersuite”,<SSL_ctxID>[,<cipher_suites>]	If the optional parameter is omitted, query the cipher suite of the specified SSL context: +SSLCFG: “ciphersuite”,<SSL_ctxID>,<cipher_suites> OK If optional parameters are specified, configure the cipher suite of the specified SSL context: OK Or +SSL ERROR: <err>

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AT+SSLCFG=”cacert”,<SSL_ctxID>[,<cacertpath>]	If the optional parameter is omitted, query the CA certificate path of the specified SSL context: +SSLCFG: “cacert”,<SSL_ctxID>,<cacertpath> OK If optional parameters are specified, configure the CA certificate path of the specified SSL context: OK or +SSL ERROR: <err>
AT+SSLCFG=”clientcert”,<SSL_ctxID>[,<client_cert_path>]	If the optional parameter is omitted, query the client certificate path of the specified SSL context: +SSLCFG: ”clientcert” ,<SSL_ctxID>,<client_cert_path> OK If optional parameters are specified, configure the client certificate path of the specified SSL context: OK or +SSL ERROR: <err>
AT+SSLCFG=”clientkey”,<SSL_ctxID>[,<client_key_path>]	If the optional parameter is omitted, query the client key path of the specified SSL context: +SSLCFG: ”clientkey” ,<SSL_ctxID>,<client_key_path> OK If optional parameters are specified, configure the client key path of the specified SSL context: OK or +SSL ERROR: <err>
AT+SSLCFG=”seclevel”,<SSL_ctxID>[,<seclevel >]	If the optional parameter is omitted, query the authentication mode of the specified SSL context: +SSLCFG: ”seclevel” ,<SSL_ctxID>,<seclevel> OK If optional parameters are specified, configure the authentication mode of the specified SSL context: OK or +SSL ERROR: <err>

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AT+SSLCFG=”session_cache”,<SSL_ctxID>[,<session_cache_enable>]	If the optional parameter is omitted, query the session resumption function of the specified SSL context: +SSLCFG: ”session_cache”,<SSL_ctxID>,<session_cache_enable> OK If optional parameters are specified, configure the session resumption function of the specified SSL context: OK or SSL ERROR: <err>
AT+SSLCFG=”sni”,<SSL_ctxID>[,<SNI>]	If the optional parameter is omitted, query the server name indication function of the specified SSL context: +SSLCFG: ”sni”,<SSL_ctxID>,<SNI> OK If optional parameters are specified, configure the server name indication function of the specified SSL context: OK or +SSL ERROR: <err>
AT+SSLCFG=”ignorelocaltime” ,<SSL_ctxID>[,<ignore_ltime >]	If the optional parameter is omitted, the query specifies whether to ignore certificate validity validation for the SSL context: +SSLCFG: ”ignorelocaltime” ,<SSL_ctxID>,<ignore_ltime> OK If you specify the optional parameter, the specified SSL context is turned on or off to ignore certificate validation validation: OK or +SSL ERROR: <err>
Max response time	300 ms

10.1.2 Field

Parameters	**Description**
<SSL_ctxID>	Integer type SSL context identifier. Range:0~5
<SSL_version>	Integer type 0 SSL 3.0 1 TLS 1.0 2 TLS 2.0 3 TLS 3.0 4 all support

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<cipher_suites>	Hexadecimal number. SSL cipher suites 0xc0ad TLS_ECDHE_ECDSA_WITH_AES_256_CCM 0xc09f TLS_DHE_RSA_WITH_AES_256_CCM
<cacertpath>	String type CA certification path
<client_cert_path>	String type client certification path
<client_key_path>	String type client key path
<seclevel>	Integer type. authentication mode 0 No authentication 1 Perform server authentication 2 If required by the remote server, perform server and client authentication
<session_cache_enable>	Integer type 0 Close the session resumption function 1 Open the session resumption function
<SNI>	Integer type 0 Close the server name indication function 1 Open the server name indication function
<ignore_ltime >	Integer type 0 is not ignored Certificate time validity verification 1 Ignore Certificate time validity verification
<err>	Error code.

10.2 AT+SSLOPEN - Connect to the remote server and handshake

This command is used to establish an SSL socket connection with a remote server. During the client and server handshake, use the AT+SSLCFG configuration parameter. After the handshake is complete, the module can send or receive data through this SSL connection.

10.2.1 Format

Command	**Response**

AT+SSLOPEN=?	+SSLOPEN: (range of supported <PDP_ctxID>), (range of supported <SSL_ctxID>), (range of supported <clientID>),<serveraddr>,<server_port>[,(range of supported <access_mode>)] OK
AT+SSLOPEN=<PDP_ctxID>,<SSL_ctxID>,<clientID>,<serveraddr>,<server_port>[,<access_mode>]	If the SSL connection is successfully established: if the <access_mode> is direct push mode: OK If the <access_mode> is transparent mode: CONNECT If there is any error, response: +SSL ERROR: <err>
Max response time	150 s

10.2.2 Field

Parameters	Description
<PDP_ctxID>	Integer type PDP context identifier. Range: 1~15
<clientID>	Integer type SSL client identifier. Range: 0~3
<SSL_ctxID>	Integer type SSL context identifier. Range: 0~5
<serveraddr>	String type The server address
<server_port>	Integer type The server port
<access_mode>	Integer type 0 Direct push mode 1 Transparent access mode
<err>	Error code.

10.3 AT+SSLSEND - Send data

After the SSL connection is established, the module can send data.

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10.3.1 Format

Command	Response
AT+SSLSEND=?	+SSLSEND: (range of supported <clientID>),(range of supported <sendlen>) OK
AT+SSLSEND=<clientID>,<sendlen>	If the parameter format is correct > Enter the data until the length of the data reaches the <sendlen>. If the data is sent successfully: OK If sending fails: +SSL ERROR: <err>
Max response time	30 s

10.3.2 Field

Parameters	Description
<clientID>	Integer type SSL CLIENT INDEX. Range: 0~3.
<sendlen>	Integer type Length of data. Range: 1~1460. Unit: byte.
<err>	Integer type Error code.

10.4 AT+SSLCLOSE - Close the SSL connection

This command is used to close the SSL connection.

10.4.1 Format

Command	Response
AT+SSLCLOSE=?	+SSLCLOSE: (range of supported <clientID>) OK

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AT+SSLCLOSE=<clientID>	If the SSL connection is successfully closed: OK If there is any error, response: +SSL ERROR:<err>
Max response time	25 s

10.4.2 Field

Parameters	Description
<clientID>	Integer type SSL CLIENT INDEX. Range: 0~3.
<err>	Integer type Error code.

10.5 AT+SSLSTATE - Query the status of the socket connection

This command is used to query the status of the socket connection. If the SSL connection has been disconnected, only OK will be returned.

10.5.1 Format

Command	Response
AT+SSLSTATE=?	OK
AT+SSLSTATE=<client_ID>	+SSLSTATE:<IP_address>,<remote_port>,<socket_state>,<SSL_ctxID> OK If there is any error, response: +SSL ERROR:<err>
Max response time	25 s

10.5.2 Field

Parameters	Description
<IP_address>	String type Remote server IP address.

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<remote_port>	Integer type: Remote server port.
<socket_state>	Integer type 0 “Init” 1 “Connected” 2 “Disconnected” 3 “Close”
<SSL_ctxID>	Integer type SSL context identifier. Range: 0~5.
<err>	Integer type Error code.

10.6 +SSLURC:”recv” - The received data

10.6.1 Format

Command	Response
	+SSLURC: <clientID>,<recvlength>,<data>

10.6.2 Field

Parameters	Description
<clientID>	Integer type SSL CLIENT INDEX. Range: 0~3.
<recvlength>	Integer type The length of the data received. Unit: byte.
<data>	String type: Data received.

10.7 +SSLURC: ”closed” - Notify disconnect

10.7.1 Format

Command	Response

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	+SSLURC: ”closed”,<clientID>

10.7.2 Field

Parameters	Description
<clientID>	Integer type SSL CLIENT INDEX. Range: 0~3.
Code’s of <err>	0 Operation successful 1 Parameter error 2 configure error 3 MBEDTLS error 4 Connect error 5 Already has a connect 6 Has no connect 7 Network not ready

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11 FILE Command(Y7S/Y7SW/Y6 not support)

11.1 AT+FOPEN

This command is used to open a file and get a file handle·

11.1.1 Format

Command	Response
AT+FOPEN=<filename>[,<mode>]	+FOPEN:<filehandle> OK If there is any error, response: ERROR
AT+FOPEN=?	+FOPEN:<filename>[,<mode>] OK
AT+FOPEN?	+FOPEN:<filename>,<filehandle>,<mode>,<size> [+FOPEN:<filename>,<filehandle>,<mode>,<size> […]] OK If there is any error, response: ERROR
Max response time	300 ms

11.1.2 Field

Parameters	Description
<filename>	String type File name，the max length is 62 bytes.
<filehandle>	Integer type File handle

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<mode>	Integer type 0 If the file does not exist, a new file will be created; if the file exists, the file will be opened directly. The files in both cases are both readable and writable 1 If the file does not exist, a new file will be created; if the file exists, the file will be cleared and overwritten. The files in both cases are both readable and writable 2 If the file exists, the file is opened, and the file is only readable; if the file does not exist, an error is responded
<size>	Integer type file data length

11.2 AT+FREAD

This command is used to read data from file.

11.2.1 Format

Command	Response
AT+FREAD=<filehandle>[,<length>]	CONNECT <read_length> OK Note: Switch to data mode, when the total size of data exceeds <length>, will switch back to command mode The size of the data read at one time cannot exceed 143360 bytes, or if the file size exceeds 143360 bytes, read it in segments. If there is any error, response: ERROR
AT+FREAD=?	+FREAD:<filehandle>[,<length>] OK
Max response time	5 s

11.2.2 Field

Parameters	Description

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<filehandle>	Integer type The file handle
<length>	Integer type Want to read data length,Range 0-8000
<read_length>	Integer type The length of the file actually read

11.3 AT+FWRITE

This command is used to write data·

11.3.1 Format

Command	**Response**
AT+FWRITE=<filehandle>[,<length>[,<tim eout>]]	CONNECT Switch to data mode, when the total size of written data exceeds <length> or writing time exceeds <timeout>, it will switch back to command mode +FWRITE: <written_length>,<total_length> OK If there is any error, response: ERROR
AT+FWRITE=?	+FWRITE:<filehandle>[,<length>[,<timeout>]] OK
Max response time	30 s

11.3.2 Field

Parameters	**Description**
<filehandle>	Integer type File handle.
<length>	Integer type The expected length of the written file, the default length is 1000, the default maximum length is 10K

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<timeout>	Integer type Waiting time for input data, 0-100s default is 5s.
<written_length>	Integer type The length of the data actually written
<total_length>	Integer type Total length of file

11.4 AT+FSEEK

This command is used to set file pointer.

11.4.1 Format

Command	**Response**
AT+FSEEK=<filehandle>,<offset>[,<position>]	OK If there is any error, response: ERROR
AT+FSEEK=?	+FSEEK:<filehandle>,<offset>[,<position>] OK
Max response time	300 ms

11.4.2 Field

Parameters	**Description**
<filehandle>	Integer type File handle.
<offset>	Integer type The offset of the file pointer from the initial position
<position>	Integer type 0 File initial location 1 The current position of the file pointer 2 File final location

11.5 AT+FPOSITION

This command is used to get file position offset.

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11.5.1 Format

Command	Response
AT+FPOSITION=<filehandle>	+FPOSITION:<offset> OK If there is any error, response: ERROR
AT+FPOSITION=?	+FPOSITION:<filehandle> OK
Max response time	300 ms

11.5.2 Field

Parameters	Description
<filehandle>	Integer type File handle
<offset>	Integer type The offset of the file pointer from the initial position

11.6 AT+FTUCAT

This command is used to tucat a file.

11.6.1 Format

Command	Response
AT+FTUCAT=<filehandle>	OK If there is any error, response: ERROR
AT+FTUCAT=?	+FTUCAT:<filehandle> OK
Max response time	5 s

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11.6.2 Field

Parameters	Description
<filehandle>	Integer type File handle

11.7 AT+FERASE

This command is used to erase a file.

11.7.1 Format

Command	Response
AT+FERASE=<filehandle>	OK If there is any error, response: ERROR
AT+FERASE=?	+FERASE:<filehandle> OK
Max response time	5 s

11.7.2 Field

Parameters	Description
<filehandle>	Integer type File handle

11.8 AT+FRENAME

This command is used to rename a file.

11.8.1 Format

Command	Response

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AT+FRENAME=<filename>,<newfilename>	OK If there is any error, response: ERROR
AT+FRENAME=?	+FREAME:<filename>,<newfilename> OK
Max response time	5 s

11.8.2 Field

Parameters	Description
<filename>	String type File name，the max length is 62 bytes.
<newfilename>	String type New File name.

11.9 AT+FCLOSE

This command is used to close file.

11.9.1 Format

Command	Response
AT+FCLOSE=<filehandle>	OK If there is any error, response: ERROR
AT+FCLOSE=?	+FCLOSE:<filehandle> OK
Max response time	5 s

11.9.2 Field

Parameters	Description
<filehandle>	Integer type File handle

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11.10 AT+FDELETE

This command is used to delete file.

11.10.1 Format

Command	Response
AT+FDELETE=<filehandle>	OK If there is any error, response: ERROR
AT+FDELETE=?	+FDELETE:<filehandle> OK
Max response time	5 s

11.10.2 Field

Parameters	Description
<filehandle>	Integer type File handle

11.11 FILE error codes

Parameters	Description
<err_code>	Integer type 1 FILE_PARAM_ERROR 2 FILE_OPERATION_NOT_SUPPORT 3 FILE_OPEN_ERROR 4 FILE_READ_ERROR 5 FILE_WRITE_ERROR 6 FILE_CLOSE_ERROR 7 FILE_SEEK_ERROR 8 FILE_POSITION_ERROR 9 FILE_TUCAT_ERROR 10 FILE_ERASE_ERROR 11 FILE_DELETE_ERROR

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12 LYNQ MQTT(Y7S/Y7SW/Y6 not support)

12.1 AT+MCONFIG - Related Parameters Configuration for MQTT

This command is used to set parameters configuration.

12.1.1 Format

Command	Response
AT+MCONFIG=?	+MCONFIG: ,,,(0-1),(0-2),(0-1),, OK
AT+MCONFIG=<clientid>[,<username>],[<password>][,<will_flag>,<will_qos>,<will_retain>,<will_topic>,<will_message>]]	OK Or ERROR
AT+MCONFIG?	+MCONFIG:<clientid>[,<username>,<password>[,<will_flag>,<will_qos>,<will_retain>,<will_topic>,<will_message>]] OK
Reference	Note This command you should send before "AT+MIPCONFIG" command.If the server is not a user name and password, please do not enter the user name and password, otherwise will logon failure.
Max response time	300 ms

12.1.2 Field

Parameters	Description
<clientid>	This parameter is used to allow the server to identify the client identity information. The maximum length of 128
<username>	This parameter is used to login server. The maximum length of 128
<password>	This parameter is used to login server. The maximum length of 128

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<will_flag>	Value of will flag: 1 If the Will flag is set 1, the Will QoS and Will Retain fields must be present in the Connect flags byte, and the Will Topic and Will Message fields must be present in the payload. 0 Without using will Qos, will retain, will topic, will message, please set 0.
<will_qos>	Quality of Service: 0 At most once delivery for will message 1 At least once delivery for will message 2 Exactly once delivery for will message
<will_retain>	Retain Flag: 0 the Server must not store the will message and must not remove or replace any existing retained message. 1 the Server must store the will Message and its QoS.
<will_topic>	The will topic of the will message. The maximum length is 128.
<will_message>	The will message content. The maximum length is 1024.

12.2 AT+MIPSTART - Set address and port and version

12.2.1 Format

Command	**Response**
AT+MIPSTART=?	+MIPSTART:,(0-65535),(3-4) OK
AT+MIPSTART=<address>,<port>[,<version>]	OK +MIPSTART: <result> Or ERROR
AT+MIPSTART?	+MIPSTART:<address>,<port>,<version> OK
Max response time	30 s

12.2.2 Field

Parameters	**Description**
<ipaddress>	The MQTT server domain name or IP address. The maximum length is 256

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<port>	The MQTT Server port.The value is between 0 and 65535, default value is 0.
<version>	The MQTT version.The value is between 3 and 4, default value is 3. 3 stands for mqtt version 3.1, 4 stands for mqtt version 3.1.1.
<result>	The result of opening socket operation: SUCCESS FAILURE

12.3 AT+MCONNECT - Client requests a connection to a Server

12.3.1 Format

Command	Response
AT+MCONNECT=?	+MCONNECT:(0-1),(30-1800) OK
AT+MCONNECT=<clean_session>,<keepalive>	OK +MCONNECT: <result> Or ERROR
AT+MCONNECT?	+MCONNECT:<clean_session>,<keepalive> OK
Max response time	30 s

12.3.2 Field

Parameters	Description

<clean_session>	This parameter specifies the handling of the Session state. 0 If CleanSession is set to 0, the Server must resume communications with the Client based on state from the current Session (as identified by the Client identifier). If there is no Session associated with the Client identifier the Server must create a new Session. The Client and Server MUST store the Session after the Client and Server are disconnected. After the disconnection of a Session that had CleanSession set to 0, the Server MUST store further QoS 1 and QoS 2 messages that match any subscriptions that the client had at the time of disconnection as part of the Session state. It may also store QoS 0 messages that meet the same criteria. 1 If CleanSession is set to 1, the Client and Server must discard any previous Session and start a new one. This Session lasts as long as the Network Connection. State data associated with this Session must not be reused in any subsequent Session.
<keepalive>	30-1800 The Keep Alive is a time interval measured in seconds.
<result>	The result of connectingmqtt server operation: SUCCESS FAILURE

12.4 AT+MPUB - Publish message

This command is used to publish message.

12.4.1 Format

Command	Response
AT+MPUB=?	+MPUB: ,(0-2),(0-1), OK
AT+MPUB=<topic>,<qos>,<retain>,<message>	OK or ERROR There are two kinds of results depending on the Settings: a.AT+MQTTMIX=0 +MPUB:<result> b.AT+MQTTMIX=1 If qos=0, nothing returned; if qos>0, return +MPUBID: <packetid> first,then return as follows: +MPUBACK: <packetid> qos = 1 +MPUBREC: <packetid> qos = 2 +MPUBCOMP:<packetid>

AT+MPUB?	+MPUB:<topic> ,<qos>,<retain>,<message> OK
Reference	Note This command is sent from a Client to a Server or from Server to a Client to transport an Application Message.
Max response time	20 s

12.4.2 Field

Parameters	Description
<topic>	The topic of the Application message. The maximum length is 256
<qos>	Quality of Service: 0 At most once delivery for Application message 1 At least once delivery for Application message 2 Exactly once delivery for Application message
<retain>	Retain Flag: 0 The Server must not store the Application message and must not remove or replace any existing retained message. 1 The Server must store the Application Message and its QoS.
<message>	The Application message content. The maximum length of 512
<result>	The result of publishing operation: SUCCESS FAILURE
<packetid>	The Packet Identifier field (1~65535).

12.5 AT+MPUBEX - Publish a long message

This command is used to publish message

12.5.1 Format

Command	Response
AT+MPUBEX=?	+MPUBEX: ,(0-2),(0-1),(1-4096) OK

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Write Command AT+MPUBEX=<topic>,<qos>,<retain>,<msgLen>	Response OK Or ERROR There are two kinds of results depending on the Settings: a.AT+MQTTMIX=0 +MPUBEX:<result> b.AT+MQTTMIX=1 If qos=0, nothing returned; if qos>0, return +MPUBID: <packetid> first,then return as follows: +MPUBACK: <packetid> qos = 1 +MPUBREC: <packetid> qos = 2 +MPUBCOMP: <packetid>
AT+MPUBEX?	+MPUBEX:<topic> ,<qos>,<retain>,<msgLen> OK
Reference	Note 1. In this mode, data in any format can be sent, such as JSON. 2 Input mode will not sent keepalive, so the input data interval cannot be longer than keepalive, or the connection may be disconnected. 3 Input mode should not exceed 4096 bytes at a time or some data will be lost, and each input should be at least 25 milliseconds apart.
Max response time	30 s

12.5.2 Field

Parameters	**Description**
<topic>	The topic of the Application message. The maximum length is 256
<qos>	Quality of Service: 0 At most once delivery for Application message 1 At least once delivery for Application message 2 Exactly once delivery for Application message
<retain>	Retain Flag: 0 The Server must not store the Application message and must not remove or replace any existing retained message. 1 The Server must store the Application Message and its QoS.
<msgLen>	The message content length. The maximum length of65535.
<result>	The result of publishing operation: SUCCESS FAILURE
<packetid>	The Packet Identifier field (1~65535).

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12.6 AT+MSUB - Subscribe to topics

This command is used to subscribe to topics

12.6.1 Format

Command	Response
AT+MSUB=?	+MSUB:,(0-2) OK
AT+MSUB?	+MSUB:<topic>,<qos> OK
AT+MSUB=<topic>,<qos>	OK Or ERROR There are two kinds of results depending on the Settings: a. AT+MQTTMIX=0 +MSUB:<result> b. AT+MQTTMIX=1 +MSUBID: <packetid> +MSUBACK: <packetid> This command is used to request to subscribe a topic.Themaximum topics is 5. The command is asynchronous.If has succeeded for subscribing a topic, the client will be to start to receive message from other client. Receive format: +MSUB: <topic>,<len> bytes,<message>
Reference	Note This command is sent from the Client to the Server to create one or more Subscriptions.
Max response time	5 s

12.6.2 Field

Parameters	Description
<topic>	The topic of the Application message. The maximum length is 256
<qos>	Quality of Service: 0 At most once delivery for Application message 1 At least once delivery for Application message 2 Exactly once delivery for Application message

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<len>	The received message size.
<message>	Message content.Max length is 1024.
<result>	The result of subscribingtopic operation: SUCCESS FAILURE
<packetid>	The Packet Identifier field (1~65535).

Example:

AT Command	**Response**
Subscribe: If AT+MQTTMIX=0 : AT+MSUB=mobiletek/topic,0	OK +MSUB: SUCCESS
If AT+MQTTMIX=1, AT+MSUB=mobiletek/topic,0	OK +MSUBID:5 +MSUBACK:5

12.7 AT+MUNSUB - UnSubscribe from topics

This command is used to Unsubsrcibe from topic

12.7.1 Format

Command	**Response**
AT+MUNSUB=<topic>	OK Or ERROR There are two kinds of results depending on the Settings: a. AT+MQTTMIX=0 +MUNSUB:<result> b. AT+MQTTMIX=1 +MUNSUBID: <packetid> +MUNSUBACK: <packetid>
Max response time	30 s

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12.7.2 Field

Parameters	Description
<topic>	The topic of the will message. The maximum length is 256
<result>	The result of unsubscribing topic operation: SUCCESS FAILURE
<packetid>	The Packet Identifier field (1~65535).

Example

AT commands	Response
Unsubscribe: If AT+MQTTMIX=0 : AT+MUNSUB=mobiletek/topic	OK +MUNSUB: SUCCESS
If AT+MQTTMIX=1, AT+MUNSUB=mobiletek/topic	OK +MUNSUBID:6 +MUNSUBACK:6

12.8 AT+MDISCONNECT - Close MQTT connection

This command is used to close MQTT connection.

12.8.1 Format

Command	Response
AT+MDISCONNECT	OK +MDISCONNECT: <result> Or Already disconnect ERROR
Max response time	5 s

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12.8.2 Field

Parameters	Description
<result>	The result of disconnecting operation: SUCCESS FAILURE

12.9 AT+MIPCLOSE - Release MQTT resources

This command is used to close TCP connection.

12.9.1 Format

Command	Response
AT+MIPCLOSE	OK +MIPCLOSE:<result> Or ERROR
Max response time	5 s

12.9.2 Field

Parameters	Description
<result>	The result of releasing operation: SUCCESS FAILURE

12.10 AT+MQTTSTATU - Query the MQTT connection status

This command is used to Query the MQTT connection status.

12.10.1 Format

Command	Response

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AT+MQTTSTATU	+MQTTSTATU:<statu> OK
Max response time	5 s

12.10.2 Field

Parameters	Description
<statu>	0 MQTT connection is not established 1 The MQTT connection is successful

12.11 AT+MQTTSSL – MQTT SSL support switch

This command is used to Query the MQTT connection status.

12.11.1 Format

Command	Response
AT+MQTTSSL=?	+MQTTSSL: (0-1),(0-1) OK
AT+MQTTSSL?	+MQTTSSL: <action>,<cert> OK
AT+MQTTSSL=<action>[,cert]	OK Or ERROR
Reference	Note The command must be used before AT+MIPSTART
Max response time	5 s

12.11.2 Field

Parameters	Description
<action>	The switch for SSL support. 0 close SSL support 1 open SSL support The default value is 0.

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<cert>	Whether to use the certificate. 0 Ignore the certificate 1 Use the certificate

12.12 AT+MQTTMIX - Set additional configuration parameters

This command is used to set additional configuration parameters.

12.12.1 Format

Command	Response
AT+MQTTMIX=?	+MQTTMIX: (0-1) OK
AT+MQTTMIX?	+MQTTMIX: <mode> OK
AT+MQTTMIX=<mode>	OK Or ERROR
Max response time	5 s

12.12.2 Field

Parameters	Description
<mode>	0 No packetid Mode(Default Mode) 1 Packet ID Mode

12.13 MQTT error codes

Parameters	Description

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<err_code>	Integer type 200 MQTT_OK 201 MQTT_ERR 202 MQTT_NETWORK_ERR 203 MQTT_CONTEXT_ERR 204 MQTT_PARAM_ERR 205 MQTT_SOCKET_ERR 206 MQTT_SOCKET_TIME_ERR 207 MQTT_MQTT_CONN_ERR 208 MQTT_TASK_ERR 209 MQTT_RECONNECT 210 MQTT_CLIENT_ERR 211 MQTT_ALI_ENCRYP_ERR 212 MQTT_BUSY_ERR 213 MQTT_CONTINUE 214 MQTT_MAX_ERR

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13 LYNQ HTTP(Y7S/Y7P/Y7SW/Y6 not support)

13.1 AT$HTTPOPEN - Open HTTP service

The command is used to open HTTP service.

13.1.1 Format

Command	Response
AT$HTTPOPEN	OK
AT$HTTPOPEN?	$HTTPOPEN: <open_or_not> OK
Max response time	300 ms

13.1.2 Field

Parameters	Description
<open_or_not>	Return HTTP service is open or not. 1:HTTP service is opened 0:HTTP service is not opened

13.2 AT$HTTPCLOSE - Close HTTP service

This command is used to close HTTP service. After executing this command, HTTP will be closed after 1 minute and the http will be unavailable.

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13.2.1 Format

Command	Response
AT$HTTPCLOSE	OK
AT$HTTPCLOSE?	$HTTPCLOSE: <closed_or_not> OK
Max response time	300 ms

13.2.2 Field

Parameters	Description
<closed_or_not>	Return HTTP service is closed or not. 1:HTTP service is closed 0:HTTP service is not closed

13.3 AT$HTTPPARA - Set HTTP request URL and port

The command is used to Set HTTP request URL and port.

13.3.1 Format

Command	Response
AT$HTTPPARA=?	$HTTPPARA:"",(0-65535),(0-1),(0-1) OK
AT$HTTPPARA=<url>,<port>[,<type>][,<cert>]	OK Or ERROR
AT$HTTPPARA?	Return current HTTP request's host,uri,and port,such as: AT$HTTPPARA? Host : "182.150.28.206" URI : "/httpdemo/http" Port : 8182. Cert : 1 OK
Max response time	300 ms

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13.3.2 Field

Parameters	Description
<url>	0-255 HTTP client URL:"http://server/path:port". If set IPV6 address,we need to add [ ],such as : "http://[ipv6 path]:port"
<port>	The HTTP request's port. The HTTP default port is 80, and HTTPS is 443.
<type>	HTTP request type. Default is 0. 0: HTTP request 1: HTTPS request
<cert>	Indicates whether the SSL connection ignores the certificate, and default is 0. 0 - ignores certificate 1 - use certificate

13.4 AT$HTTPACTION - Send HTTP request

The command is used to send HTTP request. Support request include GET, POST and HEAD.

13.4.1 Format

Command	Response
AT$HTTPACTION=<request>	$HTTPRECV:DATA,<len> ..... $HTTPRECV:DATA,2 <\r><\n> $HTTPRECV:DATA,<len> or $HTTPERROR:<errno> ERROR This command will return HTTP response header fields and file path which storage HTML text or download file if request success. If request fail, this command just return response header fields. Specially, the HEAD request only return response header fields. For POST, must set Conten-Length header item and POST's content data. Note If a 202 error code appears and the request fails, execute the AT$HTTPOPEN instruction after the AT$HTTPCLOSE command.

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AT$HTTPACTION=?	$HTTPACTION:(0-3) OK
Max response time	150 s

13.4.2 Field

Parameters	Description
<request>	HTTP request type, available data include 0,1,2,3. 0: GET request 1: POST request 2: HEAD request 3: POST request (Content length must be less than 500 and set content in the first.)

13.5 AT$HTTPDATA - Set HTTP post request's data

This command is used to set http post request's data.

13.5.1 Format

Command	Response
AT$HTTPDATA=<data_len>	OK Or ERROR
AT$HTTPDATA=?	$HTTPDATA:(0-1024) OK
AT$HTTPDATA?	$HTTPDATA:<data_len> OK
Max response time	25 s

13.5.2 Field

Parameters	Description
<data_len>	The post request's content length.The length is between 0 and 1024. 0 mean data write end. When data length reach <data_len>, it's auto exit inputing. After this,must use $HTTPSEND to send data every time

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13.6 AT$HTTPSEND - Send HTTP post content data

This command is used to send HTTP post content data

13.6.1 Format

Command	Response
AT$HTTPSEND	This command effective only to POST. After data send complete, will receive response. OK
Max response time	25 s

13.7 AT$HTTPDATAEX - Set HTTP post request's data

This command is used to set HTTP post request's data.

13.7.1 Format

Command	Response
AT$HTTPDATAEX=<data_len>,<data>	OK Or ERROR
AT$HTTPDATAEX=?	$HTTPDATAEX:(0-500) ,"" OK
AT$HTTPDATAEX?	$HTTPDATAEX:<data_len> OK
Max response time	25 s

13.7.2 Field

Parameters	Description
<data_len>	The post request's content length.The length is between 0 and 500. Before thismust be set "Content-Length" request head, and must be the same as the value.

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<data>	The post request's content.The length must be same as the <data_len>.

13.8 AT$HTTPRQH - Set HTTP header fields

The command is used to set HTTP request header fields and entity header fields.

13.8.1 Format

Command	Response
AT$HTTPRQH=?	$HTTPRQH:"","" OK
AT$HTTPRQH?	Return current HTTP request header fields and entity header fields.
AT$HTTPRQH=<ParamKey>,<ParamValue>	The common request header: "Host" : The server's host.Must be matched with URL,If not set,will get from URL. "Content-Length" : The content length which will be send.This only for POST. Refer to : " IETP-RFC 2616 "
Max response time	25 s

13.8.2 Field

Parameters	Description
<ParamKey>	HTTP request or entity header field's Key. If there are special characters, please add quotes.The max length is 50. Support parameter: "accept","accept-charset","accept-encoding","accept-language", "authorization","expect","from","host","if-match","if-modified-since", "if-none-match","if-range","if-unmodified-since","max-forwards", "proxy authorization","range","referer","te","user-agent","allow", "content-encoding","content-language","content-length", "content-location","content-md5","content-range","content-type", "expires","last-modified","connection"
<ParamValue>	HTTP request or entity header field's Value. If there are special characters, please add quotes. The max length is 255.

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13.9 AT$HTTPTYPE - Set HTTP receive content data save location

Set HTTP recv content data save in local file or output to TE.

13.9.1 Format

Command	Response
AT$HTTPTYPE=<type>	OK
AT$HTTPTYPE=?	$HTTPTYPE:(0-1) OK
AT$HTTPTYPE?	$HTTPTYPE:<type> OK
Max response time	5 s

13.9.2 Field

Parameters	Description
<type>	HTTP recv data set save the way,available data include 0,1 0: data output to TE. 1: data save to FS. Note:set value after open HTTP service,and set value before AT$HTTPACTION.Its default value is 0.

13.10 AT$HTTPREAD - Read content data from local file

This command is used to read file content from local file, but AT$HTTPTYPE=1 effective.

13.10.1 Format

Command	Response
AT$HTTPREAD=?	$HTTPREAD:(0-1),(0-512000) OK

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AT$HTTPREAD=<read_type>[,<offset>]	$HTTPREAD:<offset>,<data_size> <data> OK Or $HTTPREAD:<read_type>,<file_size> OK Or ERROR Or $HTTPERROR:<errno> ERROR
AT$HTTPREAD?	$HTTPREAD:<read_type>,<offset> OK Note: when <read_type>=1, <offset> value is 0.
Max response time	25 s

13.10.2 Field

Parameters	Description
<read_type>	0 Read File 1 File size
<offset>	Start read file position, The range is from 0 to 512000, its value should less than file size. Note: when <read_type>=1, set<offset>value will return error. When <read_type>=0, <offset> default value is 0.
<data_size>	Every time the length of the read from local file.Its value is 1024 byte.

13.11 HTTP error codes

Parameters	Description

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<err_code>	Integer type 0 Operation successful 1 Parameter error 2 Input timeout 3 URL is not set 4 Operation not support 5 No GET/POST requests 6 Request processing 7 Filename too long 8 File not found 9 Memory not enough 10 Ssl config error 11 URL parse error 12 DNS resolution failed 13 Protocol error 14 Socket error 15 PDP context bind failed 16 Socket connect timeout 17 Socket connect error 18 Socket connect close 19 Ssl error 20 Request timeout 21 Internal error 22 URL format error 23 Ssl is not supported 100 output complete 300 not init http

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14 FTP(Y7S/Y7SW/Y6 not support)

14.1 AT+CFTPPORT - Set FTP Server Port

This command is used to set ftp server port

14.1.1 Format

Command	Response
AT+CFTPPORT=?	+CFTPPORT: (list of supported <port>s) OK Or ERROR
AT+CFTPPORT=<port>	OK or ERROR
AT+CFTPPORT?	+CFTPPORT: <port> OK
Max response time	300 ms

14.1.2 Field

Parameters	Description
<port>	The FTP server port, from 1 to 65535, and default value is 21.

14.2 AT+CFTPUN - Set User Name for FTP Access

This command is used to set ftp server username

14.2.1 Format

Command	Response
AT+CFTPUN=?	OK

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AT+CFTPUN="<name>"	OK Or ERROR
AT+CFTPUN?	+CFTPUN: "<name>" OK
Max response time	300 ms

14.2.2 Field

Parameters	Description
<name>	The user name for FTP server access. The maximum length is 30

14.3 AT+CFTPPW - Set User Password for FTP Access

This command is used to Set User Password for FTP Access

14.3.1 Format

Command	Response
AT+CFTPPW=?	OK
AT+CFTPPW="<password>"	OK Or ERROR
AT+CFTPPW?	+CFTPPW: "<password>" OK
Max response time	300 ms

14.3.2 Field

Parameters	Description
<password>	The user password for FTP server access. The maximum length is 40.

14.4 AT+CFTPTLS - Set FTP Security Mode

This command is used to set FTP Security Mode

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14.4.1 Format

Command	Response
AT+CFTPTLS=?	+CFTPTLS: (0-2),(0-1) OK
AT+CFTPTLS=<mode>[,<cert>]	OK or ERROR
AT+CFTPTLS?	+CFTPTLS: <mode>,<cert> OK
Max response time	300 ms

14.4.2 Field

Parameters	Description
<mode>	Default is 0.When mode is non 0, FTP will over TLS use SSL. 0 - None 1 - Implict 2 - Explicit
<cert>	Whether to ignore a certificate,Default is 0 0 - Ignore 1 - Don't ignore

14.5 AT+CFTPTYPE - Set FTP Type

This command is used to set FTP type. Default is binary

14.5.1 Format

Command	Response
AT+CFTPTYPE=?	+CFTPTYPE: (list of supported <type>s) OK
AT+CFTPTYPE=<type>	OK Or ERROR
AT+CFTPTYPE?	+CFTPTYPE: <type> OK

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Max response time	300 ms

14.5.2 Field

Parameters	Description
<type>	I – binary type. A – ASCII type

14.6 AT+CFTPCONMODE-Toggle the connection FTP mode

This command is used to switch modes connected to the FTP server. Default passive mode

14.6.1 Format

Command	Response
AT+CFTPCONMODE=?	+CFTPCONMODE:(0-1) OK
AT+CFTPCONMODE="<mode>"	OK Or ERROR
AT+CFTPCONMODE?	+CFTPCONMODE: <mode> OK
Max response time	25 s

14.6.2 Field

Parameters	Description
<mode>	0: Passive mode 1: Active mode

14.7 AT+CFTPGETFILE - Get a file from FTP server to file

This command is used to get a file from FTP server to file

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14.7.1 Format

Command	Response
AT+CFTPGETFILE=?	+CFTPGETFILE: ,,(0-2147483647) OK
AT+CFTPGETFILE="<remote_path>","<local_path>"[,<rest_size>]	OK +CFTPGETFILE: SUCCESS,<length> Or +CFTPGETFILE: FAIL,<err> ERROR
AT+CFTPGETFILE?	+CFTPGETFILE:“<remote_path>”,“<local_path>” ,<rest_size> OK
Max response time	25 s

14.7.2 Field

Parameters	Description
<remote_path>	The remote file path. The maximum length is 252. If the directory contains non-ASCII characters, the <remote_path>parameter should contain a prefix of {non-ascii}.
<local_path>	The file path. The maximum length is 62. Local file name string notsopport non-ascii and cannot contain: / \ : , * ? ">< |
<rest_size>	The value for FTP “REST” command which is used for broken transfer when transferring failed last time. The range is from 0 to 2147483647.
<length>	The size of the download file

14.8 AT+CFTPPUTFILE - Upload a file to FTP Server

This command is used to upload a file to FTP Server

14.8.1 Format

Command	Response
AT+CFTPPUTFILE=?	+CFTPPUTFILE: ,,(0-2147483647) OK

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AT+CFTPPUTFILE="<remote_path>","<local_path>"[,<rest_size>]	OK +CFTPPUTFILE: SUCCESS Or +CFTPPUTFILE: FAIL,<err> ERROR
AT+CFTPPUTFILE?	+CFTPGETFILE:"<remote_path>","<local_path>",<rest_size> OK
Max response time	25 s

14.8.2 Field

Parameters	Description
<remote_path>	The remote file path. The maximum length is 252. If the directory contains non-ASCII characters, the <remote_path>parameter should contain a prefix of {non-ascii}.
<local_path>	The file path. The maximum length is 62. Local file name string not support non-ascii and cannot contain: / \ : , * ? ">< |
<rest_size>	The value for FTP “REST” command which is used for broken transfer when transferring failed last time. The range is from 0 to 2147483647.

14.9 AT+CFTPLIST - List the items in the directory on FTP server

This command is used to list the items in the directory on FTP server

14.9.1 Format

Command	Response
AT+CFTPLIST=?	OK
AT+CFTPLIST=<dir>	OK +CFTPLIST: … +CFTPLIST: SUCCESS Or CFTPLIST:FAIL,<err> ERROR

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AT+CFTPLIST?	+CFTPGETFILE:"<dir>" OK
Max response time	25 s

14.9.2 Field

Parameters	Description
<dir>	The directory to be listed, The maximum length is 128. If the directory contains non-ASCII characters, the <remote_path>parameter should contain a prefix of {non-ascii}

14.10 AT+CFTPGET - Get a file from FTP server to UART

This command is used to get a file from FTP server to UART.

14.10.1 Format

Command	Response
AT+CFTPGET=?	+CFTPGET: ,(0-2147483647) OK
AT+CFTPGET="<remote_path>"[,<rest_size>]	OK +CFTPGET: DATA,<len> … +CFTPGET: DATA, <len> … +CFTPGET: SUCCESS, <length> Or +CFTPGET:FAIL,<err> ERROR
AT+CFTPGET?	+CFTPGET:"<remote_path>", <rest_size> OK
Max response time	25 s

14.10.2 Field

Parameters	Description

<remote_path>	The remote file path. The maximum length is 255. If the directory contains non-ASCII characters, the <remote_path > parameter should contain a prefix of {non-ascii}.
<rest_size>	The value for FTP “REST” command which is used for broken transfer when transferring failed last time. The range is from 0 to 2147483647.
<len>	Every time the length of the read from the server
<length>	The size of the download file

14.11 AT+CFTPMKD - Create a new directory on FTP server

This command is used to create a new directory on FTP server

14.11.1 Format

Command	Response
AT+CFTPMKD=?	OK
AT+CFTPMKD="<dir>"	OK Or +CFTPMKD:FAIL,<err> ERROR
AT+CFTPMKD?	+CFTPMKD: “<dir>” OK
Max response time	5 s

14.11.2 Field

Parameters	Description
<dir>	The directory to be created, The maximum length is 128. If the directory contains non-ASCII characters, the <remote_path> parameter should contain a prefix of {non-ascii}.

14.12 AT+CFTPRMD - Delete a directory on FTP server

This command is used to delete a directory on FTP server

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14.12.1 Format

Command	Response
AT+CFTPRMD=?	OK
AT+CFTPRMD ="<dir>"	OK +CFTPRMD:FAIL,<err> Or ERROR
AT+CFTPRMD?	+CFTPRMD:"<dir>" OK
Max response time	5 s

14.12.2 Field

Parameters	Description
<dir>	The directory to be deleted, The maximum length is 128. If the directory contains non-ASCII characters, the <remote_path> parameter should contain a prefix of {non-ascii}.

14.13 AT+CFTPDELE - Delete a file on FTP server

This command is used to delete a file on FTP server

14.13.1 Format

Command	Response
AT+CFTPDELE=?	OK
AT+CFTPDELE="<filename>"	OK +CFTPDELE: SUCCESS Or +CFTPDELE:FAIL,<err> ERROR
AT+CFTPDELE?	+CFTPDELE:"<filename>" OK
Max response time	5 s

14.13.2 Field

Parameters	Description
<filename>	The file to be deleted, The maximum length is 128. If the directory contains non-ASCII characters, the <remote_path> parameter should contain a prefix of {non-ascii}.

14.14 AT+CFTPSERV - Set FTP server domain name or IP address

This command is used to set FTP server domain name or IP address.

14.14.1 Format

Command	Response
AT+CFTPSERV=?	OK
AT+CFTPSERV="<address>"	OK Or ERROR
AT+CFTPSERV?	+CFTPSERV: "<address>" OK
Max response time	300 ms

14.14.2 Field

Parameters	Description
<address>	The FTP server domain name or IP address. The maximum length is 100.

14.15 AT+CFTPRDFILE - Read file from local file to UART

This command is used to read file from local file to UART.

14.15.1 Format

Command	Response

AT+CFTPRDFILE=?	+CFTPRDFILE:,(0-2147483647),(1-2147483647) OK
AT+CFTPRDFILE="<local_file>",<read_pos>,<read_len>	OK +CFTPRDFILE:DATA,<len> +CFTPRDFILE:SUCCESS,<length> Or +CFTPRDFILE: FAIL,<err_code>
AT+CFTPRDFILE?	+CFTPRDFILE: <local_file>,<read_pos>,<read_len> OK
Max response time	5 s

14.15.2 Field

Parameters	Description
<local_file>	The local file name. The maximum length is 62.
<read_pos>	Start read file position, The range is from 0 to 2147483647.
<read_len>	Read file length, The range is from 1 to 2147483647.
<len>	Every time the length of the read from the server
<length>	The size of the download file

14.16 AT+CFTPPUT - Upload the data from UART to FTP server

This command is used to upload the DATA from serial port to FTP server as a file. Single <Ctrl+Z> means end of the FTP data.

14.16.1 Format

Command	Response
AT+CFTPPUT=?	+CFTPGET: ,(0-2147483647),(1-1500) OK
AT+CFTPPUT="<remote_path>"[,<rest_size>][,<put_len>]	OK >> +CFTPPUT:SUCCESS Or +CFTPPUT: FAIL,<err_code>
AT+CFTPPUT?	+CFTPPUT:"<remote_path>",<rest_size>,<put_len> OK
Max response time	25 s

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14.16.2 Field

Parameters	Description
<remote_path>	The remote file path. The maximum length is 252. If the directory contains non-ASCII characters, the <emote_path> parameter should contain a prefix of {non-ascii}.
<rest_size>	The value for FTP “REST” command which is used for broken transfer when transferring failed last time. The range is from 0 to 2147483647.
<put_len>	The length is 1~1500.In this mode, all characters can be transferred. Must enter the specified data length before exiting the mode. In this mode <rest_size> cannot be omitted.

14.17 AT+CFTPSIZE - Get file size from FTP server

14.17.1 Format

Command	Response
AT+CFTPSIZE=?	+CFTPSIZE:<remote_path> OK
AT+CFTPSIZE=<remote_path>	OK +CFTPSIZE: <file_size> Or +CFTPSIZE: FAIL,<err_code>
Max response time	5 s

14.17.2 Field

Parameters	Description
<remote_path>	The remote file path. The maximum length is 252. If the directory contains non-ASCII characters, the <emote_path> parameter should contain a prefix of {non-ascii}.
<file_size>	The size of the file.

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14.18 FTP error codes

Parameters	Description
<err_code>	1 Parameter error 2 Operation not support 201 Unknown error for FTP 202 FTP task is busy 203 Failed to resolve server address 204 FTP timeout 205 Failed to read file 206 Failed to write file 207 Not allowed in current state 208 Failed to login 209 Failed to logout 210 Failed to transfer data 211 FTP command rejected by server 212 Memory error 213 Invalid parameter 214 Network error 215 Failed to connect socket 216 Failed to send data using socket 217 Failed to receive data using socket 219 Socket connect timeout 220 File does not exist 221 file has exist

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15 LYNQ TCP/IP(Y7S/Y7P/Y7SW/Y6 not support)

15.1 AT+CIPOPEN - Establish Connection in Multi-socket Mode

This command is used to establish a connection with TCP server and UDP server. The sum of all connections is 6.

15.1.1 Format

Command	Response
AT+CIPOPEN=?	+CIPOPEN: (list of supported <link_num>s), (list of supported <type>s) OK
AT+CIPOPEN?	+CIPOPEN: <link_num>[,<type>,<serverIP>,<serverPort>,<index>] OK Or ERROR
AT+CIPOPEN=<link_num>,<type>,<serverIP>,<serverPort>[,<localPort>]	OK +CIPOPEN: <err>,<link_num> Or ERROR
Max response time	150 s

15.1.2 Field

Parameters	Description
<link_num>	Identifies a connection, from 0 to 5, and default value is 0. If AT+CIPMODE=<mode> is set, the <link_num> is restricted to be only 0, +++ is exit send and receive mode.

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<type>	Identifies the type of transmission protocol. TCP: Transfer Control Protocol UDP: User Datagram Protocol
<serverIP>	Identifies the IP address of server. If type is UDP serverIP set to empty, except AT+CIPMODE=1.
<serverPort>	Identifies the port ofserver, from 0 to 65535, and default value is 0. If type is UDP serverPort set to empty, except AT+CIPMODE=1
<localPort>	Identifies the port of local socket, from 0 to 65535, and default value is 0.
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure if AT+CIPMODE=1: >: connect establish and can be send or receive data. OK: exit online data send and receive mode.

15.2 AT+CIPSEND - Send data through TCP or UDP Connection

This command is used to send data through TCP or UDP connection on non_transparent mode.

15.2.1 Format

Command	**Response**
AT+CIPSEND=?	+CIPSEND: (list of supported <link_num>s), (list of supported <length>s) OK
AT+CIPSEND?	OK Or ERROR

AT+CIPSEND=<link_num>,[<length>] AT+CIPSEND=<link_num>,,,,<data> AT+CIPSEND=<link_num>,[<length>],<serverIP>,<serverPort> (This format is for UDP connect)	OK +CIPSEND: <err>,<link_num>,<reqSendLength>,<cnfSendLength> ERROR This command is used to send data to remote side on non_transparent mode. Single <Ctrl+Z>is start send. Single <ESC> is used to cancel the sending. Single<Ctrl+D>means exit the sending mode. <Ctrl+Z> is 0x1A, <ESC> is 0x1B, <Ctrl+D> is x04. These signals take effect when no length is specified Note 1. In transparent mode <length> will be ignore or the <length> is empty, when input data length reach to 2500 will trigger send out. 2. Small data can be sent as follows: TCP type: AT+CIPSEND=<link_num>,,,,<data> UDP type: AT+CIPSEND=<link_num>,,<serverIP>,<serverPort>,<data>
Max response time	30 s

15.2.2 Field

Parameters	Description
<link_num>	Identifies a connection, from 0 to 5, and default value is 0.
<length>	Indicates the length of sending data, from 1 to 2500, and default value is 0. TCP: Transfer Control Protocol UDP: User Datagram Protocol Identifies the IP address of server.The IP address format consists of 4 octets,separated by decimal point: "AAA.BBB.CCC.DDD"
<serverIP>	Identifies the IP address of server.
<serverPort>	Identifies the port ofserver, from 0 to 65535, and default value is 0.
<data>	Identifies the contents for sending, and The maximum length is 512.
<reqSendLength>	A numeric parameter that requested number of data bytes to be transmitted.

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<cnfSendLength>	A numeric parameter that confirmed number of data bytes to be transmitted. PS: When set to hexadecimal by the MCIPCFGPL, <cnfSendLength>=<reqSendLength>/2.
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure

15.3 AT+CIPRXGET- Get the network data manually

This command is used to get the network data manually.

15.3.1 Format

Command	Response
AT+CIPRXGET=?	+CIPRXGET: (list of supported <mode>s), (list of supported <cid>s),(list of supported <len>s) OK
AT+CIPRXGET?	+CIPRXGET:<link0_mode>...,<link5_mode> OK Or ERROR
AT+CIPRXGET=<mode>,<cid>[,<len>]	<mode> = 2: +CIPRXGET:SUCCESS,<mode>,<cid>,<read_len>,<remaining data len>, <data> OK Others: OK +CIPRXGET:SUCCESS,<mode>,<cid>[,<read_len>,<data>] ERROR
Max response time	5 s

15.3.2 Field

Parameters	Description

<mode>	Indicate how to get the network data, from 0 to 4, and default value is 0. 0: set the way to get the network data automatically. 1: set the way to get the network data manually. 2: read data, the max read length is 1500. 3: read data in HEX form, the max read length is 750. 4: get the rest data length.
<cid>	Identifies a connection, from 0 to 5, and default value is 0.
<len>	The data length to be read.
<read_len>	The length of the data that have read.
<remaining datalen>	The length of remaining data.
<data>	The read data.
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure

15.4 AT+CIPCLOSE - Close TCP or UDP socket connection

This command is used to close TCP or UDP socket.

15.4.1 Format

Command	Response
AT+CIPCLOSE=?	+CIPCLOSE: (list of supported <link_num>s) OK
AT+CIPCLOSE=<link_num>	OK +CIPCLOSE: <err>,<link_num> Or ERROR
Max response time	25 s

15.4.2 Field

Parameters	Description
<link_num>	Identifies a connection, from 0 to 5, and default value is 0.

<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure

15.5 AT+CIPMODE - Select TCP/IP Application Mode

This command is used to select transparent mode (data mode) or non-transparent mode (command mode) before network open.

15.5.1 Format

Command	Response
AT+CIPMODE=?	+CIPMODE: (list of supported <mode>s) OK
AT+CIPMODE?	+CIPMODE: <mode> OK
AT+CIPMODE=<mode>	OK Or ERROR
Max response time	5 s

15.5.2 Field

Parameters	Description
<mode>	Indicates to select transparent mode or non-transparent mode from 0 to 1, and default value is 0. 0: non-transparent mode 1: transparent mode(similar to the data entry pattern) Note: in transparent mode, it is recommended to limit the rate of sending data to more than 1 second. Suggest that the maximum data size is 1024 bytes at a time

15.6 AT+CIPOPQUERY - Inquire the specific link connect status

This command is used to inquire the specific link connect status.

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15.6.1 Format

Command	Response
AT+CIPOPQUERY=?	+CIPOPQUERY: (list of supported <link_num>s) OK
AT+CIPOPQUERY?	+CIPOPQUERY: < link0_state>,…,< link5_state> OK
AT+CIPOPQUERY=<link_num>,[<link_num>,...]	+CIPOPQUERY: <link_num_state>[,…,<link_n_state>] OK Or ERROR
Max response time	5 s

15.6.2 Field

Parameters	Description
<link_num>	Identifies a connection. The range of permitted values is 0 to 5.
<state>	0:disconnected 1:connected

15.7 AT+SERVERSTART - Startup TCP Server

This command starts up TCP server, and the server can receive the request of TCP client. After the command executes successfully, an unsolicited result code is returned when a client tries to connect with module and module accepts request. The unsolicited result code is +CLIENT: < link_num >,<client_IP>:<port>.

15.7.1 Format

Command	Response
AT+SERVERSTART=?	+SERVERSTART: (list of supported <port>),(list of supported < backlog >, list of supported <iptype>) OK

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AT+SERVERSTART?	+ SERVERSTART: <port>,<backlog>,<iptype> OK
AT+SERVERSTART= < port>, < backlog>, <iptype>	OK +SERVERSTART: <err> Or ERROR
Max response time	5 s

15.7.2 Field

Parameters	Description
<port>	The server port, from 1 to 65535.
<backlog>	The maximum connections can be queued in listen queue, from 1 to 6.
<iptype>	0:IPV4 1:IPV6
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure

15.8 AT+SERVERSTOP - Stop TCP Server

This command stops TCP server.

15.8.1 Format

Command	Response
AT+SERVERSTOP=?	OK
AT+SERVERSTOP	OK +SERVERSTOP:<err> Or ERROR
Max response time	5 s

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15.8.2 Field

Parameters	Description
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure

15.9 AT+IPADDR - Inquire Socket PDP Address

This command returns a list of PDP addresses for the specified context identifiers. If no <cid> is specified, the addresses for all defined contexts are returned.

15.9.1 Format

Command	Response
AT+IPADDR?	OK OR ERROR
AT+IPADDR	+IPADDR:<err>,<ip_address> OK OR ERROR
Max response time	5 s

15.9.2 Field

Parameters	Description
<err>	Indicate the result of operation. SUCCESS: is success FAIL: is failure
<ip_address>	This command inquires the IP address of current active socket PDP

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15.10 +CIPRECV - Received data on a socket

This is an unsolicited message to show data has been received on a socket.

15.10.1 Format

Command	Response
	+CIPRECV: <mode>,<link_num>[,<length>]

15.10.2 Field

Parameters	Description
<mode>	Indicate how to get the network data, from 0 to 4, and default value is 0. 0: set the way to get the network data automatically. 1: set the way to get the network data manually. 2: read data, the max read length is 1500. 3: read data in HEX form, the max read length is 750. 4: get the rest data length.
<link_num>	Identifies a connection, from 0 to 5, and default value is 0.
<length>	Integer type Received data length(bytes)

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16 CTWing command(Y7S/Y7SW/Y6 not support)

16.1 AT+CTWPARAM - Initialize module information

16.1.1 Format

Command	Response
AT+CTWPARAM?	+CTWPARAM: <softVersion>,<module>,<chipType> OK Or ERROR
AT+CTWPARAM=<softVersion>,<module>,<chipType>	OK Or ERROR
Max response time	300 ms

16.1.2 Field

Parameters	Description
<softVersion>	String type Software version.
<module>	String type Module model.
<chipType>	String type Chip type.

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16.2 AT+CTWMQTTCFG - Initialize MQTT config

16.2.1 Format

Command	Response
AT+CTWMQTTCFG?	+CTWMQTTCFG: <autoRegEnableFlag>,<uri>,<port>,<keepAlive>,<mqttsFlag>,<autoRegStatus> OK Or ERROR
AT+CTWMQTTCFG=<autoRegEnableFlag>,[<uri>],[<port>],[<keepAlive>],<mqttsFlag>,[<autoRegStatus>]	OK Or ERROR
Max response time	300 ms

16.2.2 Field

Parameters	Description
<autoRegEnableFlag>	Integer type 0 Disable 1 Enable
<uri>	String type Platform server IP,max len is 128,default value is 0 ”mqtt.ctwing.cn”.
<port>	Integer type The port of the server. The range is 0-65535,default value is 1883.
<keepAlive>	Integer type The Keep Alive is a time interval measured in seconds. The range is 3-300,default value is 300.
<mqttsFlag>	Integer type 0 close SSL support 1 open SSL support
<autoRegStatus>	Integer type 0 not regist 1 have regist

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16.3 AT+CTWMQTTREGPARAM - Initialize the device parameters

16.3.1 Format

Command	Response
AT+CTWMQTTREGPARAM?	+CTWMQTTREGPARAM: <clientId>,<userName>,<password> OK Or ERROR
AT+CTWMQTTREGPARAM=<clientId>,<userName>,<password>	OK Or ERROR
Max response time	300 ms

16.3.2 Field

Parameters	Description
<clientId>	String type The device ID provided when the platform adds the device.Max len is 64.
<userName>	String type This parameter is used to login server. The maximum length of 64.
<password>	String type Feature strings.You can get this value from the platform device details. The maximum length of 72.

16.4 AT+CTWMQTTREG - Register platform

16.4.1 Format

Command	Response
AT+CTWMQTTREG?	+CTWMQTTREG: <RegStatus> OK Or ERROR

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AT+CTWMQTTREG	OK Or ERROR
Max response time	300 ms

16.4.2 Field

Parameters	Description
<RegStatus>	The value indicates the reg status.

16.5 AT+CTWMQTTDEREG - Deregister from platform

16.5.1 Format

Command	Response
AT+CTWMQTTDEREG	OK Or ERROR
Max response time	300 ms

16.6 AT+CTWMQTTSEND - Send data to platform

16.6.1 Format

Command	Response
AT+CTWMQTTSEND=<qos>,<topic>,<payload>	OK Or ERROR
Max response time	300 ms

16.6.2 Field

Parameters	Description

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<qos>	Integer type.Quality of Service: 0 At most once delivery for Application message 1 At least once delivery for Application message 2 Exactly once delivery for Application message
<topic>	String type The topic of the Application message,max len is 128.
<payload>	String type The Application message content. The maximum length of 1024.

16.7 +CTW ERROR– Indicates an error

16.7.1 Format

Command	Response
	+CTW ERROR:<errCode>

16.7.2 Field

Parameters	Description
<errCode>	1 CTW_PARAM_ERROR 2 CTW_NOT_SUPPORT 3 CTW_HTTP_CONNECT_FAIL 4 CTW_REG_FAIL 5 CTW_NO_SET_PARAM 6 CTW_NO_REGISTER 7 CTW_SEND_FAIL 8 CTW_CLOSE_FAIL

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17 CTM2M Command(Y7S/Y7SW/Y6 not support)

17.1 AT+CTM2MVER – View CTM2M information

17.1.1 Format

Command	Response
AT+CTM2MVER?	+CTM2MVER:<lwm2m>,<ctm2m>,<ctmd>,<ctmt>,<ctmv> OK

17.1.2 Field

Parameters	Description
<lwm2m>	LwM2M version.
<ctm2m>	CTM2M version.
<ctmd>	Mode id.
<ctmt>	Chip type.
<ctmv>	Software version.

17.2 AT+CTM2MSETMOD - Set UE working mode

17.2.1 Format

Command	Response
AT+CTM2MSETMOD=<MOD_ID>,<MOD_DATA>	OK Or +CTM2M ERROR:<err>

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AT+CTM2MSETMOD?	+CTM2MSETMOD:<idAuth_Mode>,<auto_Update>,<on_UQMode>,<auto_Heartbeat>,<wakeup_Policy>,<protocol_Mode>,<send_Type>,<receive_data_length> OK
AT+CTM2MSETMOD=?	+CTM2MSETMOD:(list of supported <MOD_ID>s),(list of supported <MOD_DATA>s) OK

17.2.2 Field

Parameters	Description
<MOD_ID>	Integer type 1--- IDAuth_Mode 2--- NAT_TYPE 3--- ON_UQMode 4--- Auto_Heartbeat 5--- Wakeup_Notify 6--- Protocol_Mode 7--- Send_Type 8--- Receive_data_length

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<MOD_DATA>	Integer type If MOD_ID=1: 0: URN IMEI 1: default IMEI value, no authentication string 2: Auth mode1, urn:imei+<Auth token str> 3: Auth mode2, according to AT+CTLWSETAUTH settings to use urn:imei-simid or urn:imei-sm9 6: IMEI and IMSI string 7: IMSI string If MOD_ID=2: 0: DISABLE_NAT 1: ENABLE_NAT If MOD_ID=3: 1: UQ mode off 2: UQ mode on If MOD_ID=4: 1: default auto heartbeat 2: no auto heartbeat If MOD_ID=5: 1: default notify to MCU 2: not notify to MCU If MOD_ID=6: 1: normal 2: enhance If MOD_ID=7: 0: hex type 1: string type If MOD_ID=8: 0: no length 1: display data length

17.3 AT+CTM2MSETPM - Set LwM2M parameters to register

17.3.1 Format

Command	Response
AT+CTM2MSETPM?	+CTM2MSETPM:<Sever_IP>,<Port>,<Lifetime> OK
AT+CTM2MSETPM=<Sever_IP>,<Port>,<Lifetime>	OK Or +CTM2M ERROR:<err>

AT+CTM2MSETPM=?	+CTM2MSETPM:(list of supported <Sever_IP>),(range of supported <Port>),(range of supported <Lifetime>),(range of supported <bsMode>) OK

17.3.2 Field

Parameters	Description
<Sever_IP>	String, LwM2M platform server IP.
<Port>	Integer, LwM2M platform server port (1-65535).
<Lifetime>	Integer, LwM2M Client to the LwM2M platform register update interval (s), must be more than 300 seconds, max value is 2147483647.
<bsMode>	Integer, bootstrap mode (0-1).

17.4 AT+CTM2MIDAUTHPM - Set LwM2M auth string

17.4.1 Format

Command	Response
AT+CTM2MIDAUTHPM=<IDAuth_STR>	OK Or +CTM2M ERROR:<err>
AT+CTM2MIDAUTHPM=?	+CTM2MIDAUTHPM: <IDAuth_STR> OK

17.4.2 Field

Parameters	Description
<IDAuth_STR>	String, LwM2M Auth token string, max length is 128.

17.5 AT+CTM2MSETPSK - Set LwM2M PSK parameter

17.5.1 Format

Command	Response

AT+CTM2MSETPSK?	+CTM2MSETPSK:<Security_Mode> OK
AT+CTM2MSETPSK=<Security_Mode>,<PSKID>,<PSK>	OK Or +CTM2M ERROR:<err>
AT+CTM2MSETPSK=?	+CTM2MSETPSK: (1,2),"<PSKID>","<PSK>"] OK

17.5.2 Field

Parameters	Description
<Security_Mode>	Integer, Security mode. 1 No PSKID and PSK 2 Use PSKID and PSK
<PSKID>	String, LwM2M PSK ID, max length is 32.
<PSK>	String, LwM2M PSK, hex format, max length is 32.

17.6 AT+CTM2MREG - Register platform

17.6.1 Format

Command	Response
AT+CTM2MREG?	+CTM2MREG:<Status> OK
AT+CTM2MREG	OK Or +CTM2M ERROR:<err>

17.6.2 Field

Parameters	Description

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<Status>	Integer, indicate login status: 0 Not logined 1 Logined 2 Logining 3 Login failed 4 No ip 5 Log out

17.7 AT+CTM2MUPDATE – Update binding mode

17.7.1 Format

Command	Response
AT+CTM2MUPDATE	OK Or +CTM2M ERROR:<err>

17.7.2 Field

Parameters	Description
<MsgID>	Message Id.

17.8 AT+CTM2MDEREG – Deregister from platform

17.8.1 Format

Command	Response
AT+CTM2MDEREG	OK Or +CTM2M ERROR:<err>

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17.9 AT+CTM2MSEND – Send data to platform

17.9.1 Format

Command	Response
AT+CTM2MSEND=<data>[,<mode>]	+CTM2MSEND:<MsgID> OK Or +CTM2M ERROR:<err>
AT+CTM2MSEND=?	+CTM2MSEND:(list of supported"<data>"), (list of supported <mode>) OK

17.9.2 Field

Parameters	Description
<data>	Hex format data, Length should be less than 1024. String format data, Length should be less than 512.
<mode>	Integer type 0---CON mode 1---NON mode 2---NON with RAI flag 3---CON with RAI flag
<MsgID>	Message Id.

17.10 AT+CTM2MRMODE – Set display mode of downlink data

This command set the display mode of the downlink data. When set to 1, the received downlink data will be displayed immediately, such as +CTM2MRECV: <data>. When set to 2, UE will cache data and display +CTM2MRECV, the data can be read by AT+CTM2MREAD. If there is cached data, UE will not enter Hibernate or Sleep2 mode. UE buffers up to 8 downlink data, after exceeding this limit, the oldest data will be discard first. 1 is default value, and restore to 1 when reset.

17.10.1 Format

Command	Response

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AT+CTM2MRMODE=<mode>	OK Or +CTM2M ERROR:<err>
AT+CTM2MRMODE?	+CTM2MRMODE: <mode> OK

17.10.2 Field

Parameters	Description
<mode>	Integer type 0---Received the downlink data, buffer it and no indication. 1---Received the downlink data, immediately display +CTM2MRECV: [<Length>,]<data>, 1 is default value 2---Received the downlink data, buffer it and display +CTM2MRECV

17.11 AT+CTM2MREAD – Read data from buffer

17.11.1 Format

Command	Response
AT+CTM2MREAD	<datalen>,<data> OK Or +CTM2M ERROR:<err>

17.11.2 Field

Parameters	Description
<datalen>	Cached downstream data length
<data>	Hex format data

17.12 +CTM2MRECV – Receive data from platform

17.12.1 Format

Command	Response

	+CTM2MRECV:[<Length>,]<Data>

17.12.2 Field

Parameters	Description
<Data>	Hex format data, max length is 512 bytes.

17.13 +CTM2M – Receive response from platform

17.13.1 Format

Command	Response
	+CTM2M:<Operation>,<status code>,[<MsgID>]

17.13.2 Field

Parameters	Description
<Operation>	Operation description: conn, reg, update, dereg, ping, send, obsrv, lwstatus.
<status code>	Integer, 0---success 1---timeout 2---not send out packet 9---receive platform RST packet and mean can't send UL to platform 10---parameter error 11---other errors 13---authentication error 14---UE not login 22---iot protocol or lwm2m version mismatch 24---lwm2m session invalid 25---session load failure when quitting from sleep or after reboot 26---Engine abnormal, need reboot by MCU 28---TAU is due 29---Module exit sleep mode 31---packet is already sent out 32---object 19 not exist
<MsgID>	Message Id.

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17.14 +CTM2MCMD – Notification message from platform

17.14.1 Format

Command	Response
	+CTM2MCMD:<msgid>,<cmdtype>,<token>,<uri_str>[,<observe>[,<dataformat>,<data>]]

17.14.2 Field

Parameters	Description
<msgid>	Integer type Message ID
<cmdtype>	Integer type 0---Read 1---Observe 2---Write 3---Write-Partial 4---Write-Attribute 5---Discover 6---Execute 7---Create 8---Delete
<token>	Hex String type Async message response ID
<uri_str>	String type /objectid/instanceid/resourceid
<observe>	Integer type It is must when cmdtype=1 0---Set 1---Cancel
<dataformat>	Integer type 1---TLV 6---CoAP Param 7---text/plain 8---JSON
<data>	String type It is must when cmdtype=2、 3、 4、 6、 7 It is null when execute cmd has no arguments

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17.15 +CTM2M ERROR – Indicates an error

17.15.1 Format

Command	Response
	+CTM2M ERROR:<errCode>

17.15.2 Field

Parameters	Description
<errCode>	1 Other error 3 Parameter error 5 Psk error 8 Parameter not initalized 15 Object not observed 17 Data length not even 33 Already login 303 Operation not supported 954 Not login 955 Login processing 957 Connect useless 958 No session 969 Lifetime less than 300 972 Uri error

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18 OneNET(Y7S/Y7SW/Y6 not support)

18.1 AT+MIPLCONFIG – Config some parameters

This command is used to configure some parameters.
Note: the command should be executed before the communication suite instance is created with AT+MIPLCREATE command.

18.1.1 Format

Command	Response
AT+MIPLCONFIG=?	+MIPLCONFIG: (0-7),<parameter1>[,<parameter2>] OK
AT+MIPLCONFIG=<Mode>,<parameter1>[,<parameter2>]	OK Or +CIS ERROR:<err> Note:parameters<Mode>, <parameter1>, <parameter2>correspond as follows <mode> <parameter1> <parameter2> 0 <IP> <port> 1 <IP> <port> 2 1 <rsp_timeout> 3 <obs_autoack> 4 <auth_enable> [<auth_code>] 5 <DTLS_enable> [<psk>] 6 <write_format> 7 <buf_cfg> <buf_URC_mode>

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AT+MIPLCONFIG?	+MIPLCONFIG: 0/1,"<ip>","<port>" +MIPLCONFIG: 2,<rsp_timeout> +MIPLCONFIG: 3,<obs_autoack> +MIPLCONFIG: 4[,<auth_enable>][,<auth_code>] +MIPLCONFIG: 5,<dtls_enable>[,<psk>] +MIPLCONFIG: 6,<write_format> +MIPLCONFIG: 7,<buf_cfg>,<buf_urc_mode> OK
Max response time	300 ms

18.1.2 Field

Parameters	Description
<Mode>	Config mode 0 Disable bootstrap mode and config server ip, port and endpoint type 1 Enable bootstrap mode and config bootstrap server ip, port and endpoint type 2 Config response timeout 3 Config observe auto send ack 4 Config authtication info 5 Config DTLS and psk 6 Config write format 7 Config buf cfg and urc mode
<ip>	Server IP
<port>	Server port
<endpoint_type>	Endpoint type 0 IMEI;IMSI (default) 1 IMEI
<rsp_timeout>	Response timeout (2-20) second
<obs_autoack>	Auto send observe ack packet 0 Disable (default) 1 Enable
<auth_enable>	Authertication 0 Disable (default) 1 Enable
<auth_code>	Authertication code, max length is 17 bytes
<dtls_enable>	DTLS 0 Disable (default) 1 Enable
<psk>	DTLS psk code, max length is 17 bytes

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<write_format>	Write format (0-1)
<buf_cfg>	Received data save to buffer config 0 None (default) 1 Write data 2 Execute data 3 Write and execute data
<buf_urc_mode>	Buffer URC mode 0 Not send URC 1 Send URC (default)

18.2 AT+MIPLCREATE - Create OneNET instance

18.2.1 Format

Command	Response
AT+MIPLCREATE	+MIPLCREATE: <ref> OK Or +CIS ERROR:<err>
Max response time	300 ms

18.2.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID.

18.3 AT+MIPLDELETE - Delete OneNET instance

18.3.1 Format

Command	Response
AT+MIPLDELETE=<ref>	OK Or +CIS ERROR:<err>
AT+MIPLDELETE=?	+MIPLDELETE: (list of supported <ref>) OK

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Max response time	5 s

18.3.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.

18.4 AT+MIPLOPEN - Register to OneNET

18.4.1 Format

Command	Response
AT+MIPLOPEN=<ref>,<lifetime>[,<timeout>]	OK Or +CIS ERROR:<err>
AT+MIPLOPEN=?	+MIPLOPEN:(list of supported <ref>),(range of supported <lifetime>),(range of supported <timeout>) OK
Max response time	10 s

18.4.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<lifetime>	Integer, lifetime to register ONENET server (Unit is second, the range is 15-268435455.).
<timeout>	Integer type Timeout of registration. Range: 30-65535. unit: second

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18.5 AT+ MIPLCLOSE - Deregister to OneNET

18.5.1 Format

Command	Response
AT+MIPLCLOSE=<ref>	OK Or +CIS ERROR:<err>
AT+MIPLCLOSE=?	+MIPLCLOSE: (list of supported <ref>) OK
Max response time	5 s

18.5.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.

18.6 AT+MIPLADDOBJ - Add an object

18.6.1 Format

Command	Response
AT+MIPLADDOBJ=<ref>,<objectid>,<instancecount>,<instancebitmap>,<attributecount>,<actioncount>	OK Or +CIS ERROR:<err>
AT+MIPLADDOBJ=?	+MIPLADDOBJ: (list of supported <ref>),(range of supported <objected>),(range of supported <instancecount>),"<instanceBitmap>",(range of supported <attributecount>),(range of supported <actioncount>) OK
Max response time	5 s

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18.6.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<objectid>	Integer, object id, range: 0-65535.
<instancecount>	Integer, instance count, range: 0-65535.
<instancebitmap>	Bitmap string, instance bitmap, ex: “00101” (5 instances, only instance 1 & 3 are available).
<attributecount>	Integer, attribute count, range: 0-65535.
<actioncount>	Integer, action count, range: 0-65535.

18.7 AT+MIPLDELOBJ - Delete an object

18.7.1 Format

Command	Response
AT+MIPLDELOBJ=<ref>,<objectid>	OK Or +CIS ERROR:<err>
AT+MIPLDELOBJ=?	+MIPLDELOBJ: (list of supported <ref>),(range of supported <objected>) OK
Max response time	5 s

18.7.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<objectid>	Integer, object id, range: 0-65535.

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18.8 AT+MIPLUPDATE - Update registration

18.8.1 Format

Command	Response
AT+MIPLUPDATE=<ref>,<lifetime>,<withObjectFlag>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLUPDATE=?	+MIPLUPDATE: (list of supported <ref>),(range of supported <lifetime>),(list of supported <withobjectflag>),(list of supported <raiflag>) OK
Max response time	5 s

18.8.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<lifetime>	Integer, lifetime to update registration (Unit is second, the range is 15-268435455.).
<withObjectFlag>	Integer, whether to update objects. 1 Update object info at the same time 0 Don't update object info
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received

18.9 +MIPLREAD - Read request to user

18.9.1 Format

Command	Response

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	+MIPLREAD:<ref>,<msgid>,<objectid>,<instanceid>,<resourceid>

18.9.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id.
<resourceid>	Integer, resource id, read all resources of the instance if resourceid equals -1.

18.10 AT+MIPLREADRSP - Read response from user

18.10.1 Format

Command	Response
AT+MIPLREADRSP=<ref>,<msgid>,<result>[,<objectid>,<instanceid>,<resourceid>,<valuetype>,<len>,<value>,<index>,<flag>[,<raiflag>]]	OK Or +CIS ERROR:<err>
AT+MIPLREADRSP=?	+MIPLREADRSP:(list of supported <ref>),(range of supported <msgid>),(range of supported <objecteid>),(range of supported <instanceid>),(range of supported <resourceid>),(range of supported <valuetype>),(range of supported <len>),"<value>",(list of supported <index>), (list of supported <flag>),(list of supported <raiflag>) OK
Max response time	5 s

18.10.2 Field

Parameters	Description

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<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to +MIPLREAD.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id.
<resourceid>	Integer, resource id, read all resources of the instance if resourceid equals -1.
<valuetype>	Integer, read data value type 1 string; 2 opaque; 3 integer; 4 float; 5 bool.
<len>	Integer, read data length, max length is 10. String string len Opaque byte len Integer The number of bytes occupied by the integer. Optional 2, 4, 8 Float The number of bytes occupied by the float. Optional 4, 8 Bool 1
<value>	String, read data value. valuetype value string string opaque Hex string interger interger float string bool 0:false; 1:true
<index>	Integer, serial number, for one command may not be able to send the entire content. It's from N~0. 0 is last part of content.
<flag>	Integer, message flag 1 first message; 2 middle message; 0 last message.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received
<result>	Integer type; Result of read command 1 205 content, read success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed 15 406 not acceptable

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18.11 AT+MIPLWRITERSP - Write response from user

18.11.1 Format

Command	Response
AT+MIPLWRITERSP=<ref>,<msgid>,<result>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLWRITERSP=?	+MIPLWRITERSP: (list of supported <ref>),(range of supported <msgid>),(list of supported <result>),(list of supported <raiflag>) OK
Max response time	5 s

18.11.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to + MIPLWRITE.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received
<result>	Integer type; Result of write command 2 204 changed, write success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed

18.12 +MIPLWRITE - Write request to user

18.12.1 Format

Command	Response

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	+MIPLWRITE:<ref>,<msgid>,<objectid>,<instanceid>,<resourceid>,<valuetype>,<len>,<value>,<flag>,<index>

18.12.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id.
<resourceid>	Integer, resource id.
<valuetype>	Integer, read data value type 1 string; 2 opaque; 3 integer; 4 float; 5 bool.
<len>	Integer, read data length.
<value>	String, read data value.
<index>	Integer type Serial number, for one command may not be able to send the entire content. It's from N~0. 0 is last part of content.
<flag>	Integer, message flag 1 first message; 2 middle message; 0 last message.

18.13 +MIPLEXECUTE - Execute request to user

18.13.1 Format

Command	Response
	+MIPLEXECUTE:<ref>,<msgid>,<objectid>,<instanceid>,<resourceid>,<len>,<arguments>

18.13.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.

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<objectid>	Integer, object id.
<instanceid>	Integer, instance id.
<resourceid>	Integer, resource id.
<len>	Integer, parameter length.
<arguments>	String, parameter string.

18.14 AT+MIPLEXECUTERSP - Execute response from user

18.14.1 Format

Command	Response
AT+MIPLEXECUTERSP=<ref>,<msgid>,<result>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLEXECUTERSP=?	+MIPLEXECUTERSP: (list of supported <ref>),(range of supported <msgid>),(list of supported <result>),(list of supported <raiflag>) OK
Max response time	5 s

18.14.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to +MIPLEXECUTE.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received
<result>	Integer type; Result of read command 2 204 changed, execute success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed

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18.15 +MIPLOBSERVE - Observe request to user

18.15.1 Format

Command	Response
	+MIPLOBSERVE:<ref>,<msgid>,<flag>,<objectid>,<instanceid>,<resourceid>

18.15.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<flag>	Integer, observe flag, 1 indicates observe, 0 indicates cancel observe.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id, observe all resources of all instances of the object if instanceid equals -1.
<resourceid>	Integer, resource id, observe all resources of the instance if resourceid equals -1.

18.16 AT+MIPLOBSERVERSP - Observe response from user

18.16.1 Format

Command	Response
AT+MIPLOBSERVERSP=<ref>,<msgid>,<result>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLOBSERVERSP=?	+MIPLOBSERVERSP: (list of supported <ref>),(range of supported <msgid>),(list of supported <result>),(range of supported <raiflag>)
Max response time	5 s

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18.16.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to +MIPLOBSERVE.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received
<result>	Integer 1 205 content 2 204 changed, observe success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed 15 406 not acceptable

18.17 +MIPLDISCOVER - Discover request to user

18.17.1 Format

Command	Response
	+MIPLDISCOVER: <ref>,<msgid>,<objectid>

18.17.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<objectid>	Integer, object id.

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18.18 AT+MIPLDISCOVERRSP - Discover response from user

18.18.1 Format

Command	Response
AT+MIPLDISCOVERRSP=<ref>,<msgid>,<result>,<length>,<valuestring>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLDISCOVERRSP=?	+MIPLDISCOVERRSP: (list of supported <ref>),(range of supported <msgid>),(list of supported <result>),(range of supported <length>),\"<valuestring>\", (list of supported <raiflag>) OK
Max response time	5 s

18.18.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to +MIPLDISCOVER.
<length>	Integer, length of value string.
<valuestring>	String, value string (attribute;attribute;…;attribute;action;…;action), must start with “ and end with “. Note: Attribute + action count in valuestring is the same to attribute count+ action count in AT+MIPLADDOBJ.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received

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<result>	Integer type; Result of read command 1 205 content 2 204 changed, execute success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed 15 406 not acceptable

18.19 AT+ MIPLPARAMETERRSP - Set parameter response from user

18.19.1 Format

Command	Response
AT+MIPLPARAMETERRSP=<ref>,<msgid>,<result>[,<raiflag>]	OK Or +CIS ERROR:<err>
AT+MIPLPARAMETERRSP=?	+MIPLPARAMETERRSP: (list of supported <ref>),(range of supported <msgid>),(list of supported <result>),(list of supported <raiflag>) OK
Max response time	5 s

18.19.2 Filed

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id, the same to +MIPLPARAMETER.
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received

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<result>	Integer type; Result of read command 2 204 changed, execute success 11 400 bad request 12 401 unauthorized 13 404 not found 14 405 method not allowed

18.20 +MIPLPARAMETER - Set parameter request to user

18.20.1 Format

Command	Response
	+MIPLPARAMETER:<ref>,<msgid>,<objectid>,<instanceid>,<resourceid>,<len>,<parameter>

18.20.2 Field

Parameters	Description
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id, observe all resources of all instances of the object if instance id equals -1.
<resourceid>	Integer, resource id, observe all resources of the instance if resource id equals -1.
<parameter>	String, parameter string, must start with “ and end with “. pmin=xxx; pmax=xxx; gt=xxx; lt=xxx; stp=xxx
<len>	Integer, parameter length.

18.21 AT+MIPLNOTIFY - Notify data value change from user

18.21.1 Format

Command	Response

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AT+MIPLNOTIFY=<ref>,<msgid>,<objectid>,<instanceid>,<resourceid>,<valuetype>,<len>,<value>,<index>,<flag>[,<ackid>[,<raiflag>]]	OK Or +CIS ERROR:<err>
AT+MIPLNOTIFY=?	+MIPLNOTIFY: (list of supported <ref>),(range of supported <msgid>),(range of supported <objected>),(range of supported <instanceid>),(range of supported <resourceid>),(range of supported <valuetype>),(range of supported <len>),"<value>",(list of supported <index>),(list of supported <flag>),(list of supported <ackid>),(list of supported <raiflag>) OK
Max response time	300 ms

18.21.2 Field

Parameters	**Description**
<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<msgid>	Integer, message id.
<objectid>	Integer, object id.
<instanceid>	Integer, instance id.
<resourceid>	Integer, resource id.
<valuetype>	Integer, write data value type 1 string; 2 opaque; 3 integer; 4 float; 5 bool
<len>	Integer type, max length is 1024. Valuetype len String string len Opaque byte len Integer The number of bytes occupied by the integer. Optional 2, 4, 8 Float The number of bytes occupied by the float. Optional 4, 8 Bool 1
<value>	string type valuetype value string string opaque Hex string interger interger float string bool 0:false; 1:true

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<index>	Serial number, for one command may not be able to send the entire content. It's from N~0. 0 is last part of content.
<flag>	Integer type; if there are multiple msg 1 First msg 2 Middle msg 0 Last msg
<raiflag>	Integer type 0 (PS_SOCK_RAI_NO_INFO) disable RAI 1 (PS_SOCK_RAI_NO_UL_DL_FOLLOWED) enable RAI release after send 2 (PS_SOCK_ONLY_DL_FOLLOWED) enable RAI release after ACK has received
<ackid>	Integer type 0 This notify will be sent in Non-confirmable(NON) message 1-65535 This notify will be sent in Confirmable(CON) message and the value will indicated by "+MIPLEVENT"

18.22 AT+MIPLVER - Read version

18.22.1 Format

Command	**Response**
AT+MIPLVER?	+MIPLVER:<version> OK
Max response time	300 ms

18.23 +MIPLEVENT - Event indication to user

18.23.1 Format

Command	**Response**
	+MIPLEVENT :<ref>,<evtid>[,<extend>,<ackid>]

18.23.2 Field

Parameters	**Description**

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<ref>	Integer, OneNET communication suite ID returned by AT+MIPLCREATE.
<evtid>	The identifier of event: 1 BOOTSTRAP_START 2 BOOTSTRAP_SUCCESS 3 BOOTSTRAP_FAILED 4 CONNECT_SUCCESS 5 CONNECT_FAILED 6 REG_SUCCESS 7 REG_FAILED 8 REG_TIMEOUT 9 LIFETIME_TIMEOUT 10 STATUS_HALT 11 UPDATE_SUCCESS 12 UPDATE_FAILED 13 UPDATE_TIMEOUT 14 UPDATE_NEED 15 UNREG_DONE 20 RESPONSE_FAILED 21 RESPONSE_SUCCESS 25 NOTIFY_FAILED 26 NOTIFY_SUCCESS 30 NOTIFY_SUCCESS_FROM_BS
<extend>	Extended parameter. When <evtid> is 20 or 25, the <extend> is message ID of the response command. When <evtid> is 14, the <extend> is remaining time of lifetime, unit is second.
<ackid>	Integer type When the eventid is Notify success(26), the corresponding ackid issued when notify is carried.

18.24 <err> Codes

18.24.1 Format

Command	**Response**
	+CIS ERROR:<err>

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18.24.2 Field

Parameters	Description	
<err>	100	unknown error
	601	parameter error
	602	status error
	651	not support
	652	sdk error
	653	no instance
	654	malloc fail
	655	network not ready
	656	configuration error

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19 FOTA

19.1 AT+FOTA - Download fota package and run upgrade process

This command is used to download fota package and start upgrade process.

19.1.1 Format

Command	Response
AT+FOTA=<channel>,<mode>,<destination_ip/url>	OK Or +HTTP ERROR:<err>
AT+FOTA?	$MYFOTA:<Status> OK
Max response time	5 s

19.1.2 Field

Parameters	Description
<channel>	Connect id, default is 1.
<mode>	1: http Note: This parameter is not currently used.
<destination ip/url>	Server ip or address or url Eg: http://118.114.239.159:30128/v09.01b02.00_TO_v09.01b02.00_TEST.par
<status>	Download progress: 0-99 download file and progress bar 100: download and save fota package success

20 LBS (only for L511C-Y)

20.1 AT+GTPOSTEST - Obtaining longitude and latitude test instructions (This instruction is only used to show the function to customers.)

20.1.1 Format

Command	Response
AT+GTPOSTEST	+GTPOS: <Longitude>,<Latitude> OK Or ERROR
Max response time	20 s

20.1.2 Field

Parameters	Description
<Longitude>	String type longitude
< Latitude >	String type Latitude

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20.2 AT+GTPOS - Obtaining longitude and latitude test instructions (This command requires the customer to provide the KEY value himself)

20.2.1 Format

Command	Response
AT+GTPOS=<mode>,<key>	Mode = 0: OK or ERROR Mode = 1: CONNECT OK or ERROR Mode = 2 +GTPOS: <Longitude>,<Latitude> OK Or ERROR
Max response time	20 s

20.2.2 Field

Parameters	Description
<Longitude>	String type longitude
< Latitude >	String type Latitude
<mode>	0: Turn off the base station positioning function 1: Turn on the base station positioning function 2: Acquiring LBS location information
<key>	The customer applies for the key value himself

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21 Application Examples

21.1 TCP Demo

ECRDY

AT+CEREG?
+CEREG: 0,1

OK

AT+CGPADDR
+CGPADDR: 1,"100.84.81.114" //PDP is active

OK

AT+SKTCREATE=1,1,6 //Created a IPV4 TCP socket
+SKTCREATE: 0

OK

AT+SKTCONNECT=0,58.246.1.50,63661 //Connected the TCP Server
OK

AT+SKTSEND=0,5,3132333435 //Send data by hex mode
OK

+SKTRECV: 0,3,"313233" //Received the data 123 from server
AT+SKTSTATUS=0 //Check the socket status
+SKTSTATUS: 4

OK

AT+SKTDELETE=0 //close the socket
OK

AT+SKTSTATUS=0
+SOCKET ERROR: NO FIND CLIENT

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21.2 SNTP Demo

ECRDY

AT+CEREG?
+CEREG: 0,1

OK

AT+CGPADDR
+CGPADDR: 1,"100.65.224.82"

OK

AT+CCLK="19/01/01,06:50:30+32" // setting of the clock
OK

AT+CCLK?
+CCLK: "2019/01/01,06:50:31+32"

OK

AT+ECSNTP=120.25.115.20,,1 // synchronize time
OK

+ECSNTP: 2021/11/01,07:51:58
AT+CCLK?
+CCLK: "2021/11/01,07:51:59+32"

OK

21.3 ECMQTT Demo

ECRDY

AT+CEREG?
+CEREG: 0,1

OK

AT+CGPADDR
+CGPADDR: 1,"100.97.13.189"

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OK

AT+CPSMS=0 //close psm
OK

AT+ECMTOPEN=0,58.246.1.50,63661 //create a mqtt connection
OK

+ECMTOPEN: 0,0
AT+ECMTCONN=0,0,"admin","password" //connected
OK

+ECMTCONN: 0,0,0
AT+ECMTSUB=0,519,"test",2 //sub a topic
OK

+ECMTSUB: 0,519,0,1
AT+ECMTPUB=0,520,2,1,"test","30313233343536373839" //pub data,qos=2
OK

+ECMTRECV: 0,32770,"test",30313233343536373839 //receive data

+ECMTPUB: 0,520,0
AT+ECMTPUB=0,520,1,1,"test","30313233343536373839" //pub data,qos=1
OK

+ECMTPUB: 0,520,0

+ECMTRECV: 0,32771,"test",30313233343536373839 //receive data

21.4 ECHTTP Demo

ECRDY

+CEREG: 0,1

+CGPADDR: 0,"10.219.41.171"

AT+CEREG?
+CEREG: 0,1

OK

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//Send a GET request to the HTTP (S) server
AT+HTTPCFG="responseheader",1 //Enable the output HTTP response header information
OK

AT+HTTPURL=37,80 //Set the URL to access and set a timeout of 80 seconds
CONNECT
http://leshan.eigencomm.com:8888/test //write URL
OK

AT+HTTPGET //Send a HTTP GET request
OK

AT+HTTPREAD //The HTTP response information is output via the UART port
OK

CONNECT

+HTTPGET: 0,200,19
HTTP/1.1 200 OK
Date: Wed, 16 Aug 2023 05:21:03 GMT
Server: Apache/2.4.29 (Ubuntu)
Last-Modified: Thu, 17 Dec 2020 14:45:48 GMT
ETag: "13-5b6aa0c257130"
Accept-Ranges: bytes
Content-Length: 19
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive

1234567891234567 89

+HTTPGET: 100,200,19
//Send a POST request to the HTTP (S) server
AT+HTTPCFG="responseheader",0 //Turn off the output HTTP response header information
OK

AT+HTTPURL=37,80 //Set the URL to access and set a timeout of 80 seconds
CONNECT
http://leshan.eigencomm.com:8888/test //write URL
OK

AT+HTTPPOST=10,80 //Send HTTP POST request to get the P O S T request body via the UART
CONNECT
1234567890 //Enter a POST request body

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OK

AT+HTTPREAD
OK
CONNECT

+HTTPGET: 0,200,19
123456789123456789

+HTTPGET: 100,200,19

21.5 SSL Demo

ECRDY

AT+CEREG?
+CEREG: 0,1

OK

AT+CGPADDR
+CGPADDR: 1,"100.101.139.181"

OK

AT+SSLCFG="ciphersuite",0,0xffff
OK

AT+SSLCFG="ciphersuite",0
+SSLCFG: "ciphersuite",0,0XFFFF

OK

AT+SSLCFG="seclevel",0,0
OK

AT+SSLOPEN=1,0,0,58.246.1.50,63756
OK

AT+SSLSEND=0,5
>
12345
OK

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+SSLURC: "recv",0,8
00123333
AT+SSLCLOSE=0
OK

21.6 FILE Demo

ECRDY

AT+FOPEN="/test.txt",0 //open file
+FOPEN: 1

OK

AT+FWRITE=1,10,60 //write data to file
CONNECT
write data…
+FWRITE: 10,10

OK

AT+FCLOSE=1 //close file
OK

AT+FOPEN="/test.txt",0 //open file
+FOPEN: 1

OK

AT+FREAD=1 //read file
CONNECT 10
1234567890

OK

21.7 LYNQ MQTT Demo

ECRDY

AT+CEREG?
+CEREG: 0,1

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OK

AT+CGPADDR
+CGPADDR: 1,"10.143.223.70"

OK

AT+MCONFIG=clientid,username,password
OK

AT+MIPSTART=118.114.239.159,30128
OK

+MIPSTART: SUCCESS
AT+MCONNECT=0,30
OK

+MCONNECT: SUCCESS
AT+MSUB=mobiletek/topic,1
OK

+MSUB: SUCCESS
AT+MPUB=mobiletek/topic,1,0,12345
OK

+MPUB: SUCCESS

+MSUB: "mobiletek/topic",5 bytes,"12345"
AT+MIPCLOSE
OK

+MIPCLOSE: SUCCESS

21.8 LYNQ HTTP Demo

HTTP GET demo :

ECRDY

AT+CEREG?
+CEREG: 0,1

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OK
AT+CGPADDR
+CGPADDR: 1,"10.143.223.70"
OK

AT$HTTPOPEN
OK

AT$HTTPPARA=http://182.148.114.87/file/zipaa,6600,0,0
OK

AT$HTTPACTION= 0
OK

$HTTPRECV:DATA,2048
HTTP/1.1 200 OK
Accept-Ranges: bytes
Cache-Control:
no-cache
Connection: keep-alive
Content-Length: 14615
Content-Type: text/html

AT$HTTPCLOSE
OK

HTTP POST demo:
ECRDY

AT+CEREG?
+CEREG: 0,1

OK
AT+CGPADDR
+CGPADDR: 1,"10.143.223.70"
OK

AT$HTTPOPEN
OK

AT$HTTPPARA=http://182.148.114.87/file/zipaa,6600,0,0
OK

AT$HTTPRQH=Content-Length,26

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OK

AT$HTTPRQH=Connection,keep-alive
OK

AT$HTTPACTION=1
OK

AT$HTTPDATA=13
>> name=mobilete
OK

AT$HTTPSEND
OK

AT$HTTPDATA=13
>> k&pass=123456
OK

AT$HTTPSEND
OK

AT$HTTPDATA=0
OK

AT$HTTPSEND

$HTTPRECV:DATA,153
HTTP/1.1 200 OK
Server: ...
Date: Tue, 20 Sep 2016
05:37:48 GMT
$HTTPRECV:DATA,2
$HTTPRECV:DATA 195
…
OK

AT$HTTPCLOSE
OK

21.9 FTP Demo

AT+CFTPPORT=6521

OK

AT+CFTPSERV="58.246.1.50"
OK

AT+CFTPUN="FTP_TST"
OK

AT+CFTPPW="FTPTST_0320"
OK

AT+CFTPLIST="/"
OK

+CFTPLIST:
-rw-rw-rw- 1user group 10 Jan 7 18:47 11.txt
-rw-rw-rw- 1 user group 1360 Jan 2 13:31 12.txt
...
+CFTPLIST:SUCCESS
AT+CFTPGETFILE="/jiang/test1.txt","/1.tx",0
OK

+CFTPGETFILE: SUCCESS, 31

AT+CFTPGET="/zs.txt",0
OK

+CFTPGET:DATA,15
qwertyqwerqerq

AT+CFTPPUTFILE="/zyqqqq.txt","/1.txt",0
OK

+CFTPPUTFILE:SUCCESS

AT+CFTPDELE="/zyqqqq.txt"
OK

AT+CFTPPUT="/Z1.txt",0,10
>>
1111111111
OK

+CFTPPUT: SUCCESS

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21.10 LYNQ TCP/IP Demo

//tcp connect operation
AT+CIPOPEN=0,"TCP",118.114.239.159,30128,0
OK

+CIPOPEN: SUCCESS,0
AT+CIPSEND=0,,,,123123123123123
OK

+CIPSEND:SUCCESS,0,15,15
AT+CIPRXGET=1,0 //set the way to get the network data manually
OK

+CIPRXGET:SUCCESS,1,0

+CIPRECV:1,0 //receive data from server
AT+CIPRXGET=2,0,10 //get data
+CIPRXGET:SUCCESS,2,0,6,0,
123123
OK
AT+CIPCLOSE=0
OK

+CIPCLOSE:SUCCESS,0

//udp connect operation
AT+CIPOPEN=0,"UDP",,,0
OK

+CIPOPEN: SUCCESS,0
AT+CIPSEND=0,,118.114.239.159,30128,1234
OK

+CIPSEND:SUCCESS,0,4,4

+CIPRECV:0,0,6, //receive data from server
123123
AT+CIPCLOSE=0
OK

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+CIPCLOSE:SUCCESS,0

//TCP Server operation
AT+CGPADDR
+CGPADDR: 1,"10.154.236.30"

OK
AT+SERVERSTART=10001,6,0
OK

+SERVERSTART: SUCCESS
AT+CIPOPEN=1,"TCP",10.154.236.30,10001,0
OK

+CIPOPEN: SUCCESS,1

+CLIENT: 0,0.0.0.0,0
AT+CIPSEND=1,5 //client send data by link_num 1
>
12345
OK

+CIPSEND:SUCCESS,1,5,5

+SERVERRECV:0,5, //server receive data
12345
AT+CIPSEND=0,5 //server send data by link_num 0
>
11111
OK

+CIPSEND:SUCCESS,0,5,5

+CIPRECV:0,1,5,
11111
AT+SERVERSTOP
OK

+SERVERSTOP:SUCCESS

21.11 MQTTS Demo

//EC MQTTS

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AT+SSLCFG="cacert",0,"/cacert.pem"
OK
AT+SSLCFG="clientcert",0,"/cert.pem"
OK
AT+SSLCFG="clientkey",0,"/key.pem"
OK
AT+SSLCFG="seclevel",0,2 //perform server and client authentication
OK
AT+ECMTCFG="ssl","enable" //enable MQTTS
OK
AT+ECMTOPEN=0,"118.114.239.159",30128
OK

+ECMTOPEN: 0,0
AT+ECMTCONN=0,"clientID","username","password"
OK

+ECMTCONN: 0,0,0
AT+ECMTPUB=0,520,2,1,"test","12345"
OK

+ECMTPUB: 0,520,0
AT+ECMTCLOSE=0
OK

+ECMTCLOSE: 0,0

//LYNQ MQTTS
AT+SSLCFG="cacert",0,"/cacert.pem"
OK
AT+SSLCFG="clientcert",0,"/cert.pem"
OK
AT+SSLCFG="clientkey",0,"/key.pem"
OK
AT+SSLCFG="seclevel",0,2
OK
AT+MQTTSSL=1,1
OK
AT+MCONFIG=clientid1,username1,password
OK
AT+MIPSTART=118.114.239.159,30128
OK

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+MIPSTART: SUCCESS
AT+MCONNECT=1,30
OK

+MCONNECT: SUCCESS
AT+MPUB=mobiletek/topic,1,0,12345
OK

+MPUB: SUCCESS
AT+MIPCLOSE
OK

+MIPCLOSE: SUCCESS

21.12 HTTPS Demo

//EC HTTPS
AT+SSLCFG="cacert",0,"/ca.crt"
OK
AT+SSLCFG="clientcert",0,"/client.crt"
OK
AT+SSLCFG="clientkey",0,"/client.key"
OK
AT+SSLCFG="seclevel",0,2
OK
AT+HTTPURL=38,60
CONNECT
https://118.114.239.159:30128/test.txt
OK
AT+HTTPGET
OK
AT+HTTPREAD
OK

CONNECT

+HTTPGET: 0,200,20
HTTP/1.1 200 OK
Date: Thu, 09 Feb 2023 06:10:26 GMT
Server: Apache/2.4.54 (Win64) LibreSSL/3.5.3
Last-Modified: Wed, 01 Feb 2023 06:56:10 GMT
ETag: "14-5f39df1828227"

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Accept-Ranges: bytes
Content-Length: 20
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/plain

This is http test!!!
+HTTPGET: 100,200,20

//LYNQ HTTPS
AT+SSLCFG="cacert",0,"/ca.crt"
OK
AT+SSLCFG="clientcert",0,"/client.crt"
OK
AT+SSLCFG="clientkey",0,"/client.key"
OK
AT+SSLCFG="seclevel",0,2
OK
AT$HTTPOPEN
OK
AT$HTTPPARA=https://118.114.239.159/test.txt,30128,1,1
OK
AT$HTTPACTION=0
OK

$HTTPRECV:DATA,296
HTTP/1.1 200 OK
Date: Thu, 09 Feb 2023 06:31:46 GMT
Server: Apache/2.4.54 (Win64) LibreSSL/3.5.3
Last-Modified: Wed, 01 Feb 2023 06:56:10 GMT
ETag: "14-5f39df1828227"
Accept-Ranges: bytes
Content-Length: 20
Keep-Alive: timeout=5, max=100
Connection: Keep-Alive
Content-Type: text/plain

$HTTPRECV:DATA,20
This is http test!!!

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21.13 CTWing MQTT Demo

//Non-SSL
AT+CTWPARAM=L511C_Y_v02.01b05.00,L511C_Y,EC618
OK
AT+CTWMQTTCFG=1,mqtt.ctwing.cn,1883,30,0,0
OK
AT+CTWMQTTREGPARAM=1559273901,01,ImojaucSOMapa2l_z9wpbJwElfThAo_Ev-vHbg4cAIk
OK
AT+CTWMQTTREG
OK
AT+CTWMQTTSEND=1,data_report,{"temperature":50}
OK

+CTWMQTTRECV: set_high_temp_value,{"taskId":5,"payload":{"high_temp_value":50}}
AT+CTWMQTTDEREG
OK

//ssl
AT+CTWPARAM=L511C_Y_v02.01b05.00,L511C_Y,EC618
OK
AT+CTWMQTTCFG=1,mqtt.ctwing.cn,8883,30,1,0
OK
AT+CTWMQTTREGPARAM=1559273901,01,ImojaucSOMapa2l_z9wpbJwElfThAo_Ev-vHbg4cAIk
OK
AT+CTWMQTTREG
OK
AT+CTWMQTTSEND=1,data_report,{"temperature":50}
OK

+CTWMQTTRECV: set_high_temp_value,{"taskId":6,"payload":{"high_temp_value":60}}
AT+CTWMQTTDEREG
OK

21.14 CTM2M Demo

AT+CTM2MSETPM=lwm2m.ctwing.cn,5683,38400 //Set Server Address , Port and Lifttime
OK
AT+CTM2MREG //Register to server
OK

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+CTM2M: reg,0

+CTM2M: obsrv,0
AT+CTM2MREG? //Check status
+CTM2MREG: 1

OK

AT+CTM2MSEND=020005000a00083131313131313131 //Send data to server
+CTM2MSEND: 23526

OK

+CTM2M: send,31,23526 //Send dats successful

+CTM2MRECV: 313233 //Receive server data
AT+CTM2MDEREG //Deregister
+CTM2M: dereg,0 //Deregister succeddful

OK

AT+CTM2MREG? //Check status
+CTM2MREG: 5

OK

21.15 OneNET Demo

AT+MIPLCREATE // Create instance
+MIPLCREATE: 0

OK
AT+MIPLADDOBJ=0,3303,1,1,4,0 // Add an object
OK
AT+MIPLOPEN=0,38400 // Register to server
OK

+MIPLEVENT: 0,1

+MIPLEVENT: 0,2 // Bootstrap success

+MIPLEVENT: 0,6 // Register successful

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+MIPLOBSERVE: 0,143727,1,3303,0,-1
AT+MIPLOBSERVERSP=0,143727,1 // Observe response
OK

+MIPLDISCOVER: 0,12656,3303
AT+MIPLDISCOVERRSP=0,12656,1,19,"5513;5514;5515;5516" // Discover response
OK

+MIPLEVENT: 0,11
AT+MIPLNOTIFY=0,143727,3303,0,5513,1,5,kkkkk,0,0,1 //Send data to server
OK

+MIPLEVENT: 0,26,1
AT+MIPLCLOSE=0 // Deregister
OK

+MIPLEVENT: 0,15 // Deregister succeddful
AT+MIPLDELETE=0
OK

21.16 LBS Demo

AT+CEREG?
+CEREG: 0,1

OK
AT+CGPADDR
+CGPADDR: 1," 100.104.224.200"

OK

AT +GTPOSTEST
+GTPOS: longitude: 104.0604451, latitude: 30.6537511

OK

AT+GTPOS=1,48c3559356018bc43e20f63e164893

CONNECT OK

OKAT+GTPOS=2,48c3559356018cbc43e20f63e1648f93
+GTPOS: longitude: 104.0604457,latitude: 30.6537519

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OK

AT+GTPOS0,48c3559356018cbc43e20f63e164893

OK

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